

Evaluation of 3D and 1D Atmospheric Radiative Transfer Using EarthCARE Data

An assessment using radiative transfer models to study how the atmosphere interacts with radiation and to evaluate different radiative transfer approximations.

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Abstract

Radiative transfer models (RTMs) are important tools for understanding atmospheric interactions. In this thesis, radiative assessments were performed using simulated fluxes from 3D and 1D RTMs in the libRadtran software package. The simulations were based on atmospheric data from the EarthCARE satellite and were used to study shortwave (SW) and longwave (LW) fluxes at the top-of-atmosphere (TOA) and at the surface. A key goal was to evaluate the EarthCARE input data (profiles of trace gases, clouds, aerosols, and surface albedo) by performing simulations with the 3D RTM MYSTIC and the 1D RTM DISORT and comparing the results to satellite observations at the TOA. A small but representative set of EarthCARE swaths was selected for the study. Overall, the simulations met the EarthCARE radiative-closure target of $\pm 10 \text{ W/m}^2$, with biases below 5 W/m^2 . Another objective was sensitivity studies of TOA fluxes with respect to aerosol loading and ice crystal habits. The model simulations showed that LW TOA fluxes could change by up to 25 W/m^2 when aerosols were included in the LW region. Similarly, sensitivity studies of ice clouds demonstrated that the choice of ice crystal habit led to local SW TOA flux differences of up to 60 W/m^2 and LW differences of around 4 W/m^2 . Simulations of surface SW fluxes, based on EarthCARE atmospheric input data, were also evaluated using ground-based pyranometer observations across Norway. Many of the satellite overpasses occurred at high latitudes and in regions with complex topography and surface albedo. On average, the biases between simulations and observations were within $25\text{-}30 \text{ W/m}^2$ for both 1D and 3D RTMs. Overall, model-vs.-observation agreement was better at TOA than at the surface. Finally, 3D and 1D RTM results were compared by studying selected flux fields that varied with altitude. The results showed that horizontal transport of radiation played an important role in modifying SW fluxes in selected inhomogeneous cloud scenes. The identified 3D effects included side illumination, slanted shadows, upward and downward escape, upward trapping, and entrapment. Overall, the study showed that EarthCARE data and RTMs provided a useful framework for radiative assessment and highlighted the importance of accurate atmospheric input data, model parameter choices, and 3D radiation transport.

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Chapter 1

Introduction

Looking out the window, it is clear that the atmosphere is far from simple. Clouds, aerosols, and gases scatter, absorb, and redirect radiation in ways that are not easy to see. What looks simple to the eye is actually a complex and constantly changing system. Even the calmest sky is full of hidden activity. The complex radiative system of Earth includes the external source of radiation from the sun, which drives weather, climate, and atmospheric chemistry, as well as thermal emission from the Earth’s surface and the atmosphere. The process of radiative transfer (RT) in the atmosphere is modified by many atmospheric components, ranging from gases to particles such as aerosols and clouds, as well as by interactions with the surface.

A wide range of RT applications is performed using radiative transfer models (RTMs) under different approximations. In many cases, RT in the atmosphere is represented as a one-dimensional (1D) process that depends only on the vertical coordinate, as used in climate models. W. Wiscombe put it nicely ([Wiscombe, 1983](#); [Stamnes et al., 2017](#)):

“[...] our job is not to make our models as complicated as nature herself; it is to simplify and idealize, in order to gain understanding. Plane-parallel cloud modeling is an entirely acceptable way of to do this.”

However, the Earth’s atmosphere, especially clouds, varies in all three spatial dimensions, which can only be represented in three-dimensional (3D) RTMs. The inhomogeneous nature of the atmosphere is most evident in broken cloud fields, where clouds can form and dissipate rapidly, but also in the non-uniform distribution of aerosols, for example around cities or volcanic eruptions. Neglecting horizontal inhomogeneities can be a good approximation in clear-sky conditions, but it is less accurate for partly cloudy scenes, highlighting an important limitation of 1D RTMs. Instead, a realistic treatment of an inhomogeneous atmosphere should use 3D RT ([Stamnes et al., 2017](#)). However, since there is no simple solution to the 3D radiative transfer equation (RTE) ([Mayer, 2009](#)), one approach is the Monte Carlo method, where photons are randomly sampled throughout the atmosphere ([Noebauer and Sim, 2019](#)). While 3D RT provides a more realistic description of atmospheric RT by including horizontal energy transport, fully resolving it comes at a high computational cost, forcing atmospheric models to balance accuracy and speed ([Veerman et al., 2020](#)). In addition, the limited availability of 3D atmospheric data further restricts the practical use of 3D RTMs ([Mayer, 2009](#)). So far, the high computational cost of Monte Carlo methods has limited their use in operational contexts. However, 3D RT methods are no longer considered unreasonable for solving RT in atmospheric models, although their operational use remains a long-term perspective ([Villefranche et al., 2023](#)). The Earth Cloud, Aerosol and Radiation Explorer

(EarthCARE) mission therefore marks a significant step forward in this direction as the first mission to use 3D RTMs operationally (Cole et al., 2023).

The EarthCARE satellite mission was launched in 2024 with the objective of improving our understanding of the role of clouds, aerosols, and radiation in the climate system (Wehr et al., 2023). Validation of 3D RTMs is challenging, as it requires realistic atmospheric conditions together with co-located radiative observations (Mayer, 2009). EarthCARE directly addresses this challenge by providing 3D atmospheric scenes extending across-track, together with co-located shortwave (SW) and longwave (LW) flux observations (Wehr et al., 2023). The main goal of EarthCARE is to measure clouds and aerosols well enough to simulate top-of-atmosphere (TOA) SW and LW fluxes within $\pm 10 \text{ W/m}^2$ (Barker et al., 2025). This makes it a unique and powerful framework for applying and evaluating 3D RTMs, and for addressing differences with 1D models.

Through the *Norwegian initiative for EarthCARE Validation of Aerosol uncertainties and Radiation products in the Arctic* (NEVAR) (NILU, 2022–2027), the research institute NILU supports the geophysical validation of EarthCARE products. In line with this, NEVAR aims to improve the understanding of cloud-aerosol-radiation interactions and the Earth’s radiative balance. As a part of the NEVAR project, this Master’s project aims to study interactions between radiation, clouds, and aerosols using different parameterizations and model setups in RTMs. In this thesis, we study how well 3D and 1D RTMs perform under different atmospheric conditions by comparing with satellite observations and ground-based measurements. We also compare the two approaches directly, investigating 3D radiative effects of clouds. The aim is to quantify differences between 3D and 1D RT results and to understand how horizontal transport of radiation modifies radiative properties. The thesis also quantifies the radiative impact of aerosol loading in the LW region and examines how the choice of ice crystal habit affects TOA fluxes in both the SW and LW regions. These efforts aim to improve the assessment and understanding of cloud and aerosol radiative effects and to advance RT modeling in the atmosphere.

The RT simulations were performed using the *libRadtran* RT library (Mayer and Kylling, 2005; Emde et al., 2016), with the 1D RTM DISORT (Stamnes et al., 2000) and the 3D Monte Carlo RTM MYSTIC (Mayer, 1999; Mayer and Kylling, 2005; Mayer, 2009). The simulations were used for comparisons with observations and for comparisons between RTMs. The use of a 3D RTM in this work was motivated not only by its more realistic representation of the atmosphere, but also by its relevance for future developments in atmospheric modeling.

The structure of this thesis is as follows: Chapter 2 presents the theoretical background for the radiative assessments conducted in this study. The methodology is described in Chapter 3. The results are presented and discussed in Chapter 4. Finally, the main findings are summarized in Chapter 5.

Chapter 2

Theory

2.1 Radiative Transfer Theory

Accurate simulation of radiative transfer (RT) is important because solar (shortwave; SW) and terrestrial (longwave; LW) radiation drive atmospheric circulation and influence weather and climate. Such simulations are used in many applications, including satellite remote sensing, ultraviolet forecasting, radiative forcing, photolysis rates, and radiative heating and cooling (Mayer and Kylling, 2005). It is also important for biological effects, as received radiative energy can initiate processes such as skin damage from ultraviolet radiation (Stamnes et al., 2017). Radiative transfer models (RTMs) are also used to determine the total ozone column (Svendby et al., 2021) and for radiative assessment through model-model or model-observation comparisons, such as in the EarthCARE mission (Wehr et al., 2023; Cole et al., 2023; Barker et al., 2025) and in this Master's project.

2.1.1 Fundamentals of Radiative Transfer

The current section describes the fundamentals of RT and is adapted from Stamnes et al. (2017). Basic variables are introduced, and the roles of thermal emission, absorption, and scattering in shaping the radiation field are presented.

Basic State Variables

Radiative energy flow is described by the information carrier of the radiation field, the (scalar) radiance I_ν , also referred to as intensity. The radiance is defined as the energy per unit projected area, per unit time, per unit frequency, per unit solid angle:

$$I_\nu = \frac{d^4E}{dA \cos \theta dt d\nu d\Omega} \quad [\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}], \quad (2.1)$$

where dE is the radiant energy transported through area dA during the time interval dt . The angle θ denotes the angle between the beam direction and the surface normal of dA (hence the factor $\cos \theta$ for projected area). θ is commonly referred to as the zenith (or polar) angle, see Fig. 2.1. The terms $d\nu$ and $d\Omega$ denote the frequency interval and the solid-angle element, respectively.

Spectral net flux (or irradiance) F_ν is obtained by angular integration of the radiance:

$$F_\nu = \int_{4\pi} I_\nu(\Omega) \cos \theta d\Omega \quad [\text{W m}^{-2} \text{Hz}^{-1}]. \quad (2.2)$$

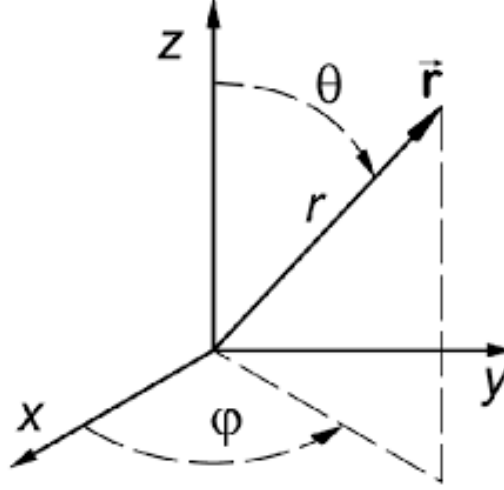


Figure 2.1: Schematic illustration of the zenith angle θ (polar angle) and the azimuth angle ϕ . Figure taken from [Mayer \(2009\)](#).

The spectral net flux can be separated into two spectral *hemispherical fluxes*: the upward (or upwelling) flux ($F_\nu^\uparrow = F_\nu^+$) and the downward (or downwelling) flux ($F_\nu^\downarrow = F_\nu^-$). These represent the positive energy flows per unit area within the spectral interval $\nu \in [\nu, \nu + d\nu]$ during the time interval $t \in [t, t + dt]$. The hemispherical fluxes are defined by integrating the radiance over their respective hemispheres:

$$F_\nu^\pm = \pm \int_{\pm} I_\nu(\Omega) \cos \theta d\Omega = \frac{d^3 E^\pm}{dA dt d\nu}. \quad (2.3)$$

Here, $\int_+$ denotes integration over the upper hemisphere and $\int_-$ over the lower hemisphere.

The total flux is obtained by integrating the spectral net or hemispherical flux over all frequencies, yielding F , $F^\uparrow$, and $F^\downarrow$. This removes the explicit frequency-dependence. The flux can also be limited to the shortwave (SW) or longwave (LW) region by integrating over the corresponding frequency intervals, yielding F_{SW} and F_{LW} .

Radiative Transfer Equation

The primary task of RTMs is to solve the radiative transfer equation (RTE) under various approximations and parameterizations. The RTE describes how radiance changes along a radiation path due to extinction, emission, and scattering. The RTE can be written in compact form with two terms: one that removes (sink) and one that adds (source) radiance along the radiation path:

$$\frac{dI_\nu(\tau, \Omega)}{d\tau} = \underbrace{-I_\nu(\tau, \Omega)}_{\text{sink (extinction)}} + \underbrace{S_\nu(\tau, \Omega)}_{\text{source (emission + scattering)}}, \quad (2.4)$$

where the change in radiance I_ν depends on the loss of radiation due to extinction, and the source of radiation from the source function S_ν , which includes both thermal emission and scattering. The radiance I_ν at frequency ν in direction Ω is defined at optical depth τ . In differential form, the optical depth is defined as

$$d\tau = \kappa_{\text{ext}}(\nu) ds, \quad (2.5)$$

where ds is the differential geometrical path length, and the extinction coefficient $\kappa_{\text{ext}}(\nu)$ describes the proportional attenuation of radiance per unit path length. The extinction coefficient has units of m^{-1} , making the optical depth τ a dimensionless quantity.

It is useful to build intuition for the RTE by examining its two parts: the sink and source terms. The sink term in the RTE appears as the extinction law (Beer-Lambert law) when going from the optical-depth form ($d\tau$) to the path-length form (ds) using the relation in Eq. 2.5:

$$\frac{dI_\nu}{ds} = \underbrace{-\kappa_{\text{ext}} I_\nu}_{\text{extinction law}} + \dots \quad (2.6)$$

This term describes the loss of radiation due to absorption and scattering.

The source term represents processes that add radiance in the direction of interest due to thermal emission (S_{th}) and scattering (S_{sc}). The source function S_ν can be written as

$$S_\nu(\tau, \Omega) = \underbrace{(1 - \omega(\nu)) B_\nu(T)}_{S_{th}} + \underbrace{\frac{\omega(\nu)}{4\pi} \int_{4\pi} p_\nu(\Omega, \Omega') I_\nu(\Omega') d\Omega'}_{S_{sc}}. \quad (2.7)$$

Here, the thermal contribution S_{th} includes the factor $(1 - \omega) = \sigma_a / (\sigma_a + \sigma_s)$, which represents the fraction of total extinction due to absorption. σ_a and σ_s are the absorption and scattering coefficients, and the total extinction is given by $\kappa_{\text{ext}} = \sigma_a + \sigma_s$. The Planck function $B_\nu(T)$ describes thermal emission at temperature T . The term $(1 - \omega)B_\nu$ therefore gives the local thermal emission of the medium following Kirchhoff's law for volume absorption and emission (for further details, see e.g. [Stamnes et al. \(2017\)](#)).

The scattering term S_{sc} represents the scattered contribution from all incoming directions Ω' (the angular integral) that is scattered into the direction of interest Ω . This term is weighted by the fraction of total extinction due to scattering, given by the single-scattering albedo $\omega(\nu) = \frac{\sigma_s}{\sigma_a + \sigma_s}$. The scattering phase function $p_\nu(\Omega, \Omega')$ describes the probability of scattering from direction Ω' into Ω . It is normalized such that $\int_{4\pi} \frac{p_\nu(\Omega, \Omega')}{4\pi} d\Omega' = 1$.

The RTE in full form can be written as

$$\frac{dI_\nu}{d\tau} = \underbrace{-I_\nu}_{\text{sink (extinction)}} + \underbrace{(1 - \omega(\nu)) B_\nu(T) + \frac{\omega(\nu)}{4\pi} \int_{4\pi} p_\nu(\Omega, \Omega') I_\nu(\Omega') d\Omega'}_{\text{source (emission + scattering)}}. \quad (2.8)$$

Key Radiative Transfer Variables

Some key variables in atmospheric RT are emittance, reflectance, absorptance, and the asymmetry parameter, introduced below. Kirchhoff's Law states that the emittance ϵ is equal to the absorptance α , an identity that can be considered exact for planetary surfaces:

$$\epsilon(\nu, \Omega', T_{\text{surface}}) = \alpha(\nu, -\Omega', T_{\text{surface}}), \quad (2.9)$$

where T_{surface} is the surface temperature.

For an opaque surface, i.e., a surface that does not transmit radiation, energy conservation gives a general relation between the flux reflectance (albedo) ρ , the absorptance α , and the emittance ϵ :

$$\rho(\nu, -\Omega', 2\pi) = 1 - \alpha(\nu, -\Omega', T_{\text{surface}}) \quad (2.10)$$

$$= 1 - \epsilon(\nu, -\Omega', T_{\text{surface}}), \quad (2.11)$$

where the second equality follows from Kirchhoff's law (Eq. 2.9).

The asymmetry parameter g is derived from the scattering phase function $p(\cos \Theta)$:

$$g = \langle \cos \Theta \rangle = \frac{1}{2} \int_{-1}^1 \cos \Theta p(\cos \Theta) d \cos \Theta, \quad (2.12)$$

which represents the mean cosine of the scattering angle Θ . It characterizes the degree of forward versus backward scattering and thus quantifies the angular asymmetry of scattering. A value of $g = 1$ indicates purely forward scattering, $g = 0$ indicates isotropic scattering (i.e., independent of direction), and $g = -1$ indicates purely backward scattering.

2.1.2 1D Radiative Transfer

The following description of 1D RT is adapted from [Stamnes et al. \(2017\)](#).

In the most commonly used geometry in RT, namely the plane-parallel approximation, the atmosphere is represented as a set of horizontally homogeneous slabs, referred to as “slab geometry”. The optical properties depend only on the vertical coordinate and are therefore referred to as a 1D approximation. Figure 2.2 shows an illustration of the 1D RT with slab geometry. The plane-parallel approximation may seem coarse, but gravity produces a density stratification in the atmosphere, which provides a physical basis for the slab-geometry assumption. The independent pixel (column) approximation (IPA/ICA) is obtained by applying the plane-parallel approximation independently to each vertical column in the atmosphere, thereby accounting for local atmospheric heterogeneity within each pixel ([Várnai and Davies, 1999](#)).

It is often convenient to express the radiance in terms of half-range quantities, separating it into upward (I_ν^+) and downward components (I_ν^-). The polar angle dependence is described by $u = \cos \theta$, and $\mu = |u|$ is used when only the magnitude is needed. The slant optical depth along a ray τ is related to the vertical optical depth τ_z by

$$\tau_z = \mu \tau, \quad (2.13)$$

using the geometrical relation $dz = \mu ds$ and that $d\tau_z = \kappa_{\text{ext}}(\nu) dz$. Consequently, one can use the vertical optical depth τ_z as the independent variable in the RTE with the μ factor for angular dependency.

In compact $\pm$ notation, the two half-range equations for a plane-parallel atmosphere are

$$\pm \mu \frac{dI_\nu^\pm}{d\tau_z} = I_\nu^\pm - S_\nu^\pm. \quad (2.14)$$

The corresponding source function is given by

$$\begin{aligned} S_\nu^\pm &= S_\nu(\tau, \pm\mu, \theta) \\ &= (1 - \omega) B_\nu \\ &\quad + \frac{\omega}{4\pi} \int_{\pm} p_\nu(\Omega, \Omega') I_\nu^\pm d\Omega' \\ &\quad + \frac{\omega}{4\pi} \int_{\pm} p_\nu(-\Omega, \Omega') I_\nu^\mp d\Omega'. \end{aligned} \quad (2.15)$$

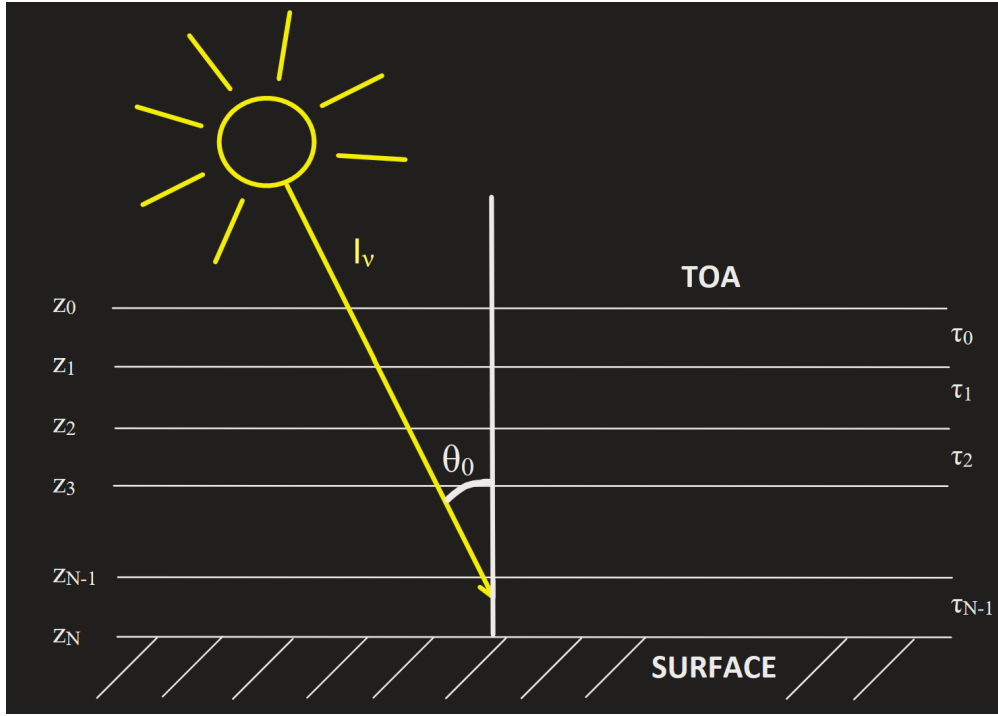


Figure 2.2: Illustration of a slab geometry for 1D RT in the atmosphere. The atmosphere is divided into N homogeneous layers, extending from the top-of-atmosphere (TOA) to the surface. The vertical coordinate is altitude z and the corresponding vertical optical depth τ . The yellow line shows the solar radiance I_ν at solar zenith angle θ_0 .

The hemisphere integrals are defined as

$$\begin{aligned} \int_+ d\Omega' &\equiv \int_0^{2\pi} \int_0^1 d\mu' d\phi', \\ \int_- d\Omega' &\equiv \int_0^{2\pi} \int_{-1}^0 d\mu' d\phi', \end{aligned}$$

with directions specified by the polar angle $\mu = |\cos \theta|$ and azimuthal angle ϕ (see Fig. 2.1). The quantities $I_\nu^\pm = I_\nu(\tau_z, \pm\mu, \theta)$ represent the half-range radiances, with the convention I_ν^+ for $u > 0$ (upward directions) and I_ν^- for $u < 0$ (downward directions).

It is worth noting that under the assumption of slab geometry, the flux and mean radiance depend only on azimuthally averaged quantities. This reduces the RTE to its azimuthal independent form, effectively removing the ϕ -variable by introducing the azimuthal averaged radiance

$$I_\mu^\pm(\tau, \mu) \equiv \frac{1}{2\pi} \int_0^{2\pi} d\phi' I_\mu^\pm(\tau, \mu, \phi'), \quad (2.16)$$

which also transforms related quantities in the RTE, such as the phase function and the source function, into their azimuthal averaged forms.

When the solar zenith angle (SZA) is below about 70° , the plane-parallel approximation is generally reasonable (Dahlback and Stamnes, 1991; Mayer and Kylling, 2005). For larger SZA, the pseudospherical approximation provides a solution by treating the direct solar beam in spherical geometry to account for Earth's curvature while retaining a plane-parallel treatment for multiple scattering (Mayer and Kylling, 2005).

The N-stream formulation of the RTE is an approximation that simplifies the angular dependence of the radiance by discretizing it into an even number N of propagation directions, each representing an angular average obtained by Gaussian quadrature. The angular integral of the radiance is approximated by a finite sum over discrete directions, i.e., “streams”, using numerical quadrature formula:

$$\int_{-1}^1 I_\nu(\tau, u) du \approx \sum_{j=1}^N w_j I_\nu(\tau, u_j), \quad (2.17)$$

where w_j are quadrature weights and u_j are the discrete ordinates.

As mentioned in Stamnes et al. (2017), the Discrete-Ordinates Method was first introduced by Wick (1943) and further developed by Chandrasekhar (1960). The method uses a quadrature approximation (Stamnes et al., 2000), as shown in Eq. 2.17. This way, the RTE can be expressed compactly in a matrix formulation for an N-stream system, reducing the RTE to a set of coupled first-order differential equations. Solving the resulting matrix form of the RTE becomes an eigenvalue problem of order $N \times N$.

2.1.3 3D Radiative Transfer

The atmosphere is, perhaps unsurprisingly, 3D and so are clouds. The 1D approximation assumes an atmosphere that varies only with height (see Sect. 2.1.2) and therefore has clear limitations. To account for spatial variability in all directions, 3D RTMs are needed to simulate how radiation propagates through the atmosphere (Várnai and Davies, 1999). The following description of the Monte Carlo method for 3D RT in the atmosphere is based primarily on Mayer (2009).

With no simple closed-form solution to the 3D RTE, the Monte Carlo method is one of the most widely used approaches. Unlike 1D RTMs, 3D Monte Carlo methods have the ability to represent nearly arbitrarily complex media and radiative processes in the atmosphere and at the surface without simplifying assumptions. A simple illustration of RT in a 3D Monte Carlo model is shown in Fig. 2.3.

Monte Carlo RTMs trace individual photons randomly through the atmosphere, in contrast to the deterministic 1D approach that solves the RTE as differential equations described in Sect. 2.1.2. The random sampling of photon paths is propagated in physical space based on the 3D distribution of the extinction coefficient. Each photon interaction is governed by probability functions that describe interactions with the surface and with atmospheric constituents such as molecules, aerosols, and cloud particles. In this context, “photons” are test particles that behave analogously to real photons (Noebauer and Sim, 2019).

The 3D computational domain used for photon propagation is, unfortunately, not infinite in size. To account for this, periodic horizontal boundary conditions are applied: when a photon exits the domain through one side, it re-enters from the opposite side. These boundary conditions ensure energy conservation and are appropriate for sufficiently large domains where edge effects on the simulated radiative processes become negligible.

The stochastic tracing of photons makes Monte Carlo simulations probabilistic in nature, which leads to noisy results, referred to as Monte Carlo noise (Noebauer and Sim, 2019). This noise, quantified by the standard deviation σ , scales as

$$\sigma \propto \frac{1}{\sqrt{N}}, \quad (2.18)$$

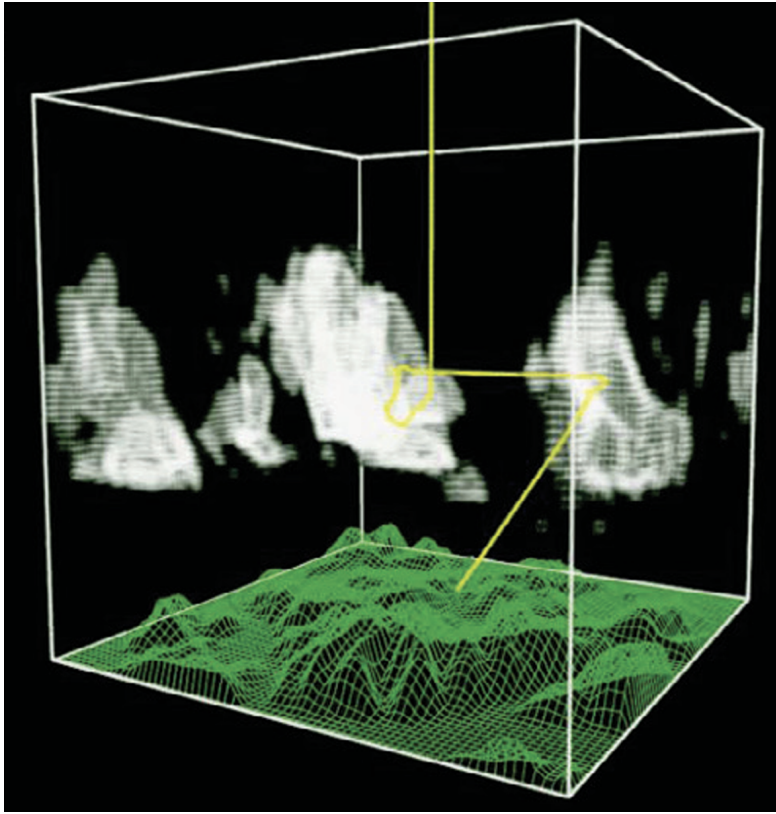


Figure 2.3: Illustration of photon tracing (yellow line) in a 3D RTM with a gridded atmosphere. Figure taken from [Mayer \(2009\)](#).

for a large number of photons N . Thus, doubling the number of photons reduces the noise by a factor of $1/\sqrt{2} \approx 0.71$, corresponding to an improvement of about 30%.

The Monte Carlo approach comes at a price. While it is more realistic, it is often computationally more expensive than 1D methods. The computational cost depends on several factors:

1. The computational cost increases with the number of interactions per photon and, naturally, with the number of photons ([Noebauer and Sim, 2019](#)). Since photons are traced explicitly, higher optical thickness increases the computational cost ([Noebauer and Sim, 2019](#)). For example, clouds lead to more scattering events and thus increase the computational time ([Mayer, 2009](#)).
2. More sampled grid boxes of the 3D computational domain increases the computational cost ([Mayer, 2009](#)).
3. The requested output also affects the computational cost. Domain-averaged quantities can be computed relatively efficiently, sometimes comparable to 1D methods. In contrast, spatially resolved fields (e.g., horizontal distributions or spatially resolved radiances) can become very expensive ([Mayer, 2009](#)).

The computational cost can be reduced by lowering the number of photons, but at the expense of increased statistical noise in the simulated radiative quantities ([Noebauer and Sim, 2019](#)).

The versatile Monte Carlo approach is not necessarily the preferred method for all applications. Simpler 1D methods, such as the independent pixel approximation

(IPA), can be sufficient for quantities averaged over large areas (Cahalan et al., 1994). In contrast, 3D RTMs are required when the radiation interaction is more complex. This includes situations with horizontal photon transport (Várnai and Davies, 1999), spatially resolved radiances in 3D cloud fields, high-resolution remote sensing of clouds, topographic effects on radiation, and passive remote sensing over inhomogeneous surfaces (Mayer, 2009).

2.2 Radiative Transfer in the Atmosphere

Radiative transfer in Earth System Models (ESMs) is usually calculated using the independent pixel approximation (IPA) (Singer et al., 2021). ESMs are numerical frameworks that simulate physical, chemical, and biological processes of the Earth system and extend beyond traditional Global Climate Models (GCMs). The IPA neglects horizontal transfer of radiation between adjacent columns, and scattering is confined to each isolated vertical 1D column (Singer et al., 2021), yet it is widely used because it often obtains averaged results over multiple individual columns that closely match full 3D calculations across diverse cloud types and conditions (Cahalan et al., 2005). However, multiple studies have shown that neglecting horizontal photon transport and 3D cloud morphology can introduce significant radiative biases (Di Giuseppe and Tompkins, 2003; Barker et al., 2012; Okata et al., 2017; Gristey et al., 2020; Singer et al., 2021; Ademakinwa et al., 2024). It is also becoming more realistic that Monte Carlo 3D RTMs could be used to solve 3D RT in atmospheric models (Villefranque et al., 2023), underscoring their growing importance for accurate RT simulations.

Clouds interact with radiation in complex ways and are often important for many applications (Mayer, 2009). Clouds cover roughly 70% of Earth’s surface and account for about half of the reflected solar flux back to space (Singer et al., 2021). Clouds have a large impact on Earth’s radiation budget, producing a cooling effect of about 20 W/m^2 . This is much larger than the radiative forcing from a doubling of CO_2 , estimated at around 4 W/m^2 (IPCC, 2021). Radiative forcing is defined as the difference between incoming and outgoing top-of-atmosphere (TOA) fluxes (Stamnes et al., 2017). The SW reflection by clouds depends on their location, optical properties (i.e., extinction coefficient, scattering phase function, and single-scattering albedo (Mayer and Kylling, 2005)), and their 3D structure (Singer et al., 2021). Because ESMs typically operate at horizontal resolutions of 10-100 km and vertical resolutions of 100-200 m in the lower troposphere, they cannot resolve the fine-scale structure of clouds, which is on the order of tens of meters or less (Singer et al., 2021). This limits the realistic representation of clouds in global models (Winker et al., 2017) and highlights the need to quantify the radiative impact of 3D cloud structures.

Simplified radiative schemes, such as the IPA, restrict photon transport to the vertical column, omitting key processes such as cloud-side illumination (Singer et al., 2021) and slant-angle shadowing (Veerman et al., 2020). In contrast, full 3D RT allows photons to encounter clouds from the side, increasing the effective cloud cover and producing larger shadows, which increases reflectance compared to 1D methods (Singer et al., 2021). However, this is only part of the picture, as clouds interact with radiation in complex ways. More detailed discussion of 3D cloud radiative effects is given in Sect. 2.2.3.

Systematic differences arise between 3D and 1D RT treatments, especially in cloudy atmospheres and in the SW region (Cole et al., 2023). For example, Singer et al. (2021) reported a net TOA SW flux bias from neglecting 3D cloud effects, peaking at low

SZA and in deep cloud regimes, with an annual-mean tropical value of $3.1 \pm 1.6 \text{ W/m}^2$, reaching up to 6.5 W/m^2 locally.

2.2.1 Aerosol Radiative Impact

Aerosol effects are generally stronger in the SW region, but are not negligible in the LW region under high aerosol loading (Zhou and Savijärvi, 2014). This Master's project uses aerosols in the SW simulations and also examines the LW sensitivity to aerosols.

Scattering in the LW region is often neglected in most mainstream GCMs to reduce computational cost, because atmospheric LW absorption is strong and usually dominates over scattering (Kuo et al., 2020). However, following Zhou and Savijärvi (2014), aerosols also interact with terrestrial radiation and produce a LW radiative effect, although this effect is weaker than in the SW and less well known than the LW effect of clouds. Aerosols exert two opposing influences: (1) they increase downward LW radiation at the surface, contributing to atmospheric warming, while (2) simultaneously increasing planetary albedo through scattering in the SW spectrum, leading to a net cooling. Their LW impact is strongest in the atmospheric window regions (i.e., transparent spectral regions, approximately $8\text{--}12 \mu\text{m}$ (Stamnes et al., 2017)), and largely ineffective in opaque absorption bands. The combined radiative effect depends on aerosol type, vertical distribution, and concentration. Dense near-surface aerosol layers have been shown to exhibit LW radiative behavior similar to optically thin clouds at comparable altitudes, whereby absorption of LW radiation followed by thermal re-emission enhances downward LW radiation and reduces outgoing LW radiation. Although the magnitude of the LW radiative effect is generally small and difficult to observe, it depends on the aerosol concentration and becomes non-negligible under conditions of high aerosol loading.

Ångström Power-Law and Ångström Exponent

The spectral dependence of aerosol extinction can be approximated using the Ångström power-law relationship, which is used to extrapolate aerosol optical depth (AOD) within the SW spectral region (Schuster et al., 2006). The Ångström power-law, originally introduced by Ångström (1929), expresses the AOD as a function of wavelength according to

$$\tau(\lambda) = \tau(\lambda_0)\lambda^{-\alpha}, \quad (2.19)$$

where $\tau(\lambda_0)$ denotes the AOD at the reference wavelength λ_0 , and the exponent α is referred to as the Ångström exponent (Ångström, 1964; Schuster et al., 2006).

As described by Schuster et al. (2006), the Ångström exponent gives a general measure of the aerosol particle size. Values below 1 are characteristic of size distributions dominated by coarse-mode aerosols with particle radii exceeding $0.5 \mu\text{m}$, such as dust and sea salt, whereas values above 2 indicate fine-mode aerosols with particle radii below $0.5 \mu\text{m}$, such as urban pollution.

2.2.2 Radiative Impact of Ice Crystal Habit

Ice clouds are mainly composed of non-spherical particles with different habits (shapes) (Um et al., 2015). The microphysical properties, such as the ice-particle size distribution, particle roughness, and habit distribution, affect the radiative properties of ice clouds (Wang et al., 2014).

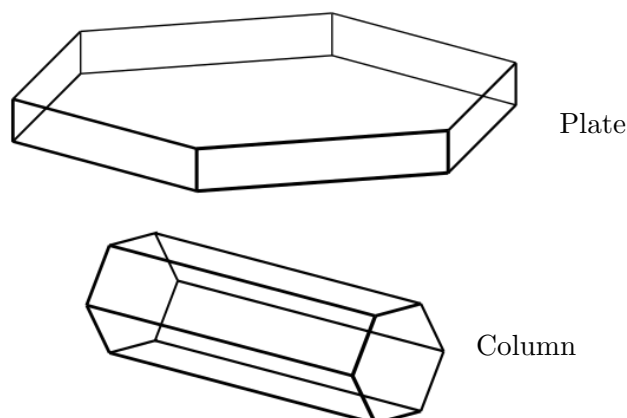


Figure 2.4: Two hexagonal ice crystals: plate and column. Image from [Wikimedia Commons](#), public domain.

In situ observations demonstrate that the relative fractions of pristine ice habits vary markedly across geographic regions such as the Arctic, mid-latitudes, and tropics (Um et al., 2015). Pristine habits are ice crystals grown by vapor deposition and include habits such as plates, columns, needles, and dendrites (Lamb and Verlinde, 2011).

Since a single pristine habit is not representative of all cloud structures and geographical conditions (Um et al., 2015), and aggregates and roughened particles are frequently retrieved in both in situ and satellite observations (Korolev et al., 1999; Cole et al., 2014; van Diedenhoven, 2014), a general habit mixture (GHM) is an approach to represent this variability in natural clouds. It provides consistent agreement with observations over a wide range of scattering angles (Cole et al., 2014), and comparisons between observed and simulated phase functions show that such mixtures can reduce differences between measurements and modeled scattering behavior (Wang et al., 2014).

Wendisch et al. (2005) reported that modeled upward and downward flux spectra above cirrus generally agreed with measurements within about 10% across multiple ice habits, although larger deviations were also found. They further show that habit signatures in reflected flux are most apparent for optically thin cirrus at small SZA, when single scattering dominates, where larger differences between habits occur. As optical thickness increases or SZA grows (longer slant paths), multiple scattering intensifies and progressively suppresses habit-specific features outside absorption bands.

As RT in the atmosphere differs between the SW and LW regions, several distinctions are worth noting. Ice absorbs less in the shortwave infrared (SWIR: $0.75\text{--}3\text{ }\mu\text{m}$) than in the LW region. Therefore, SW RT is mainly dominated by scattering and is more sensitive to ice crystal habit and surface roughness (Wang et al., 2014). Nevertheless, the LW radiative impact of ice crystal habit should not be completely overlooked. Following Wendisch et al. (2007), habit-induced effects can be significant under specific conditions, with larger differences evident for high, optically thin cirrus, while they are small for low, optically thick cirrus. The LW sensitivity is strongest in the atmospheric window (around $8\text{--}12\text{ }\mu\text{m}$) and weak to negligible in regions with strong gas absorption.

2.2.3 3D Cloud Radiative Effect

The impact of clouds on atmospheric radiation is determined by their capacity to scatter, absorb, and emit radiative energy. Through scattering and absorption, clouds

redistribute incoming SW radiation and alter the spatial distribution of direct and diffuse fluxes, while also absorbing and emitting LW radiation (Veerman et al., 2020). A glance out of the window shows that clouds are far from uniform and are naturally heterogeneous, and their radiative properties depend strongly on their 3D structure (Stephens et al., 2005). Realistic RT of inhomogeneous clouds therefore requires 3D RT modeling (Emde et al., 2022). A 1D RT approach does not provide a fully realistic representation of such inhomogeneous scenes as horizontal photon transport often plays an important role (Várnai and Davies, 1999).

Following Várnai and Davies (1999), we define the effect from horizontal transport of radiation between neighboring pixels as the difference between the 3D RTM solution and the 1D solution obtained with the IPA. We refer to this difference as the 3D effect:

$$\text{3D effect} = F_{3D} - F_{1D}, \quad (2.20)$$

where $F_{3D/1D}$ refers to the simulated fluxes, either upward or downward, from 3D or 1D simulations. This definition is consistent with Villefranque et al. (2023) and Emde et al. (2022). As described by Várnai and Davies (1999), the impact of horizontal transport on cloud reflection and transmission is generally described through three fundamental effects associated with changes in photon path:

1. The inclusion of horizontal photon transport modifies the photon paths, which affects absorption and thus the amount of radiation available for reflection and transmission.
2. Changes in photon paths can lead to fewer or more scattering events, thereby impacting the angular distribution of reflected and transmitted radiation.
3. Changes in photon paths affect where photons leave the cloud, thereby modifying the spatial distribution of reflected and transmitted radiation.

The simplest mechanism to understand 3D effects is cloud-side illumination (Hogan et al., 2019). In 3D RT, more sunlight reaches the cloud-sides, while in 1D RT with the top-down view of the IPA it does not. This increases the effective cloud cover, larger shadows, brighter cloud-sides, and enhances reflection to space in 3D RT (Singer et al., 2021). The effect increases with SZA as more incoming solar radiation intercepts the cloud (Hogan and Shonk, 2013).

The model by Várnai and Davies (1999) is presented here, which separates the 3D effect on SW radiation into four mechanisms: upward and downward trapping (M1, M2), and upward and downward escape (M3, M4). In addition, a fifth mechanism (M5), called entrapment, is included as proposed by Hogan et al. (2019).

Upward and Downward Trapping (M1, M2)

Radiation moving from thinner to thicker cloud regions can undergo additional scattering in 3D RT compared to the 1D case, which increases photon pathlength and reduces photon escape. This trapping mechanism is divided into upward trapping (UT) and downward trapping (DT). UT reduces reflection, while DT reduces transmission in the 3D simulation.

Upward and Downward Escape (M3, M4)

Photons moving from thicker to thinner cloud regions can escape the cloud with shorter pathlengths in 3D than in 1D simulations. This mechanism is called escape and is divided into upward escape (UE) and downward escape (DE). UE increased reflection, while DE increased transmission in the 3D simulation.

More specifically, in 3D RT, photons can also escape through cloud sides, unlike in the IPA, where radiation is confined within the column. Because SW cloud scattering is predominantly forward, this cloud-side escape often directs photons toward the surface and causes DE (Hogan and Shonk, 2013).

Entrapment (M5)

The effect of solar elevation on cloud reflectance differs between 3D and 1D RT. At low sun, 3D RT tends to increase cloud reflectance relative to 1D RT due to cloud-side illumination. At high sun, the opposite occurs, with 3D RT tending to reduce cloud reflectance. This high-sun behavior is explained by the entrapment mechanism, in which the inclusion of horizontal photon transport in 3D RT increases the likelihood that radiation is absorbed by the cloud or the surface, thereby reducing the overall scene reflectance. Backscattered photons from clouds or the surface have a higher chance of escaping to space in 1D as they follow the same clear-sky path previously propagated downward. In 3D, horizontal transport can trap radiation in neighboring clouds and redirect it downward (Villefranque et al., 2023) or cause it to be absorbed (Hogan et al., 2019).

The total 3D effect (Eq. 2.20) depends on SZA through the response of the underlying mechanisms (Várnai and Davies, 1999). In general, increasing SZA increases upward flux at TOA in 3D relative to 1D RT, while it reduces downward flux at the surface (Villefranque et al., 2023).

2.3 EarthCARE

The Earth Cloud, Aerosol, and Radiation Explorer (EarthCARE) is a joint ESA-JAXA satellite mission launched on 29 May 2024 to advance our understanding of cloud and aerosol radiative effects in the Earth system (Wehr et al., 2023).

With its four instruments, the ATmospheric LIDar (ATLID), the Cloud-Profiling Radar (CPR), the MultiSpectral Imager (MSI), and the BroadBand Radiometer (BBR), it provides (1) vertically resolved profiles of clouds, aerosols, and precipitation along the satellite nadir-track, (2) 3D cloud-aerosol scenes extending across-track and (3) co-located SW and LW radiances and fluxes at the TOA over all climate zones (Wehr et al., 2023). Figure 2.5 show the instrument geometry of the EarthCARE satellite.

The two satellites, CloudSat and CALIPSO (Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation), were launched together in 2006 and provided the first active profiling of clouds from space (Winker et al., 2017). These instruments collected nearly 20 years of global cloud profiles before both missions ended in 2023. The satellite missions marked major advances in understanding interaction of clouds and climate, but their records must be extended over multiple decades to detect long-term trends (Winker et al., 2017). EarthCARE will extend these previous records with improved lidar and radar, together with co-located radiative flux measurements (Wehr et al., 2023).

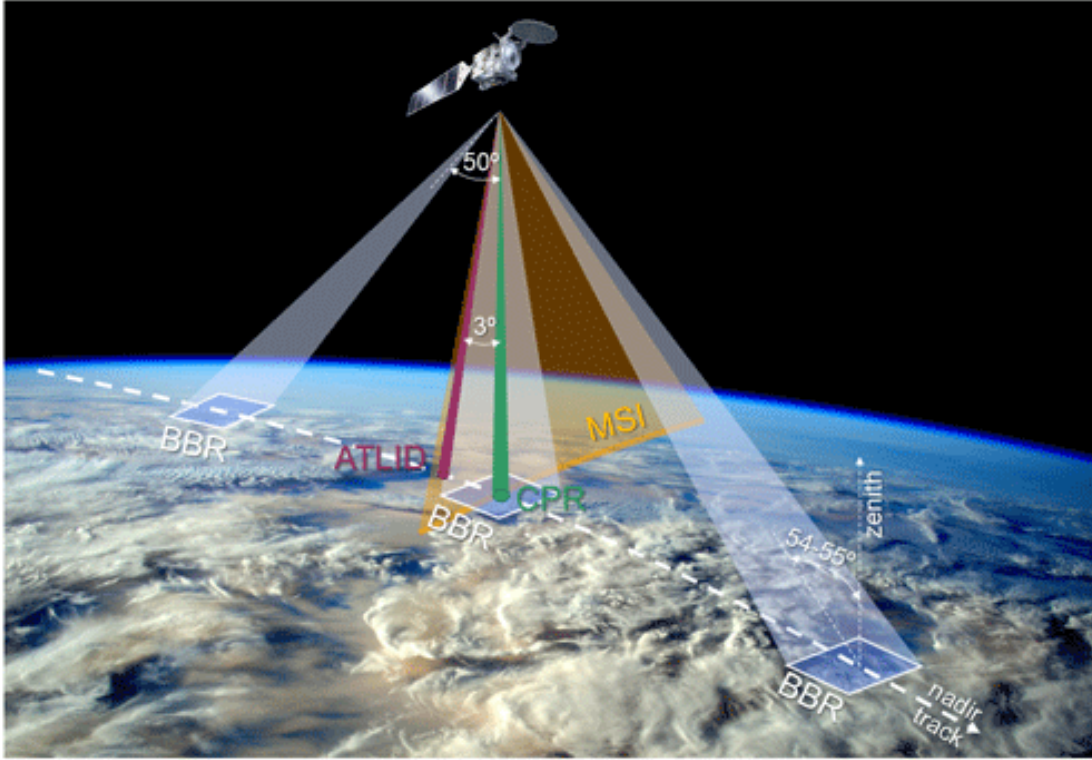


Figure 2.5: Viewing geometry of the EarthCARE satellite (not to scale). CPR observes at nadir, while ATLID is tilted 3° backward off-nadir to reduce specular reflection from ice crystals. MSI provides a 150 km across-track swath, offset to avoid sunglint. BBR observes in three directions (forward, nadir, backward), with forward and backward views inclined 50° along-track. The figure and information are taken from [Wehr et al. \(2023\)](#).

2.3.1 Relevant Products

EarthCARE data pass through a data processing chain. Level 2a provides single-instrument geophysical data, while Level 2b combines two or more of these measurements to produce more advanced cloud, aerosol, and precipitation fields ([Eisinger et al., 2024](#)). Several Level 2b products are used in this Master's project, including ACM-RT, ACM-COM, ACMB-3D (or ALL-3D), and BMA-FLX, which are introduced below. The AM-ACD dataset is used only for the Ångström exponent (see Sect. 2.2.1) and is not described further.

ACM-RT

EarthCARE's 1D and 3D RT products are provided by the ACM-RT processor, see [Cole et al. \(2023\)](#) for details. Due to the high computational cost of tracking large numbers of photons in cloudy atmospheres, 3D Monte Carlo RTMs have not been used operationally ([Villefranque et al., 2023](#)). EarthCARE therefore represents an important step forward, being the first mission to apply an operational 3D Monte Carlo RT solver, meaning that the RTMs are run routinely in the processing chain.

The 1D RT engine (ACM-RT 1D) is based on the Rapid Radiative Transfer Model for General Circulation Models (RRTMG), which is designed and commonly used in large-scale models. ACM-RT 1D applies a two-stream approach based on the correlated-k distribution method to compute SW and LW fluxes.

The SW and LW models differ in that the 1D LW RTM neglects scattering by all atmospheric constituents. This 1D LW formulation is an emissivity model, which effectively removes the scattering source term S_{sc} from the source function in the RTE (see Eq. 2.7). Clouds are often treated as purely absorptive in the LW in many GCMs since the atmosphere is optically thick over much of the thermal spectrum (Kuo et al., 2020). This supports the approach used in the LW model of ACM-RT 1D. However, this absorption approximation can introduce significant errors. Kuo et al. (2020) found deviations of up to $\pm 15\%$ relative to 128-stream DISORT results when scattering is neglected.

The 3D RT engine (ACM-RT 3D) is applied only to a subset of the ~ 1 km grid columns in the ACM-COM processor (referred to as pixels) (Eisinger et al., 2024), whereas ACM-RT 1D is applied to all of the pixels along the satellite nadir. The resulting ACM-RT 3D quantities are reported for each 21 km non-overlapping along-track pixel over 21×5 km assessment domains where RT quantities are horizontally averaged.

The restriction in the number of ACM-RT 3D results reflects the high computational cost of ACM-RT, in particular the 3D SW RTM, which requires a sufficient number of photons to achieve sufficiently low uncertainties for reliable radiative closure assessment.

The dual use of 3D and 1D RT simulations in the EarthCARE mission provides a framework to quantify the magnitude and spatial structure of 3D effects. The differences between 3D and 1D RT are a key focus in EarthCARE (Cole et al., 2023) and in this Master’s project. In this context, a key objective of EarthCARE is radiative closure, i.e., comparing measured BBR TOA fluxes with modeled fluxes, which should agree within ± 10 W/m² on average. However, Cole et al. (2023) reported that 3D - 1D differences for upward SW fluxes at 20 km are expected to exceed this threshold in about half of the cases.

ACM-COM

The ACM-COM processor (Cole et al., 2023) provides atmospheric profiles, which are used by the RTMs in ACM-RT. It provides two atmospheric configurations for RT applications: one based on the ACM-CAP “best-estimate” atmosphere and one representing the Level 2a composite atmosphere. The planned configuration in ACM-RT is to use the ACM-CAP atmosphere.

ACMB-3D

The ACMB-3D processor provides a mapping product (`index_construction`) to construct 3D domains for the 3D RTMs. This uses the Scene Construction Algorithm (SCA) to extend nadir profiles across-track (Barker et al., 2011). The method matches off-nadir pixels with similar MSI radiances and assigns them corresponding nadir pixels as substitutes.

The underlying assumption of the SCA is that neighboring pixels with nearly identical atmospheric and surface conditions, and small differences in spectral TOA radiances, have similar cloud profiles (Barker et al., 2011). However, the performance of the SCA typically degrades with increasing distance between recipient across-track pixels and their donor nadir pixels (Qu et al., 2023).

Qu et al. (2023) evaluated the impact of the imperfect atmosphere from the SCA and found that the resulting TOA flux errors are not expected to compromise EarthCARE’s ability to achieve radiative closure within ± 10 W/m² on average. They conclude that

deviations in both the SW and LW regions are generally much smaller than this target and only rarely exceed it.

BMA-FLX

The BMA-FLX processor ([Velázquez Blázquez et al., 2024](#)) provides products with observations of SW and LW upward TOA fluxes. These observations are derived from BM-RAD radiances measured at three along-track viewing directions at $10 \times 10 \text{ km}^2$ resolution, as seen in Fig. 2.5. Although these views provide information on scene anisotropy, fluxes are retrieved independently for each telescope to meet mission requirements to handle potential critical failures. The BBR telescope measures total-wave radiances from $0.25 \text{ }\mu\text{m}$ to above $50 \text{ }\mu\text{m}$, with the SW to LW separation at $4 \text{ }\mu\text{m}$. The LW channel is obtained by subtracting the SW signal from the total-wave measurement.

The conversion of radiances from a single observer geometry to hemispherical fluxes is complex. To account for scene anisotropy, BMA-FLX uses angular distribution models. The SW method applies a neural network trained on Clouds and the Earth's Radiant Energy System (CERES) and Moderate-Resolution Imaging Spectroradiometer (MODIS) data, while the LW method uses angular models based on RT simulations linked to MSI radiances.

The three independently retrieved TOA fluxes (from the three BBR views) are ideally similar, but can differ in practice. They are therefore merged into the combined SW and LW flux products to reduce uncertainties and providing a more stable and improved flux estimate.

In addition to the standard $10 \times 10 \text{ km}^2$ resolution, TOA fluxes are also provided at a $21 \times 5 \text{ km}^2$ assessment domain resolution, matching the ACM-RT 3D resolution ([Cole et al., 2023](#)). The availability of multiple spatial resolutions of TOA fluxes enables flexible radiative assessment.

Chapter 3

Method

In this Master’s project, EarthCARE atmospheric profiles of gases, clouds, and aerosols were used to perform radiative assessments using the libRadtran RTMs. Multiple assessments were conducted using observed TOA fluxes from EarthCARE and surface fluxes from ground-based measurements in Norway. Additionally, comparisons between 3D and 1D RTMs and sensitivity analyses of the atmospheric inputs, including aerosols, ice crystal habits, and different atmospheric configurations, were performed.

3.1 Radiative Transfer Models

The [libRadtran library](#) provides a collection of RT tools that support several RTMs ([Mayer et al., 2025b](#)). The RTMs from libRadtran used in this study include:

1. DISORT, a discrete-ordinate radiative-transfer solver representing the 1D RTM ([Stamnes et al., 2000](#)).
2. MYSTIC, a Monte Carlo 3D RTM capable of handling fully 3D atmospheric fields ([Mayer, 1999](#); [Mayer and Kylling, 2005](#); [Mayer, 2009](#)).

Note that the fully 3D version of MYSTIC is not included in the public libRadtran release, but is available in joint projects ([Emde et al., 2016](#); [Mayer et al., 2025b](#))¹. In addition to the libRadtran simulations, EarthCARE’s ACM-RT 1D simulations (Sect. 2.3.1) were used for comparison to our model results.

DISORT (1D RTM)

DISORT (DIScrete ORdinates Radiative Transfer) is the default RTE solver in libRadtran ([Mayer et al., 2025b](#)). It solves the RTE in plane-parallel (1D) geometry using the discrete-ordinates method, enabling accurate computation of radiances and fluxes for layered atmospheres. It is one of the most widely used and extensively validated 1D RT solvers worldwide ([Mayer et al., 2025b](#)).

DISORT applies the discrete-ordinates method using Gaussian quadrature with points and weights determined by a double-Gauss scheme ([Stamnes et al., 1988](#)). The number of streams can be selected by the user, with 6 streams used as default for flux calculations ([Mayer et al., 2025b](#)). When using a bidirectional reflectance distribution function (BRDF) to represent directional surface reflectance, the number of streams is automatically increased to 16 ([Mayer et al., 2025b](#)), as applied in this study.

¹The libRadtran user guide (see [Mayer et al. \(2025b\)](#)) does not include all MYSTIC specifications since the full 3D version is not part of the public release.

MYSTIC (3D RTM)

The Monte Carlo code for the **phYSically correct Tracing of photons In Cloudy atmospheres** (MYSTIC) is an RT solver designed for realistic simulations of atmospheric processes (Mayer, 1999; Mayer and Kylling, 2005; Mayer, 2009). It is suitable for cases with horizontal inhomogeneity, polarization, spherical geometry, or sharp scattering phase functions, where other solvers may be limited (Mayer et al., 2025b). MYSTIC can handle fully 3D atmospheres, inhomogeneous surface albedo, and topography, but the simulation is limited by the spatial resolution of the input atmosphere (Mayer, 2009). The Monte Carlo method implemented in MYSTIC follows the description outlined in Sect. 2.1.3.

3.2 Model Setup

In this section, the setup of MYSTIC (3D RTM) and DISORT (1D RTM) is described, including the configuration of the input files used for the simulations. The description focuses on the key input parameters to the RTMs. Full code details are not included due to their length and complexity, but are available in the [GitHub repository](#). Details of all input parameters used in the RT simulations are described in the libRadtran User Guide (Mayer et al., 2025b).

The atmospheric data used to drive the RT simulations were taken from EarthCARE Level 2b products. The RTM setup with libRadtran was based on Python scripts developed by Arve Kylling (personal communication, 2025) and was modified and extended to meet the requirements of this work.

SW and LW regions

The SW and LW spectral ranges used in the RT simulations were 0.25–4 μm and 4.5–100 μm , respectively. This follows from the broadband definition of the BBR instrument, where SW and LW regions are separated at 4 μm (Velázquez Blázquez et al., 2024). The LW range in our model simulations started at 4.5 μm due to the use of the fast **fu** parameterization (Fu and Liou, 1992, 1993), developed for climate models (Mayer et al., 2025b), does not include the 4–4.5 μm interval. In the SW region, we used the correlated- k distribution **kato2** (Kato et al., 1999). With these spectral ranges, the **fu** parameterization spans 12 spectral bands, while the SW setup spans 32 spectral bands.

Solar Geometry

The solar geometry for each simulated atmospheric grid column (hereafter referred to as a pixel) was specified using the SZA and solar azimuth angle, ϕ , obtained from BMA-FLX. An SZA of 0° corresponds to the Sun at zenith, while 90° corresponds to a horizontal solar direction (Mayer et al., 2025b). The solar azimuth angle ϕ was defined clockwise from north, e.g., $\phi = 90^\circ$ corresponds to east. Details are given in the [EarthCARE Product Data Handbook \(BMA-FLX\)](https://earthcarehandbook.earth.esa.int/catalogue/bma_flx_2b) (https://earthcarehandbook.earth.esa.int/catalogue/bma_flx_2b). However, the solar azimuth angle was required only for the MYSTIC simulations (Mayer et al., 2025b).

For SZAs exceeding 90° , the SW simulated pixel was terminated and returned a NaN value. These cases were considered unreliable for both the 1D and 3D SW RTMs, as the SW fluxes were close to zero.

The pseudospherical option in DISORT was not used in this work, as $\text{SZA} < 70^\circ$ covered most simulated EarthCARE swaths (except for surface validations; Sect. 3.4). For $\text{SZA} < 70^\circ$, the plane-parallel approximation is generally valid (Dahlback and Stamnes, 1991; Mayer and Kylling, 2005).

Atmospheric Construction

The atmospheric state was constructed on a pixel-by-pixel basis. Each pixel corresponds to a single column of the EarthCARE joint standard grid, with a horizontal extent of approximately 1×1 km (Eisinger et al., 2024), as used in the ACM-COM products (Cole et al., 2023). The atmosphere consists of 249 levels. The simulated TOA is defined at the second-highest altitude level, corresponding to approximately 65 km.

The atmospheric setup required generating input files describing the gaseous composition, cloud properties (liquid and ice), aerosols, and surface characteristics. For MYSTIC simulations, surface elevation was also included. An overview of the input parameters is given in Tab. 3.1. Explanations of the parameter names are provided in the libRadtran User Guide (Mayer et al., 2025b). EarthCARE products associated with the libRadtran input parameters (Tab. 3.1) are listed in Tab. 3.2.

The atmospheric description was constructed in five steps:

1. **The gaseous atmosphere** was constructed in 1D (i.e., depending only on the vertical coordinate) using altitude, pressure, temperature, air density, and molecular number densities of gases (O_3 , O_2 , CO_2 , H_2O). The retrieved gas profiles from ACM-COM were converted to molecular number densities for use in the RTMs. The procedure is described in Appendix Sect. 6.3.
2. **Liquid and ice clouds** were described by their liquid and ice water content (LWC and IWC) and effective radii. The MYSTIC simulations used 3D cloud fields, whereas the DISORT simulations used only a single pixel.

For liquid clouds, the optical properties were calculated using precomputed Mie tables available from the libRadtran website (Mayer et al., 2025b).

For ice clouds, optical properties were calculated using the `baum_v36` parameterization (Heymsfield et al., 2013; Yang et al., 2013; Baum et al., 2014). This scheme provides three ice crystal habits: (1) a general habit mixture (GHM) consisting of nine ice habits, (2) solid-columns (SC), and (3) rough-aggregates (RA; aggregates of solid columns (Baum et al., 2014)). All habits assume severely roughened and randomly oriented ice particles (Mayer et al., 2025b). Unless stated otherwise, the GHM was used in the simulations.

The radiative properties of the ice crystal habits, such as extinction, single-scattering albedo, and asymmetry parameter, depend on wavelength and ice crystal effective diameter. A detailed description is beyond the scope of this Master's thesis.

3. **Aerosols** were implemented in 1D using vertically resolved AOD profiles and the total column AOD at a given wavelength. In the SW region, the Ångström exponent was used for spectral scaling following Eq. 2.19. A default Ångström exponent of $\alpha = 1.1$ was applied for most pixels, representing a mixture of fine and coarse aerosols (Haarig et al., 2023; Schuster et al., 2006). This was required because the obtained exponent (based on the 670-865 nm spectral range) relies on

Table 3.1: Relevant input parameters for the libRadtran RTMs.

Relevance	Input Parameters
RTM Specifications	day_of_year
	mc_minphotons*
	mc_photons*
	mc_randomseed*
	mc_reference_to_nn**
	mc_sample_grid**
	mc_std**
	mol_abs_param
	output_process
	output_user
	rte_solver
	source
	wavelength
	zout
Atmospheric description	aerosol_angstrom
	aerosol_default
	aerosol_file_tau
	aerosol_set_tau_at_wvl (355/4500)
	albedo_file
	atmosphere_file
	brdf_cam
	ic_file (1D/3D)
	ic_habit
	ic_properties
	mc_albedo_spectral_file**
	mc_albedo_type**
	mc_elevation_file**
	phi0
	sza
	wc_file (1D/3D)
	wc_properties

Note: Parameters marked with * refer to MYSTIC-specific parameters, and those marked with ** are not described in the public release of the libRadtran User Guide (Mayer et al., 2025b).

MSI measurements at 865 nm and is only available over the ocean, which limits its spatial coverage (Haarig et al., 2023).

The total column AOD was defined at the ATLID wavelength of 355 nm in the SW region (Wehr et al., 2023) and at 4500 nm in the LW region. Since aerosol extinction is only provided in the SW region (Cole et al., 2023), including aerosols in the LW region required an approximation. A fixed conversion factor k was used to scale AOD from the SW to the LW region,

$$AOD_{LW} = k \cdot AOD_{SW}, \quad (3.1)$$

where k ranged from 0 (no aerosol) to 0.6. Further details are given in Section 3.4.2.

Table 3.2: EarthCARE data products used as input for the libRadtran RTMs.

Product	Variable
ALL_3D (ACMB-3D)	latitude, longitude index_construction
ACM_COM	time latitude, longitude height_level, height_layer pressure_level, pressure_layer_mean temperature_level, temperature_layer_mean specific_humidity_layer_mean volume_mixing_ratio_layer_mean_O3 volume_mixing_ratio_layer_mean_O2 volume_mixing_ratio_layer_mean_CO2 liquid_water_content liquid_effective_radius ice_water_content ice_effective_radius aerosol_extinction aerosol_classification cloud_flag surface_temperature albedo_diffuse_radiation_surface_visible albedo_diffuse_radiation_surface_near_infrared surface_emissivity_type_index surface_emissivity_table surface_albedo_classification wind_speed_at_10_meters quality_status
ACM_RT	flux_up_solar_1d_all_sky flux_up_thermal_1d_all_sky quality_status
BMA_FLX	solar_combined_top_of_atmosphere_flux solar_combined_top_of_atmosphere_flux_quality_status thermal_combined_top_of_atmosphere_flux thermal_combined_top_of_atmosphere_flux_quality_status solar_zenith_angle
AM_ACD	aerosol_angstrom_exponent

4. **Surface** properties were defined as a Lambertian reflector, meaning that reflected radiance is independent of direction ([Stamnes et al., 2017](#)), with wavelength-dependent albedo. For MYSTIC, horizontally inhomogeneous 2D spectral albedo was used, while for DISORT, only the single pixel was used.

In the SW region, data on diffuse albedo were used. In the LW region, reflectance was approximated from the emissivity using Eq. 2.11.

Ocean BRDF was only included for the 1D RTM using the Cox and Munk model ([Cox and Munk, 1954a,b](#)). BRDF options do not work with topography in libRadtran ([Mayer et al., 2025b](#)), and were therefore not used in the 3D RTM.

5. **Topography** was only included with MYSTIC. Surface elevation was prescribed on a 2D grid using the [Copernicus Digital Elevation Model](#), enabling a more realistic RT treatment of slanted surfaces ([Mayer et al., 2025b](#)).

For clarity, “1D” refers to a single vertical column (a pixel), while “3D” refers to a horizontal arrangement of pixels forming a 3D computational domain, hereafter referred to as the *3D buffer*.

All input data were obtained from the ACM-COM products, except for the Ångström exponent, which was taken from AM-ACD. The EarthCARE retrieved nadir cross-section of cloud profiles was extended to 3D cloud fields using the scene construction algorithm (SCA) from ACMB-3D (see [Qu et al. \(2023\)](#) for details). The ACM-COM products provide two atmospheric representations (see Sect. 2.3.1). Unless otherwise stated, all input data were based on the ACM-CAP “best-estimate” atmosphere ([Cole et al., 2023](#)). The composite atmosphere was used only in Sect. 4.4. Quality flags from ACM-COM were applied to the libRadtran RT simulations (MYSTIC and DISORT) unless stated otherwise. Only pixels classified as *good* or *good with warnings* were included.

Assessment Domains

To compare simulated fluxes with observed TOA upward SW and LW fluxes from BMA-FLX at $10 \times 10 \text{ km}^2$ resolution ([Velázquez Blázquez et al., 2024](#)), the simulated fluxes were averaged over assessment domains chosen to match the observed TOA flux footprint as closely as possible. For MYSTIC, TOA fluxes were averaged over an 11×11 pixel domain, with an odd number of pixels due to the 3D buffer construction around the center pixel. For DISORT, which does not provide across-track fluxes, TOA fluxes were averaged over 10 consecutive along-track pixels (5 before and 4 after the center pixel).

A single-pixel (1×1) assessment domain was also used to examine sensitivities between RTMs and their input parameters.

MYSTIC Configurations

The MYSTIC setup required additional specifications, including the 3D buffer and photon configuration, as described below.

To allow for horizontal photon transport, the 3D buffer was constructed by aggregating nadir pixels with horizontally inhomogeneous 3D cloud fields, while gases and aerosols remained plane-parallel (1D). A 21×21 pixel 3D buffer (horizontal extent) was used as a compromise between computational cost and reducing edge effects from the periodic horizontal boundary conditions (see Sect. 2.1.3). The choice of this configuration was based on two considerations:

1. A larger 41×41 buffer was tested, but was not feasible within the available computational resources (see Appendix Sects. 6.1 and 6.2).
2. To reduce edge effects in the 11×11 pixel assessment domains used for comparison with BMA-FLX, the 3D buffer must extend beyond the assessment region to keep boundary effects away from the area of interest ([Qu et al., 2023](#)). In this study, this corresponded to an extension of five pixels in both the along-track and across-track directions.

For LW simulations with output at TOA, backward photon tracing was used. In this mode, photons are traced from the detector (e.g., TOA or surface) back toward the source, in contrast to forward tracing, where photons are traced from the source through the atmosphere to the detector (Mayer, 2009). Backward tracing is efficient when results are required at only a few locations, especially in LW conditions where the atmosphere is optically dense, and can reduce computational time by several orders of magnitude (Mayer, 2009).

The number of injected photons, N_{photons} , per simulated 3D buffer was $3.2 \cdot 10^7$ in the SW region (32 spectral bands, 10^6 photons per band) and $1.2 \cdot 10^4$ in the LW region (12 spectral bands, 10^3 photons per band). For LW simulations with output at all atmospheric altitude levels, N_{photons} was reduced to $1.2 \cdot 10^3$. These values represent the highest practical numbers of injected photons given the computational constraints of this study. The setup provided comparable runtime in the SW and LW regions, while maintaining acceptable Monte Carlo noise with standard deviations on the order of 4-20 W/m² in SW and 0.1-2 W/m² in LW. Further details on computational performance and runtime are given in Appendix Sect. 6.1, and on Monte Carlo noise in Appendix Sect. 6.2.

3.3 Reference Fluxes

This section describes the reference fluxes used in this study. These include TOA observations from EarthCARE’s BMA-FLX processor, simulated TOA fluxes from EarthCARE’s ACM-RT 1D RTM, and surface fluxes from a Norwegian pyranometer network.

3.3.1 TOA Fluxes: Observations

The combined SW and LW fluxes from BMA-FLX at standard resolution (10×10 km footprint (Velázquez Blázquez et al., 2024)) were used as reference (“model truth”) at the TOA.

Quality filtering of BMA-FLX fluxes was applied consistently in all visualizations and comparisons with BMA-FLX, if not otherwise stated. Pixels with both components flagged as fully trustful were used in both SW and LW. Pixels with one fully trustful component were used only in that regime. Pixels without fully trustful components were not used.

Additional processing was required because BMA-FLX pixels do not align with the joint standard grid used by ACM-COM and our simulations. BMA-FLX follows BBR sampling at 1 km spacing (Velázquez Blázquez et al., 2024), while the simulation grid had an approximate 1 km spacing (Eisinger et al., 2024). To ensure consistency, two modifications were made:

1. For the RTM input, latitude-based matching was applied. The nearest BMA-FLX latitude grid point was used to provide solar geometry information.
2. For the flux comparison, BMA-FLX fluxes were interpolated onto the simulation latitude grid.

3.3.2 TOA Fluxes: EarthCARE Simulations

ACM-RT 1D simulations were used as an additional reference. Quality filtering was applied consistently unless stated otherwise. Only pixels flagged as very good (i.e., good

input quality with no RT adjustments) or good (i.e., good input quality with some RT adjustments) were included in the analyses.

3.3.3 Surface Fluxes: Observations

Simulated surface SW downward fluxes were evaluated using observations from 19 ground-based stations across Norway, which are shown in Fig. 3.1. Pyranometer data were obtained from the Norwegian Center for Climate Services (NCCS; seklima.met.no, last accessed: 16 March 2026).

The data consist of near-real-time 1 min averages of global radiation (direct + diffuse) over the 295-2800 nm range. To match the observations, the simulated SW surface downward flux was calculated over the same spectral range.

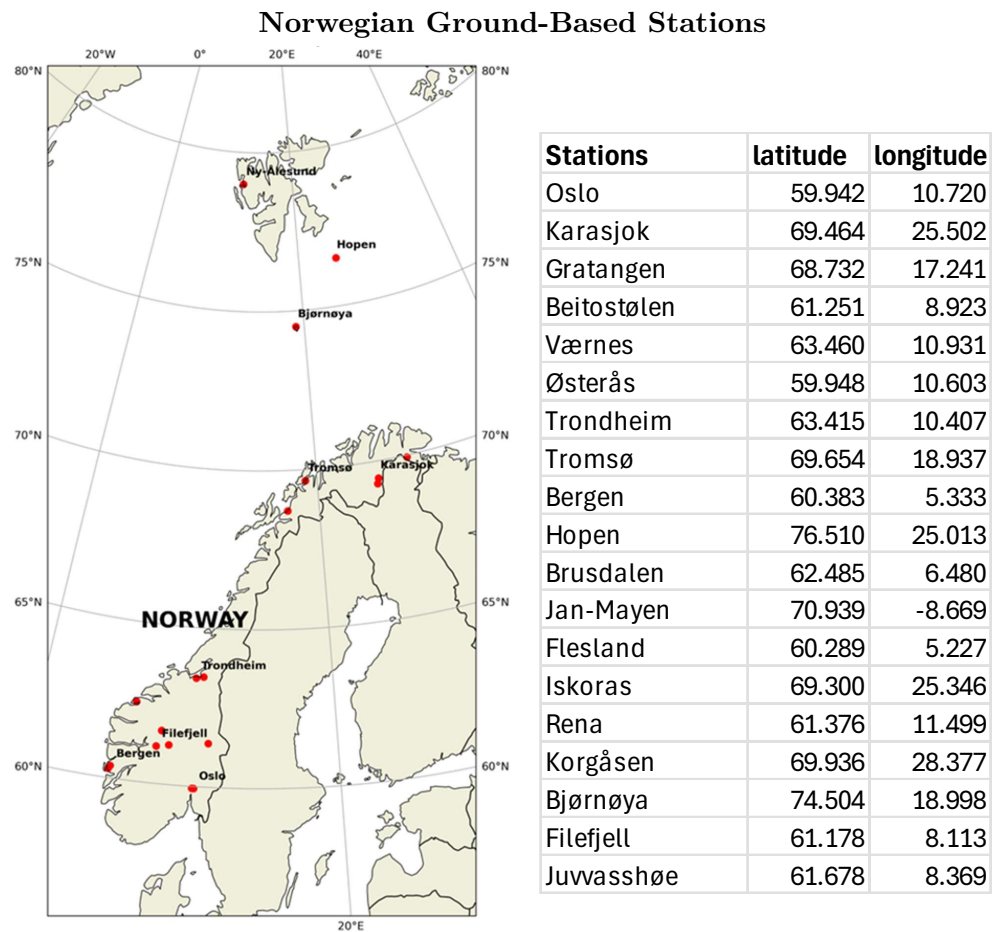


Figure 3.1: Locations of the 19 ground-based stations providing pyranometer measurements of surface SW downward flux across Norway.

Prior to the surface radiative assessment, a brief validation was performed by Tove Svendby at NILU to confirm that the pyranometer measurements provided physically reasonable fluxes. The validation was limited to two selected stations, Oslo and Juvvasshøe, where surface downward SW fluxes over a full day under clear-sky conditions were simulated using DISORT with a simplified RTM setup (standard atmosphere, no clouds or aerosols). The validation results showed smooth and temporally consistent clear-sky surface SW downward fluxes in both simulations and observations, with

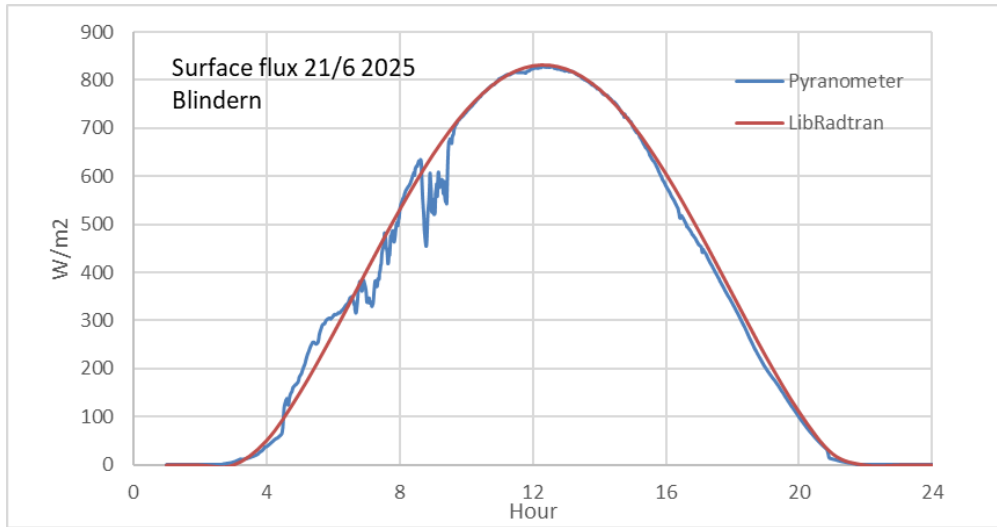


Figure 3.2: Temporal surface SW downward flux from DISORT (LibRadtran; red line) and observations (Pyranometer; blue line) from ground-based station at Blindern (Oslo) for a mostly clear-sky day.

Table 3.3: Ground-based stations and number of overpasses.

Station	Overpasses
Bergen	1
Bjørnøya	6
Filefjell	10
Flesland	1
Gratangen	1
Hopen	10
Iskoras	1
Jan-Mayen	1
Juvvasshoe	5
Karasjok	1
Korgåsen	5
Oslo	1
Rena	1
Trondheim	1
Tromsø	14
Værnes	1
Østerås	1

agreement within about $\pm 10 \text{ W/m}^2$ at noon, indicating no strong bias. An example is shown in Fig. 3.2, where DISORT simulations (red) are compared with pyranometer observations (blue) for a mostly clear-sky day at Oslo on 21.06.2025. Small deviations in the observations between hours 4 and 8 are due to cloud effects not included in the simulations. This validation was limited to a subset of stations and was used to get an initial view of the data quality and a simple consistency check before the full surface radiative closure assessment was performed. It was not intended to assess the stations themselves, for which quality control procedures for the ground-based instruments are described in [Dobler et al. \(2024\)](#).

The number of satellite overpasses over the ground-based stations depends on the allowed horizontal distance between the station and the satellite nadir-track. Increasing this distance increases the number of overpasses, but also increases the reliance on the SCA because the station pixel is reconstructed using the SCA. To limit uncertainties from the SCA at larger distances, as the performance of the SCA generally decreases with distance from the nadir-track (Qu et al., 2023), a maximum across-track distance of ± 5 pixels ($\sim \pm 5$ km) between the station and the satellite nadir-track was used. This provided enough overpasses while limiting uncertainties from the SCA.

A total of 61 satellite overpasses were obtained for the surface SW flux assessment using two criteria: (1) satellite overpasses within ± 5 pixels of ground-based stations and (2) daytime conditions. The selected stations and the number of EarthCARE overpasses are listed in Tab. 3.3. No additional quality filtering was applied to the pyranometer measurements.

3.4 Studies and Sensitivity Analyses

For this Master’s project, six different groups of studies were conducted:

1. Overall assessment of EarthCARE TOA fluxes using simulations and BMA-FLX observations.
2. LW TOA flux sensitivity to aerosols.
3. Sensitivity of SW and LW TOA fluxes to ice crystal habits.
4. Variations in TOA fluxes related to different ACM-COM atmospheric profiles.
5. Evaluation of modeled and observed surface SW fluxes.
6. 3D cloud radiative effects.

Each swath is 1/8 of an orbit and spans approximately 5000 km along-track (Eisinger et al., 2024). It is identified by its Orbit NNNNNX identifier (N: number, X: letter). The latitudinal region is indicated by the final letter (X): E denotes tropical regions, D mid-latitudes, and C Arctic regions. The swaths were selected based on two criteria:

1. Coverage of a wide range of latitudes, including tropical, mid-latitude, and Arctic regions.
2. Mostly continuous TOA flux observations in the BMA-FLX products.

The second criterion was based on visual inspection of BMA-FLX fluxes without using quality flags. As a result, some discontinuities appear in the observed fluxes.

Lower data quality is generally observed for Arctic swaths (Orbits 06331C, 06888C, 07277C). Applying ACM-COM and BMA-FLX quality flags leads to more fragmented TOA fluxes in these cases compared to lower-latitude swaths. This is consistent with the known challenges in retrieving clouds and fluxes in the Arctic (Griesche et al., 2024).

The EarthCARE data used in this study (ACM-COM, ACM-RT, AM-ACD, ACMB-3D, and BMA-FLX) were downloaded from the [ESA Simple Online Catalog - Search Form](#) (last access: 16 March 2026). Each product is associated with a processor baseline, which defines the data version and ensures traceability. An overview of releases is available at [EarthCARE Data](#) (European Space Agency (ESA), 2025). Figures of individual swaths include the processor baseline at the bottom of the figure. In principle,

Table 3.4: Selected swaths used in the analysis.

Region	Date	Orbit
Greenland	9 July 2025	06331C
Svalbard	14 August 2025	06888C
Svalbard	8 September 2025	07277C
Europe	17 October 2025	07883D
USA/Canada	21 July 2025	06518D
USA/Canada	15 August 2025	06907D
Africa	20 July 2025	06497E
Africa	14 August 2025	06886E

the most recent baseline should be used. However, the simulations were computationally expensive and reprocessing for each new release was not feasible (see Appendix Sect. 6.1). Therefore, all swaths used the BA baseline for the EarthCARE products, except ACM-RT, which used the AC baseline.

3.4.1 Overall Radiative Assessment

An overall assessment was performed for the eight swaths listed in Tab. 3.4 and shown in Fig. 4.1. TOA upward SW and LW fluxes from MYSTIC and DISORT were compared with BMA-FLX observations, with ACM-RT 1D included as an additional reference.

Three swaths (Orbits 06497E, 07883D, and 06888C), spanning tropical to Arctic regions, were selected for spatial visualization. The full set of eight swaths was evaluated using model-observation scatter plots.

3.4.2 LW Aerosol Effect from SW-Derived AOD

Aerosols were included in the LW simulations to assess their radiative impact on TOA upward LW fluxes. This was motivated by Zhou and Savijärvi (2014), which showed that dense near-surface aerosol layers can affect LW radiation in a way similar to thin low-level clouds through absorption and thermal re-emission. Since aerosol products are mainly derived in the SW (Clarisse et al., 2019), and EarthCARE provides aerosol extinction only in this range (Cole et al., 2023), approximations were required to include aerosols in LW simulations.

SW AOD was scaled to LW using a fixed conversion factor according to Eq. 3.1 and prescribed to the RTM as vertically resolved profiles and column-integrated values at wavelength $\lambda = 4500$ nm. Note that this scaling is a simplification as it approximates the spectral transition of AOD from the SW to the LW region. In reality, the scaling depends on particle size distribution and the wavelength dependence of the refractive index (Clarisse et al., 2019). For reference, reported conversion factors using this scaling approach from infrared to visible wavelengths ranged from $k' \in [0.9, 3.5]$ (Clarisse et al., 2019), corresponding to $k \in [0.3, 1.1]$ when mapped from the SW to the LW region (using $k = 1/k'$).

Different AOD-models, i.e., different ways of including aerosols in the LW simulations, were tested using DISORT. Simulations with MYSTIC was not used due to its higher computational cost, which would have limited either the number of swaths or simulated pixels.

Eight AOD-models were defined (labels used in figures are given in parentheses):

- (1) No aerosols (**None**): no aerosol included in the RTM.
- (2) Default aerosol model (**aerosol_default**): aerosol model from [Shettle \(1990\)](#) ([Mayer et al., 2025b](#)).
- (3–5) Fixed conversion factor ($\times 0.2$, $\times 0.4$, $\times 0.6$): SW AOD scaled to LW using Eq. 3.1 with conversion factor $k = 0.2, 0.4, 0.6$.
- (6–8) Fixed conversion factor for coarse-mode absorbing aerosols ($\times 0.2$, $\times 0.4$, $\times 0.6$ **absorbing**): scaling using Eq. 3.1 applied only to pixels with coarse-mode absorbing aerosol type, otherwise no aerosol was used.

The chosen conversion factors fall within reported ranges ([Clarisse et al., 2019](#)). The focus on coarse-mode absorbing aerosols reflected their LW impact through absorption and re-emission of terrestrial radiation ([Zhou and Savijärvi, 2014](#)). These aerosol types included dust, dusty smoke, dusty mix, and stratospheric ash based on ACM-COM classification.

DISORT TOA upward LW flux simulations were performed for every third atmospheric pixel to reduce computational cost. The swaths used for the analysis on LW sensitivity to aerosols were selected to cover tropical, mid-latitude, and Arctic regions. Six swaths were used, two from each region, as shown on the world map in Fig. 3.3.

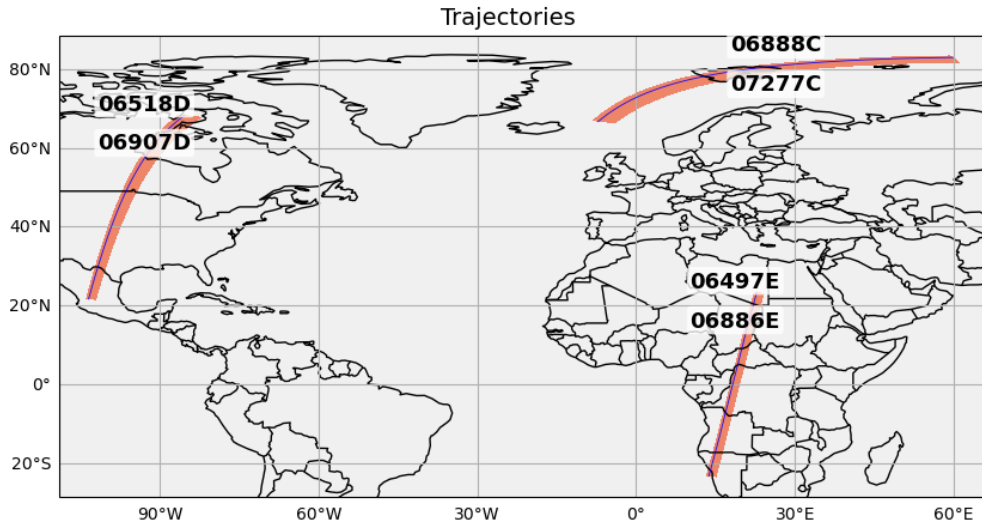


Figure 3.3: Trajectories of the swaths used for analyses of LW sensitivity to aerosols (Sect. 4.2) and ice habit sensitivity (Sect. 4.3). Blue lines show nadir tracks, and red shading indicates the across-track dimension.

The flux bias of the eight AOD-models was assessed as the swath-averaged difference between simulated TOA upward LW fluxes and the corresponding BMA-FLX flux. The simulated fluxes were averaged over 10 along-track pixels to match the standard BMA-FLX resolution ([Velázquez Blázquez et al., 2024](#)). This corresponds to averaging three consecutive simulated pixels for each BMA-FLX pixel, since only every third pixel was simulated in this sensitivity study. Quality filtering was applied for the comparison with BMA-FLX observations.

To examine spatial sensitivity to the AOD-models, differences in TOA upward LW flux between the coarse-mode absorbing AOD-model with a scaling factor of $\times 0.4$ and simulations without aerosols were analyzed together with the spatial variability of total

column AOD. In this case, simulations without aerosols were used as the reference instead of BMA-FLX. Therefore, the analysis was performed on a per-pixel basis, corresponding to a 1×1 pixel domain. No quality filtering was used to isolate the effect of the AOD-model on LW fluxes, as this is a model-model comparison with identical atmospheric input except for the AOD-model.

3.4.3 Radiative Transfer Sensitivity to Ice Crystal Habit

Interactions between clouds and radiation are complex and important in atmospheric science (Villefranque et al., 2023). In the current work, we have analyzed the sensitivity of TOA SW and LW fluxes to the choice of ice crystal habit.

In this work, greater emphasis was placed on the SW region, although sensitivities in the LW region were also examined. This focus reflects the larger scattering contribution by ice in the SW region and the resulting increased sensitivity to ice habit (Wang et al., 2014). Ice habit radiative effects in the LW are still non-negligible under certain conditions (Wendisch et al., 2007) and were therefore also included.

TOA upward SW and LW fluxes were simulated using the 1D RTM with DISORT. This enabled efficient simulations across multiple swaths and a large number of atmospheric pixels, allowing a more extensive analysis than feasible with MYSTIC. A 1D approach was considered sufficient to assess sensitivity to ice crystal habit. The RTM setup included a coarse-mode absorbing AOD-model with a scaling factor of 0.4.

The analysis was performed with the same six swaths as used in the LW aerosol sensitivity study, as shown in Fig. 3.3, covering tropical, mid-latitude, and Arctic regions. Swaths were cropped to retain segments dominated by ice clouds, resulting in six swath segments representing distinct ice cloud regions.

The general habit mixture (GHM) configuration was used as the reference in this ice habit sensitivity analysis. This choice was based on the need for a common reference and on the theoretical basis of GHM as a credible representation of natural ice clouds. Further justification is given in Sect. 2.2.2.

To assess biases from different ice habits, swath-averaged differences in simulated TOA upward SW and LW fluxes were analyzed for solid-column (SC) and rough-aggregate (RA) relative to the GHM, using the ice cloud parameterization `baum_v36` (Heymsfield et al., 2013; Yang et al., 2013; Baum et al., 2014). DISORT fluxes were computed using a 1×1 pixel assessment domain to retain as much information as possible without using quality filtering, as all simulations used identical input data except for the ice habit.

Inter-habit differences in SW and LW TOA upward fluxes were analyzed across tropical, mid-latitude, and Arctic regions (two swaths per region) by comparing maximum flux differences between ice habits. The role of SZA and ice water path (IWP) was also examined, as higher optical thickness and larger SZA increase multiple scattering, which suppresses radiative features linked to ice habit (Wendisch et al., 2005) (see Sect. 2.2.2).

3.4.4 Radiative Assessment of ACM-COM Atmospheres

RTMs are subject to uncertainties from model simplifications. For example, differences arise when using horizontally averaged plane-parallel homogeneous optical properties compared to the IPA, or when including horizontal transport of radiation in 3D RT (Várnai and Davies, 1999). In addition, large uncertainties can arise from inaccuracies in the atmospheric input fields.

Practically, all atmospheric input data used in the simulations were based on ACM-COM products, consistent with the inputs used by the ACM-RT processor (Cole et al., 2023). However, two atmospheric configurations were provided by ACM-COM, defined as:

- *Atmosphere 0*: the composite atmosphere derived from L2a products and resembles CloudSat-CALIPSO-CERES Merged product produced at NASA-Langley (Cole et al., 2023; European Space Agency, 2025). It represents a backup best estimate based on single-sensor retrieval products (Cole et al., 2023).
- *Atmosphere 1*: the atmosphere based on the ACM-CAP processor and considered EarthCARE’s best estimate, providing the most comprehensive synergistic use of observations from the individual EarthCARE instruments (Cole et al., 2023).

TOA upward SW and LW fluxes were simulated for Atmosphere 0 and 1 using both MYSTIC and DISORT over Orbit 06497E. A visualization was made for a cloudy subset of the swath over northern Congo. A single swath was selected, representing both cloud phases and providing near-continuous BMA-FLX coverage (with quality filtering). This limited selection was used to allow detailed visual inspection.

Results were compared with BMA-FLX, and with ACM-RT 1D for Atmosphere 1 (consistent with the prioritization of ACM-CAP profiles for the RTMs in ACM-RT (Cole et al., 2023)). To match the BMA-FLX footprint, an 11×11 pixel domain was used for MYSTIC, while DISORT and ACM-RT 1D fluxes were averaged over 10 consecutive pixels. No quality filtering was used, as the focus was on differences between the two atmospheric configurations.

Flux differences between MYSTIC and DISORT, relative to BMA-FLX observations, were computed for the two ACM-COM atmospheres across the full swath of Orbit 06497E to further quantify the effect on SW fluxes. Quality filtering was applied to exclude non-trustworthy regions in the obtained biases.

3.4.5 Evaluation of Simulations Against Surface Flux Measurements

This thesis largely focuses on TOA fluxes in the SW and LW regions. However, EarthCARE data can also be used to simulate surface fluxes to analysis surface radiation. For this purpose, pyranometer data from seklima.met.no were used to compare observed and simulated surface downward SW fluxes. Further details on the pyranometer observations and the satellite overpasses over the ground-based stations are provided in Sect. 3.3.3.

61 satellite overpasses over Norwegian ground-based stations, listed in Tab. 3.3, were used to compare simulated surface downward SW fluxes with pyranometer measurements. Valid cases excluded the winter season and EarthCARE overpasses at nighttime. In addition, TOA upward SW fluxes were simulated for the same 61 overpasses and compared with BMA-FLX observations to assess the correlation between surface and TOA fluxes. BMA-FLX quality status flags were applied for TOA comparisons, while no filtering was applied to the pyranometer data.

For surface comparisons (i.e., downward fluxes), the simulated pixel corresponded to a reconstructed across-track pixel using the SCA to match the station location. In contrast, for TOA comparisons (i.e., upward fluxes), the satellite nadir pixel was used to match the BMA-FLX observation. As a result, surface simulations relied more strongly on the SCA than the TOA analysis.

All comparisons used a 1×1 pixel domain. Although not optimal for BMA-FLX, this choice reflected the smallest possible resolution to match the surface observations, which was the priority for this specific analysis of surface fluxes.

SW fluxes were evaluated using model-observation scatter plots at the surface and TOA, comparing MYSTIC and DISORT against observations. A single-swath (Orbit 01690C) was also analyzed to examine temporal and spatial variability. Surface flux variability was assessed using 1-minute pyranometer data, and spatial downward SW flux fields from MYSTIC and DISORT were compared with cloud structures from ACM-COM.

3.4.6 3D Cloud Radiative Effects

Clouds are inherently inhomogeneous and require 3D RT for realistic representation, since 1D RTMs cannot capture horizontal photon transport (Várnai and Davies, 1999) (see Sect. 2.2.3). 3D effects, defined as the differences between MYSTIC and DISORT fluxes (3D - 1D) following Eq. 2.20, were analyzed, consistent with the horizontal transport effects described by Várnai and Davies (1999).

The analysis examined upward and downward SW flux fields with altitude. Several swaths were initially investigated, with two selected for detailed analysis. The nadir flux fields from MYSTIC and DISORT were analyzed together with cloud masks to identify features such as cloud-side illumination, slanted shadows, and the mechanisms downward escape (DE), upward trapping (UT), upward escape (UE) and entrapment (see Sect. 2.2.3 for descriptions). No clear cases of downward trapping (DT) were identified.

As noted by Várnai and Davies (1999), photon pathlength details may be irrelevant when only the net reflected or transmitted flux is of interest. For example, a photon may experience UT yet still exit the cloud in the upward direction with a longer pathlength in 3D RT than with the independent pixel approximation (IPA), thereby not contributing to a change in the radiative quantity of interest (Várnai and Davies, 1999). In this analysis, 3D effects refer to their *net* influence. This means that the mechanisms DE, UT, and UE refer to the portion of SW flux that changes the downward (DE) or upward flux (UT, UE) in 3D relative to 1D RT. For example, UT then refers to the portion of SW flux affected by UT that actually reduces upward flux above the cloud in 3D RT.

It should be noted that the IPA, as used in DISORT, yields accurate fluxes only when averaged over sufficiently large horizontal areas (Cahalan et al., 1994). The 3D effects examined in the flux fields therefore represented local features. Systematic 3D-1D biases at large spatial scales were not addressed in this specific analysis of 3D effects.

3.5 Resources and Tools

Data This Master’s thesis used EarthCARE products (European Space Agency (ESA), 2025), which provided atmospheric geophysical data (ACM-COM, ALL-3D, AM-ACD), reference TOA flux observations (BMA-FLX), and reference simulated fluxes (ACM-RT). Additionally, surface observations were obtained from the Norwegian Center for Climate Services seklima.met.no, which provided reference downward SW flux observations at the surface.

Code All Python scripts for data pre-processing, setting up and running the RTMs, and plotting figures are available at the [GitHub repository](#). A README provides a brief overview and instructions for using the repository.

Tools Key tools used in this work include:

- **libRadtran** ([Mayer et al., 2025a](#)): Radiative transfer library used for DISORT (1D) and MYSTIC (3D) simulations.
- **ChatGPT** ([OpenAI, 2026](#)): Assisted with language clarity and debugging code. Error messages related to code were sent to ChatGPT for interpretation and suggested improvements. All text in this Master’s thesis was written by the author. ChatGPT was only used to suggest improvements to grammar, sentence structure, and clarity. No text was generated independently without written text as input from the author, and all suggestions were carefully checked before being included.

Chapter 4

Results & Discussion

This chapter presents the main results from the libRadtran RTM simulations and their evaluation against other RTMs, EarthCARE products, and surface pyranometer data. Results from several sensitivity studies and 3D cloud radiative effects are also presented and discussed.

4.1 Radiative Assessment of TOA Fluxes

An overview of TOA performance for the RTMs relative to observations is presented first. Accurate simulations of TOA fluxes require accurate atmospheric input data. Comparisons with BMA-FLX observations (Sect. 2.3.1) and libRadtran simulations (Sect. 3.1) at the top-of-atmosphere (TOA) therefore provide valuable insight into the quality of the EarthCARE geophysical data. The results presented in this section reflect the final RTM setup used in this study and provide an overview of the overall radiative assessment.

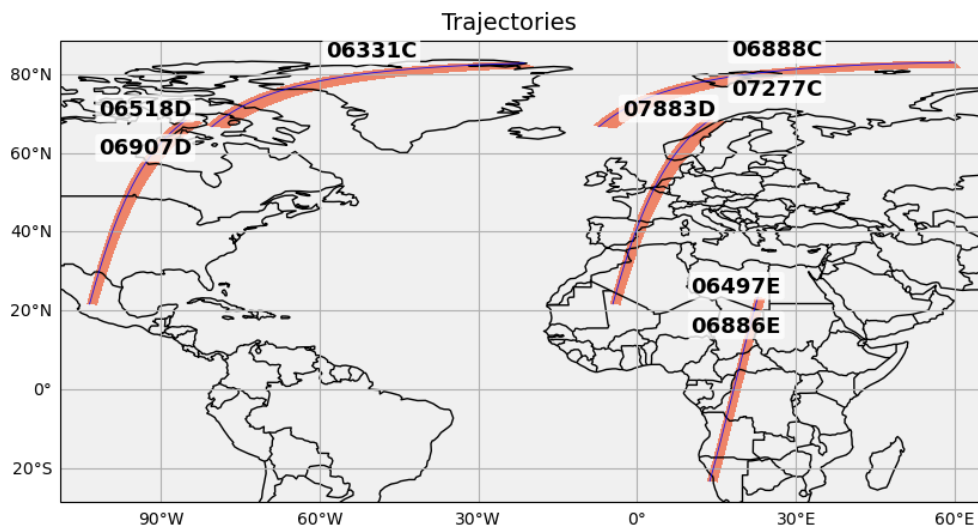


Figure 4.1: Trajectories of all simulated swaths in this study, corresponding to the eight swaths listed in Tab. 3.4. Blue lines show the nadir track, while red shading indicates the across-track extent.

Figures 4.2(a)-(c) and 4.3(a)-(c) show TOA SW and LW fluxes, respectively, for tropical (a, Orbit 06497E), mid-latitude (b, Orbit 07883D), and Arctic regions (c,

Orbit 06888C). Simulated fluxes match the BMA-FLX resolution, and quality filtering is applied to all the data. As seen in the figures, the agreement between BMA-FLX (red), MYSTIC (blue), DISORT (green), and ACM-RT 1D (grey) is generally good for all swaths. The close agreement between DISORT and the operational ACM-RT 1D results provides further confidence in the simulations.

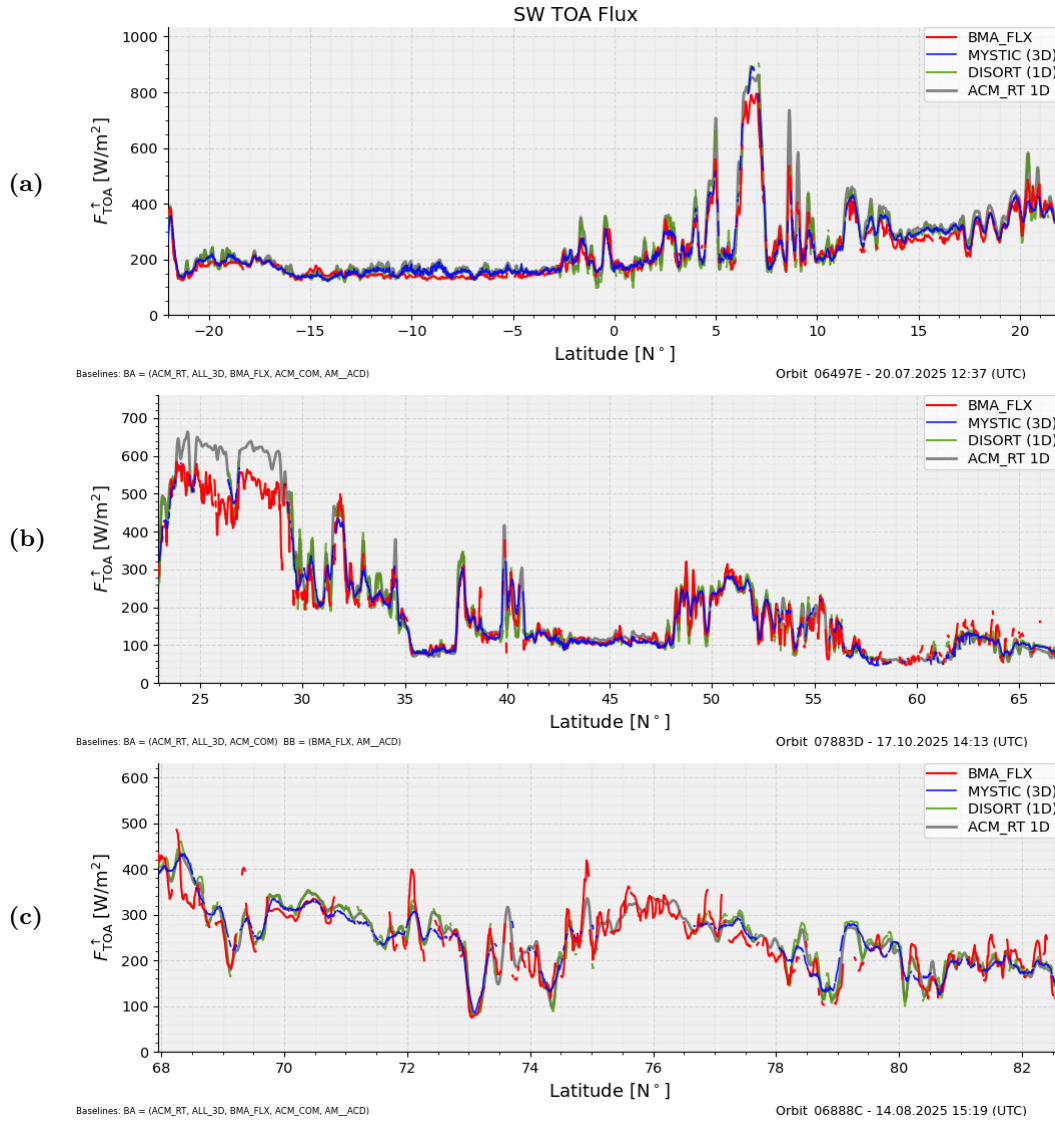


Figure 4.2: Overview of TOA upward SW flux for multiple swaths covering tropical (a), mid-latitude (b), and Arctic (c) regions. Shown are BMA-FLX (red), MYSTIC (blue), and DISORT (green), and ACM-RT 1D (gray).

The tropical swath shown in Figs. 4.2(a) and 4.3(a) (Orbit 06497E) has SZAs ranging from 28° to 53° across the swath. The cloud fraction along the satellite nadir-track is 0.72, meaning that 72% of the nadir pixels contain clouds. The cloud cover is especially visible north of the equator, where larger reflected SW fluxes from clouds are seen in Fig. 4.2(a), particularly between 5° and 10° N. The clouds are also visible through reduced LW transmission in the same regions in Fig. 4.3(a).

The mid-latitude swath 07883D in Figs. 4.2(b) and 4.3(b) has the highest SZAs

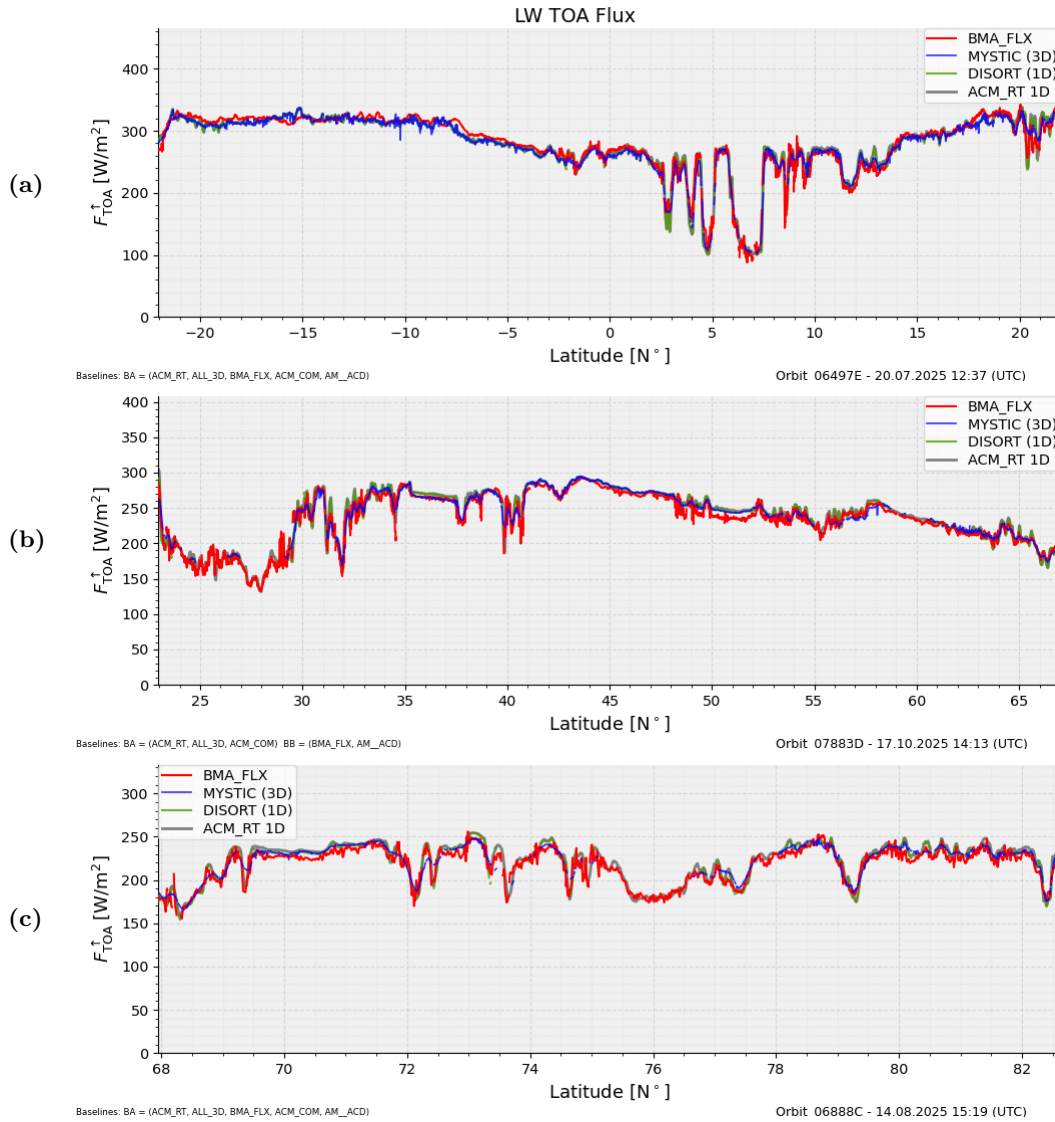


Figure 4.3: Overview of TOA upward LW flux for multiple swaths covering tropical (a), mid-latitude (b), and Arctic (c) regions. Shown are BMA-FLX (red), MYSTIC (blue), and DISORT (green), and ACM-RT 1D (gray).

among the three presented swaths, ranging from 44° to 89°, and the lowest cloud fraction with a value of 0.59. In the most cloudy region around 25-30°N, only the BMA-FLX and ACM-RT 1D fluxes are shown, and larger deviations are seen there compared with the rest of the swath. Our libRadtran simulations are not shown in this region due to the poor quality status of the ACM-COM atmospheric input data used to drive the simulations. If plotted without applying the quality status filtering, the results closely follow the ACM-RT 1D fluxes. Such deviations are communicated to the ESA EarthCARE validation teams and will be a topic of future studies.

The Arctic Orbit 06888C shown in Figs. 4.2(c) and 4.3(c) has the largest cloud cover with a cloud fraction of 0.97, indicating near-complete cloud coverage along the nadir-track. The SZAs range from 60° to 78°. This swath is shown as it provides the most continuous BMA-FLX observations among the selected Arctic EarthCARE

swaths. However, after applying BMA-FLX quality filtering, the observed SW flux is not fully continuous for Orbit 06888C (Fig. 4.2(c)). Poor quality status of the ACM-COM atmospheric input data is commonly observed for Arctic swaths, leading to fewer simulated TOA upward SW and LW fluxes in several regions compared to swaths at lower latitudes. The less continuous BMA-FLX data coverage and the poorer data quality status in Arctic regions reflect the uncertainties and challenges in retrieving clouds and radiative fluxes in the Arctic (Griesche et al., 2024).

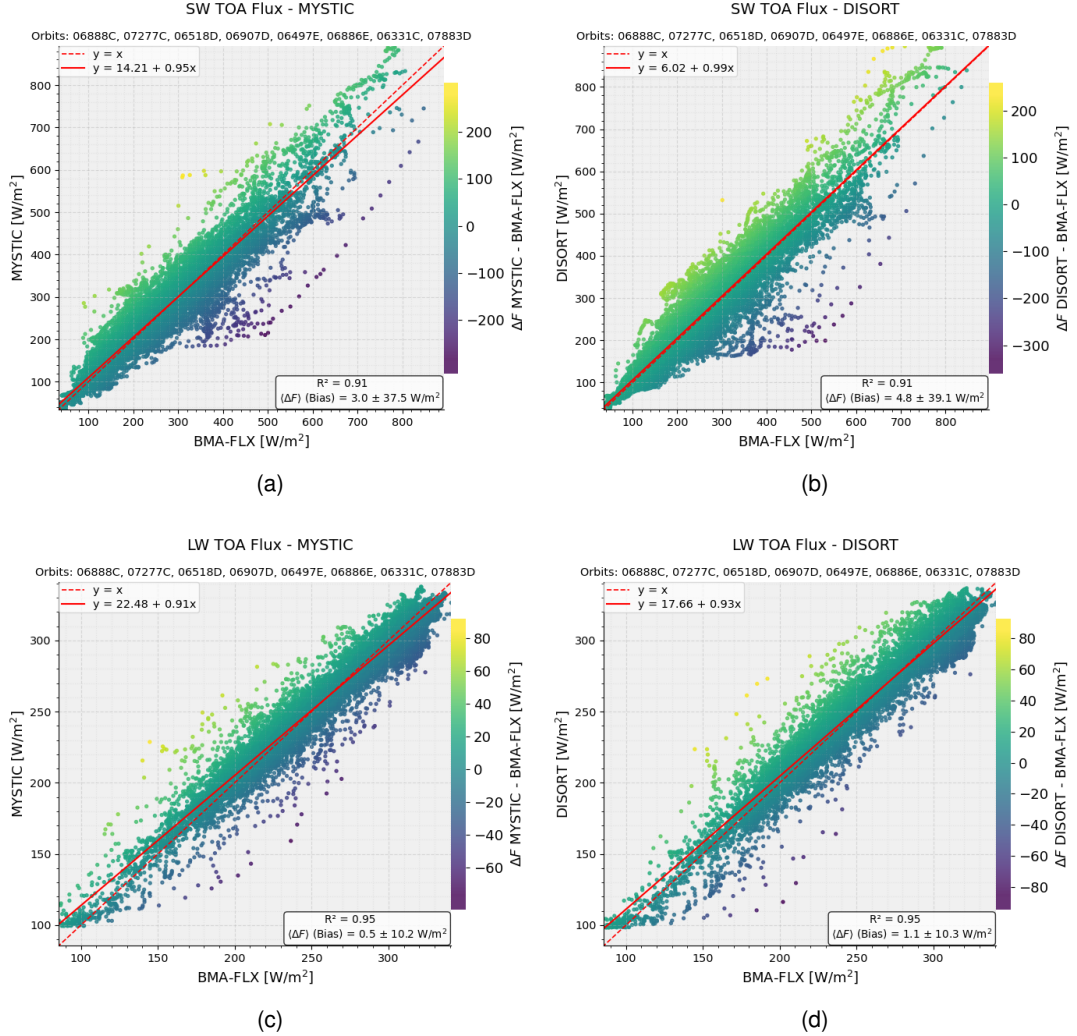


Figure 4.4: Model-observation scatter plots of TOA upward SW (upper panels) and LW (lower panels) fluxes for MYSTIC (left panels) and DISORT (right panels), across the total eight simulated swaths. The specific swaths are indicated in the small text above each panel.

Figure 4.4(a)-(d) presents a quantitative radiative assessment of libRadtran simulations against BMA-FLX observations, with SW results for MYSTIC (a) and DISORT (b), and LW results for MYSTIC (c) and DISORT (d), across the eight swaths simulated in this study (shown in Fig. 4.1). Simulated fluxes match the BMA-FLX resolution, and quality filtering is applied.

Stronger spatial variability and systematic differences are seen in the TOA fluxes in the SW region (Fig. 4.4(a)-(b)) than in the LW region (Fig. 4.4(c)-(d)), reflected in the larger biases and standard deviations against BMA-FLX observations. The mean

Table 4.1: Mean flux differences (bias $\pm$ standard deviation) between libRadtran RTMs and BMA-FLX observations in the SW and LW regimes across eight swaths (Fig. 4.1).

libRadtran RTMs	SW [W/m^2]	LW [W/m^2]
MYSTIC	3.0 ± 37.5	0.5 ± 10.2
DISORT	4.8 ± 39.1	1.1 ± 10.3

flux differences, i.e., the bias, for MYSTIC and DISORT in the SW and LW regions are listed in Tab. 4.1. These biases do indicate a small overestimation by all the models when compared with EarthCARE’s radiative-closure target of $\pm 10 \text{ W}/\text{m}^2$ (Cole et al., 2023).

The improved agreement when using more realistic 3D RTM with MYSTIC compared to the 1D approximation in DISORT is expected. The mean flux difference between the 3D and 1D models is $1.8 \text{ W}/\text{m}^2$ in the SW region and $0.6 \text{ W}/\text{m}^2$ in the LW region. This reflects that systematic flux differences between 3D and 1D RT are more pronounced in SW with cloudy atmospheres and generally much smaller in the LW (Cole et al., 2023).

4.2 Impact of Aerosols on LW Radiation

In general, SW radiation is more strongly influenced by aerosols than LW radiation, although aerosols can still have a non-negligible impact in the LW region (see Sect. 2.2.1) (Zhou and Savijärvi, 2014).

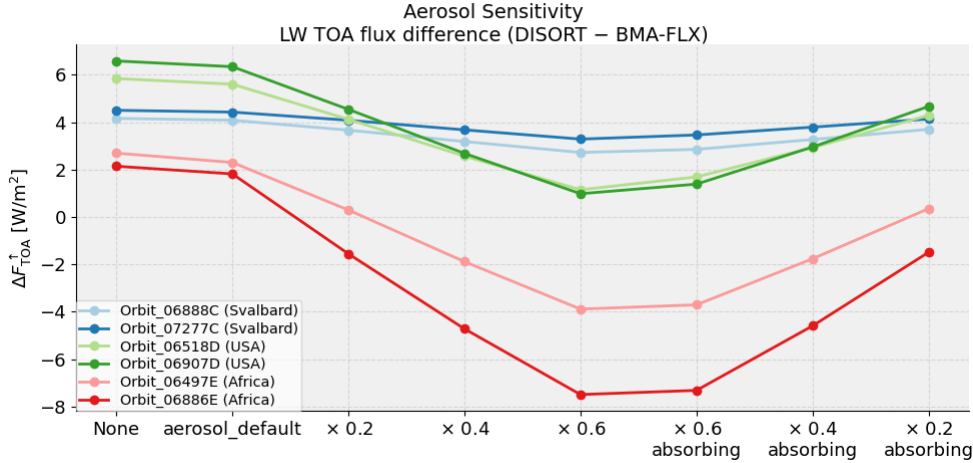


Figure 4.5: Swath-averaged difference in TOA upward LW flux between DISORT simulations, using the derived LW AOD-models (x-axis), and BMA-FLX observations, shown for six swaths. The x-axis labels denote the AOD-models used to convert AOD from the SW to the LW region using linear scaling with conversion factors k (i.e., $\times k$). The label “absorbing” indicates that LW AOD is applied only to pixels containing coarse-mode absorbing aerosols.

Figure 4.5 shows the swath-averaged impact of eight AOD-models (see Sect. 3.4.2) as the difference in TOA upward LW flux between DISORT and BMA-FLX observations for the six swaths in Fig. 3.3. Simulated fluxes match BMA-FLX resolution (see Sect. 3.2) and quality filtering is applied. As seen in Fig. 4.5, variations in TOA upward LW flux arise from differences in AOD. Nevertheless, the swath-average flux differences remain within EarthCARE’s radiative-closure target of $\pm 10 \text{ W}/\text{m}^2$ (Cole et al., 2023). This

aligns with the generally small LW radiative impact of aerosol from absorption and re-emission (Zhou and Savijärvi, 2014). A reduction in outgoing LW flux is observed as a negative bias relative to BMA-FLX for higher conversion factors, as this leads to larger LW AOD (see Eq. 3.1)

Fluxes in Fig. 4.5 are referenced to BMA-FLX observations so that results from different swaths can be compared on a common scale. The differences are defined as $\Delta F_{\text{TOA}}^{\uparrow} = F_{\text{DISORT}} - F_{\text{BMA-FLX}}$. An AOD-model that yields the smallest flux difference across all swaths does not necessarily represent the optimal choice. The comparison reflects sensitivity to the AOD-model relative to the BMA-FLX reference, and can be affected by retrieval uncertainties in both the atmospheric profiles and the observed fluxes. This analysis is therefore used to assess sensitivity rather than model performance.

The many-to-one problem (Barker et al., 2025) with mapping between aerosol properties and radiative fluxes implies that agreement with observations does not uniquely validate the aerosol representation, making the radiative assessment more difficult. Accordingly, the TOA upward LW flux computed using an “optimal” AOD-model does not necessarily perfectly match the observed fluxes from BMA-FLX when evaluated using a swath-averaged metric. Instead, the preferred AOD-model for LW applications should ideally provide the most physically consistent representation of atmospheric RT supported by theoretical considerations and observational evidence from the literature.

Clarisse et al. (2019) reported the use of a constant conversion factor of 2 between dust optical depth at 10 μm (LW) and 550 nm (SW), corresponding to a conversion factor of 0.5 when converting AOD from the SW to the LW region. The authors further noted that possibly large regional biases may be expected when applying a fixed conversion factor. Other reported conversion factors between the infrared and visible spectral ranges span approximately 0.9 to 3.5 (Clarisse et al., 2019), corresponding to SW-to-LW conversion factors of about 0.3 to 1.1. These reported values do not clearly favor a single AOD-model. However, a conversion factor of 0.4 lies well within the range of the reported estimates and is adopted as a conservative and robust choice for our standard libRadtran simulations.

LW aerosol effects are dominated by absorption and subsequent thermal re-emission, so neglecting aerosol scattering is a good approximation in the LW region (Zhou and Savijärvi, 2014). The coarse-mode absorbing aerosol representation is therefore expected to provide the most realistic representation of LW RT. The coarse-mode absorbing AOD-models (see Sect. 3.4.2) also enable a framework in which a single fixed conversion factor is not uniformly applied to all pixels, thereby addressing the risk of introducing large regional biases associated with a fixed conversion factor (Clarisse et al., 2019). Accordingly, the AOD-model applying a conversion factor of 0.4 to coarse-mode absorbing aerosols only (i.e., dust, dusty smoke, dusty mix, or stratospheric ash), referred to as the **AOD-model $\times 0.4$ absorbing**, is considered the best choice.

The impact of limiting the AOD scaling from the SW to the LW region only to coarse-mode absorbing aerosols is most clearly seen for a conversion factor of 0.6 in Fig. 4.5, which yields the largest LW AOD. This is seen from the difference between the AOD-models $\times 0.6$ and **$\times 0.6$ absorbing**. However, these differences are smaller than those between models using different conversion factors (e.g., larger flux differences between $\times 0.4$ and $\times 0.6$ than between $\times 0.6$ and **$\times 0.6$ absorbing**). This indicates that the TOA upward LW flux is more sensitive to the choice of conversion factor used here than to restricting the scaling to coarse-mode absorbing aerosols.

Following the restriction of LW AOD effects to coarse-mode absorbing aerosols, the swath-average flux differences in Fig. 4.5 depend on the aerosol classification. If most of the AOD along a swath is classified as coarse-mode absorbing aerosols, only small differences are expected between models applying a fixed conversion factor to all pixels and those applying it only to such aerosols. This dependence on aerosol classification helps explain the weaker sensitivity to the coarse-mode absorbing AOD-models over the African swaths (red lines in Fig. 4.5) compared to the swaths over the USA (green lines), as seen in the flux difference between $\times 0.6$ and $\times 0.6$ **absorbing**. This indicates that a larger fraction of the aerosols over Africa are coarse-mode absorbing aerosols, supported by the ACM-COM target classification, which shows a higher occurrence of such aerosol types, particularly dusty mix, over the two African swaths. In contrast, the African swaths show the largest sensitivity to the choice of conversion factor, seen as larger deviations from the aerosol-free case (AOD-model **None**), as expected from higher aerosol loading.

Figure 4.6 shows TOA upward LW fluxes for Orbit 06886E over Africa. DISORT simulations with the coarse-mode absorbing AOD-model ($\times 0.4$ **absorbing**; blue dots) and without aerosols (AOD(**NONE**); green dots) are shown together with BMA-FLX (red line). The figure shows that increased LW AOD reduces the outgoing flux (blue dots) compared to the aerosol-free case (green dots). The simulated fluxes are matched to the BMA-FLX resolution. No quality filtering is applied as the figure is included to provide a picture on how an AOD-model affects LW fluxes rather than providing a quantitative evaluation with BMA-FLX.

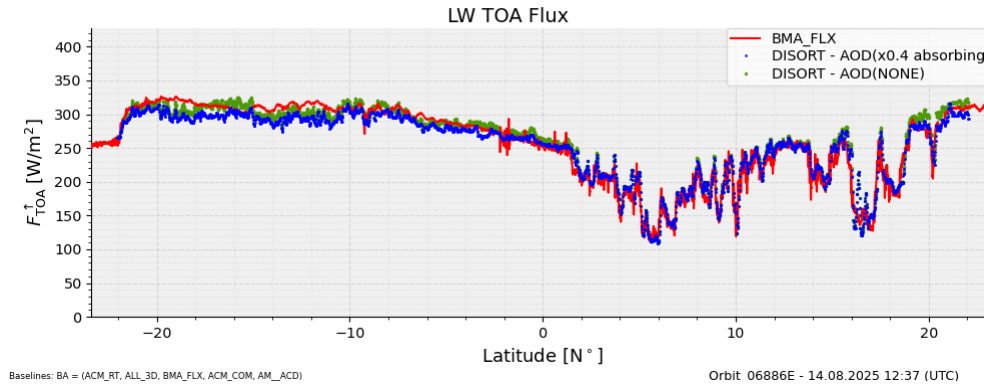


Figure 4.6: TOA upward LW flux along Orbit 06886E. The figure shows BMA-FLX (red line), DISORT simulations with the coarse-mode absorbing AOD-model ($\times 0.4$ **absorbing**; blue dots), and DISORT simulations without aerosols (AOD(**NONE**); green dots).

Figure 4.7 shows the target classification for Orbit 06886E, i.e., types of aerosols and clouds identified by EarthCARE’s ATLID and CPR instruments (AC-TC product, TC: target classification). The figure indicates a thick layer of dusty mix (purple) between altitude 2 and 5 km, and stratospheric sulfate and ash (black and gray) above 17 km. These aerosol types can have a notable effect on the absorption of LW radiation.

Figure 4.8 (upper panel) shows spatial differences in TOA upward LW flux between simulations with the coarse-mode absorbing AOD-model $\times 0.4$ **absorbing** and without aerosols for the same swath as in Figs. 4.6 and 4.7, i.e., Orbit 06886E over Africa. The lower panel shows the corresponding LW AOD. For this model-model comparison, the highest resolution is used with simulated fluxes evaluated on a 1×1 pixel domain. No quality filtering is used, as both RTMs use the same prescribed input data, and the focus

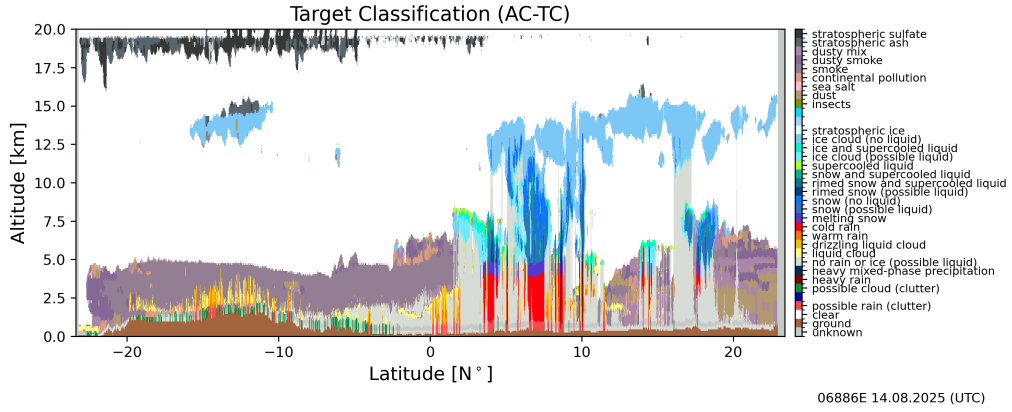


Figure 4.7: The target classification (TC) from the AC-TC product for Orbit 6886E.

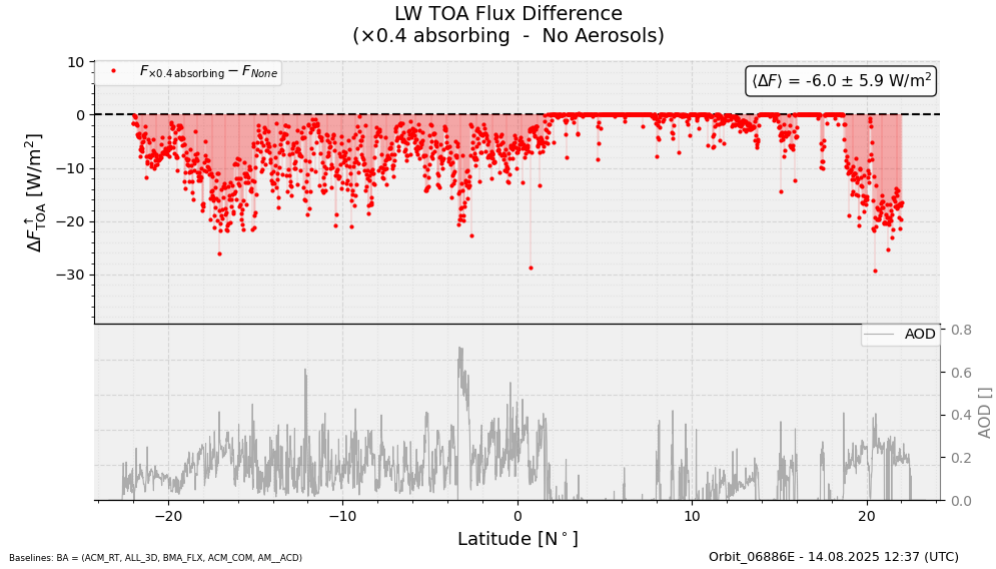


Figure 4.8: Spatial difference in TOA upward LW flux between DISORT simulations that include only coarse-mode absorbing aerosols with a conversion factor of 0.4 ($\times 0.4$ absorbing) and corresponding simulations without aerosol. The lower panel shows the approximated total column AOD at 4500nm (LW).

is on sensitivity.

As seen in Fig. 4.8, the swath-averaged effect of the selected AOD-model ($\times 0.4$ absorbing), relative to the aerosol-free case, is a reduction in TOA upward LW flux of $-6.0 \pm 5.9 \text{ W/m}^2$. However, local differences exceed 25 W/m^2 at individual pixels. This swath-average effect includes contributions from aerosol-free pixels within the swath, which reduce the magnitude of the aerosol-induced LW radiative effect when averaged over the full swath.

As seen in Fig. 4.8, the flux difference shows a nearly mirrored latitudinal pattern with the AOD distribution. The large local differences are linked to the use of a coarse-mode absorbing AOD-model and the high occurrence of coarse-mode absorbing aerosol over Africa, especially dusty mix, confirmed by the target classification in Fig. 4.7.

Combining the TOA flux visualization (Fig. 4.6) and the flux difference between the $\times 0.4$ absorbing AOD-model and the aerosol-free simulations (Fig. 4.8), with the target

classification (Fig. 4.7), helps explain the differences between the two AOD-models. The $\times 0.4$ **absorbing** model and the aerosol-free simulations differ mainly between -20° and 0° , and again around 20°N . The region from -20° to 0° contains a thick layer of dusty mix (Fig. 4.7), which is classified here as an absorbing coarse-mode aerosol type. Model differences are therefore expected in this region, and they are observed. In the adjacent cloudy region, the impact of the AOD-models is either absent or small compared with regions dominated by aerosol (Fig. 4.8). Around 20°N , differences between the AOD-models are observed, consistent with the presence of dusty mix and dusty smoke, which are absorbing coarse-mode aerosols used in the $\times 0.4$ **absorbing** AOD-model.

4.3 Sensitivity to Ice Crystal Habits

The radiative properties of ice clouds depend on their ice crystal habit (Wang et al., 2014), which varies across geographic regions such as the tropics, mid-latitudes, and the Arctic (Um et al., 2015). The following analysis examines the sensitivity of TOA upward SW and LW fluxes to ice crystal habits (Solid Column, SC; Rough Aggregate, RA; and General Habit Mixture, GHM) in tropical (E-swaths over Africa), mid-latitude (D-swaths over the USA), and Arctic regions (C-swaths over Svalbard) using DISORT simulations across six swaths shown in Fig. 3.3. The swaths are cropped to regions dominated by ice clouds. Flux sensitivity is evaluated using inter-habit differences, i.e., simulations that differ only in prescribed ice habit. A 1×1 pixel assessment domain with no quality filtering is chosen as a suitable setup for presenting the inter-habit flux differences.

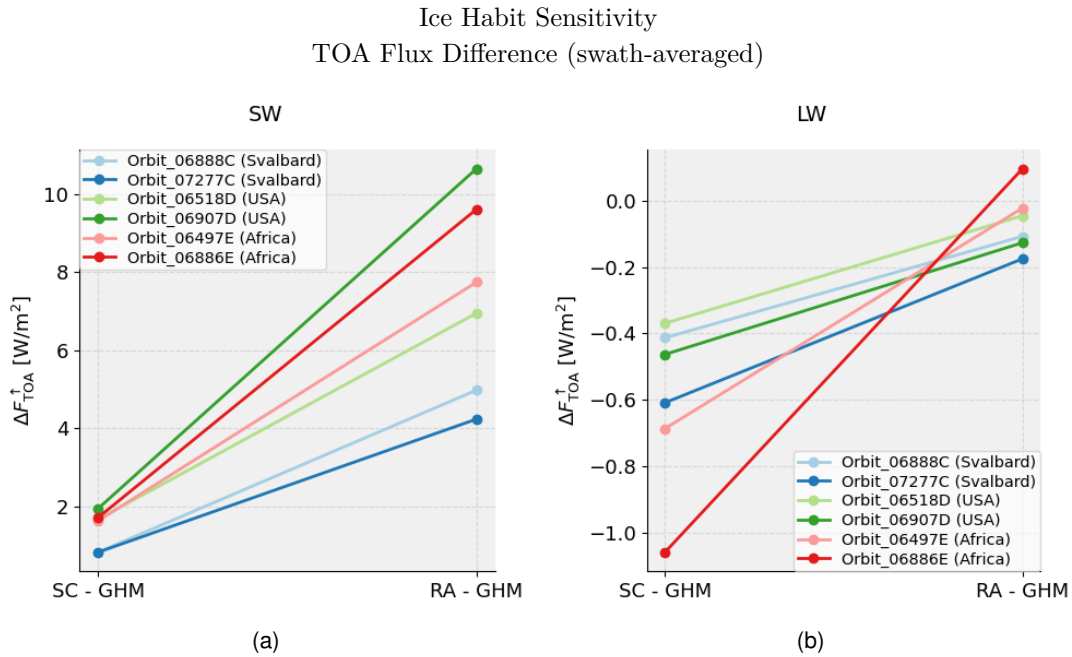


Figure 4.9: Swath-averaged TOA upward SW flux differences (a) and LW flux differences (b) from DISORT simulations with the ice habits SC and RA, relative to the GHM, as labeled on the x-axis. Colors indicate different swaths. Swaths are cropped to regions dominated by ice clouds.

Figure 4.9(a)-(b) shows swath-averaged differences in TOA upward SW (a) and LW (b) fluxes, where SC and RA are compared to the GHM (i.e., SC-GHM and RA-GHM)

across the six swaths seen in Fig. 3.3. The figure shows that the choice of ice habit introduces a systematic SW and LW flux bias as seen for both the SC-GHM and RA-GHM configurations across all six swaths spanning tropical to Arctic regions. The largest differences occur in the SW region (Fig. 4.9(a)), particularly for RA-GHM, with flux differences ranging from 4.1-10.3 W/m². In contrast, LW flux differences are relatively small (Fig. 4.9(b)), with the largest deviations between SC and GHM ranging from 0.4-1.1 W/m². The figure also shows that GHM produces the lowest TOA upward SW flux, while SC produces the lowest TOA upward LW flux.

Table 4.2: Maximum TOA upward SW and LW flux differences for RA and SC relative to the GHM from DISORT simulations for three regions: tropics, mid-latitudes, and Arctic. The regional values represent averages over two swaths in each region.

Region	SW [W/m ²]		LW [W/m ²]	
	RA-GHM	SC-GHM	RA-GHM	SC-GHM
Tropics	60.4	14.5	1.7	4.4
Mid-lat.	50.7	11.6	1.4	4.2
Arctic	18.0	3.4	0.7	2.3

Table 4.2 gives an overview of SW and LW maximum absolute inter-habit TOA flux difference for the habit configurations RA-GHM and SC-GHM. These values are shown for tropical, mid-latitude, and Arctic regions, averaged over two E-, D-, and C-swaths, respectively. The table shows that SW flux sensitivity between SC and GHM is about a factor of five lower than between RA and GHM. In contrast, in the LW region, the SC-GHM configuration yields the largest flux differences. The results in Tab. 4.2 confirm that inter-habit sensitivity is much smaller in the LW than in the SW region by more than an order of magnitude. This is consistent with the stronger role of scattering in the SW region and the resulting higher sensitivity to ice habit (Wang et al., 2014). In the following, the focus is on the SW region, where sensitivity is stronger than in the LW, so no further figures of ice habit sensitivity in the LW are shown.

Table 4.3 lists mean SZA and ice water path (IWP) at the location of maximum TOA upward SW flux difference between the RA-GHM configuration (see Tab. 4.2) for the tropical, mid-latitude, and Arctic regions. Again, these regional values are averages over the two swaths in each region and are based on the cropped ice cloud segments used in this section.

Table 4.3: Mean SZA and mean IWP at the location of the maximum TOA upward SW flux difference between the RA-GHM ice habit configurations (see Tab. 4.2 for flux values) across the tropics, mid-latitudes, and Arctic.

Region	Mean SZA [°]	IWP at max flux difference [g/m ²]
Tropics	29.5	$4.0 \cdot 10^2$
Mid-lat.	38.8	$2.4 \cdot 10^2$
Arctic	68.5	$2.7 \cdot 10^2$

The mean SZAs in Tab. 4.3 reflect the geographical separation of the tropical, mid-latitude, and Arctic regions, with values of approximately 30°, 40°, and 70°, respectively. Interestingly, the IWP at the locations of maximum RA-GHM TOA upward SW flux differences falls within a narrow range of $2 \cdot 10^2$ to $4 \cdot 10^2$ g/m² across all regions. In contrast, the overall IWP range within the swaths varies substantially, with maximum values from about $2.0 \cdot 10^3$ g/m² (Orbit 06888C) to $3.1 \cdot 10^4$ g/m² (Orbit 06518D).

The influence of SZA and IWP on the ice habit signature in the SW region is first examined by relating the maximum TOA flux differences (RA-GHM and SC-GHM; Tab. 4.2) to the corresponding SZAs in Tab. 4.3, and then by assessing how spatial variations in IWP impact the flux differences where the SZA remains approximately constant. Following the rationale that enhanced multiple scattering progressively suppresses radiative signatures specific to individual ice habits, either through larger SZAs (i.e., longer slant paths) or increased optical thickness, as discussed by [Wendisch et al. \(2005\)](#), TOA upward SW flux differences are expected to decrease with increasing SZA (see Sect. 2.2.2). Consistent with this reasoning, we observe that as SZA increases from tropical to Arctic conditions ($\text{SZA} \approx 30 \rightarrow 70^\circ$; Tab. 4.3), the maximum TOA upward SW flux differences decrease for both RA-GHM and SC-GHM configurations (RA-GHM: $\Delta F \approx 60 \rightarrow 18 \text{ W/m}^2$; SC-GHM: $\Delta F \approx 15 \rightarrow 3 \text{ W/m}^2$; Tab. 4.2).

When addressing the impact of IWP on the ice habit sensitivity of TOA upward SW fluxes, the effect of SZA cannot easily be separated as both contribute to enhanced multiple scattering that suppresses habit-specific features ([Wendisch et al., 2005](#)). This complexity complicates the interpretation of regional differences between tropical, mid-latitude, and Arctic conditions. Therefore, spatial patterns of inter-habit flux differences with approximately constant SZA are examined to isolate the influence of IWP on flux differences.

Figure 4.10(a)-(c) shows the spatial distribution of RA-GHM TOA upward SW flux differences with the corresponding IWP shown on a secondary axis for Orbit 06497E (a, tropics), Orbit 06518D (b, mid-latitude), and Orbit 06888C (c, Arctic). The RA-GHM configuration is used because it yields the largest SW flux differences (see Tab 4.2). Results for SC-GHM and the corresponding LW cases are available on [GitHub](#).

A pattern is observed in Fig. 4.10, where RA-GHM flux differences increase with IWP, indicating a direct influence of IWP on the flux differences. However, this relationship holds only up to $\text{IWP} \approx 10^2 \text{ g/m}^2$, which is also seen from Tab. 4.3, with flux differences decreasing for higher IWP. This behavior is most clearly seen in Fig. 4.10(b) around 41°N , where a continuous increase in IWP toward the south is associated with increasing flux differences up to $\text{IWP} \approx 10^2 \text{ g/m}^2$. Similar patterns are observed in Fig. 4.10(a) around 6°N and in Fig. 4.10(c) around 68.5°N .

Following the reasoning that enhanced multiple scattering progressively suppresses habit-specific features ([Wendisch et al., 2005](#)), we observe that increasing IWP from values on the order of 10^2 g/m^2 leads to reduced ice habit flux sensitivity (Fig. 4.10). In contrast, this behavior is not evident at lower IWP. This may indicate that the clouds are too optically thin to sufficiently impact the radiation, although this interpretation remains uncertain.

The observed impact of SZA and IWP on flux differences between ice habit configurations is based on a relatively small set of six cropped swaths. However, the trends are consistent across all swaths:

1. As SZA increases, the SW habit-induced flux differences decrease.
2. As IWP increases above values in the order of 10^2 g/m^2 , the SW habit-induced flux differences decrease.

In future studies, a larger set of swaths should be analyzed to draw more robust conclusions.

A note is made regarding the very high IWP values observed in Fig. 4.10. We have been in contact with the ESA EarthCARE validation teams through the NEVAR project ([NILU, 2022–2027](#)), which is part of the EarthCARE validation activities, concerning

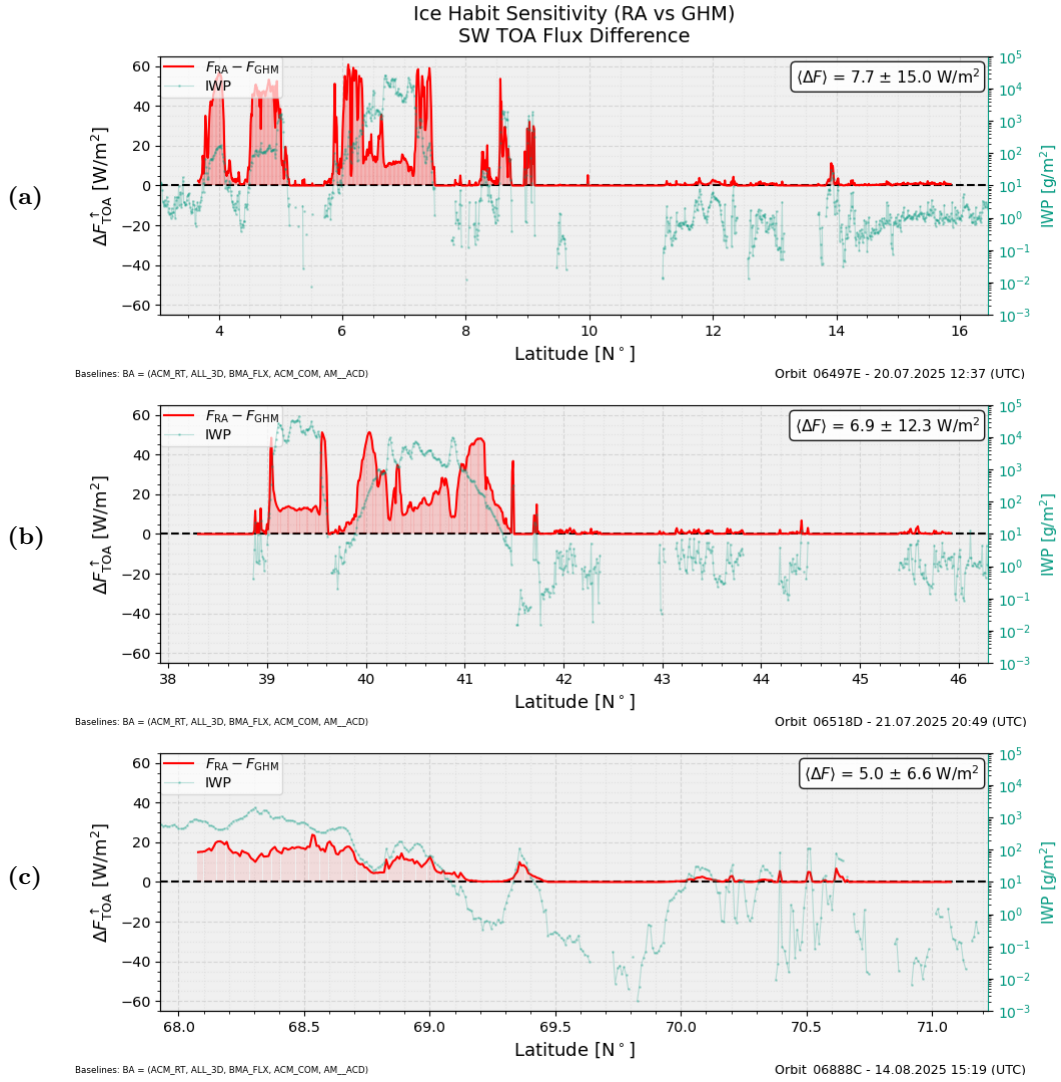


Figure 4.10: TOA upward flux difference between simulations with ice habit RA and GHM for three swaths: 06497E (a, tropics), 06907D (b, mid-latitude), and 06888C (c, Arctic). Ice cloud properties along the swaths are shown on the twin axis as IWP.

these values. The reported IWP values, reaching up to $\text{IWP} \approx 7 \cdot 10^4 \text{ g/m}^2$ (Fig. 4.10(b)), were confirmed to be non-physical. According to the validation teams, this is a known issue in the BA baseline where ACM-CAP overestimates ice extinction at the top of convective clouds, leading to the high IWP values. Further work is ongoing to improve this issue. However, it does not affect the validity of the present ice habit sensitivity study since the prescribed model input is consistent across simulations.

Throughout this Master's thesis, the GHM ice habit is used as the reference based on theoretical considerations (see Sects. 2.2.2 and 3.4.3). All figures showing flux sensitivity to ice habit therefore use GHM as a reference. Agreement with observations is not used as a metric here following the reasoning of the many-to-one problem (Barker et al., 2025), where agreement between ice cloud properties and radiative fluxes does not uniquely validate the cloud representation. For example, unrealistic high IWP values can still produce agreement in fluxes if other settings are “tuned” to match observations. The

analysis therefore focuses on flux differences between ice habits.

Overall, the sensitivity of TOA upward fluxes to ice habit, assessed from the RA-GHM and SC-GHM configurations, is small in the LW region. Swath-averaged differences are usually below 1 W/m^2 (Fig. 4.9(b)), and maximum differences (SC-GHM; Tab. 4.2) remain below 5 W/m^2 . These values are well within the EarthCARE radiative-closure threshold defined by agreement within $\pm 10 \text{ W/m}^2$ of measured and modeled TOA fluxes (Cole et al., 2023). In the SW region, swath-averaged differences are also within the $\pm 10 \text{ W/m}^2$ radiative-closure target (Fig. 4.9(a)). However, maximum differences (RA-GHM; Tab. 4.2) reach $18\text{-}60 \text{ W/m}^2$, exceeding the radiative-closure threshold. This indicates that the choice of ice habit in the SW RTM can lead to flux deviations that occasionally exceed, and in some cases substantially exceed, the EarthCARE radiative-closure goal.

4.4 Sensitivity of TOA Fluxes to ACM-COM Atmospheres

Uncertainties and biases in RT simulations arise from both model simplifications and atmospheric input, including aerosol and cloud representation as discussed in Sects. 4.2 and 4.3. This section evaluates the sensitivity of simulated TOA SW and LW fluxes using two atmospheric profiles from the ACM-COM processor (see Sect. 2.3.1 and 3.4.4 for details). Atmosphere 1 (ACM-CAP), regarded as the mission’s best estimate (Cole et al., 2023), and Atmosphere 0 (composite profiles) are analyzed over a subset of Orbit 06497E, a swath exhibiting both liquid and ice clouds (the full swath is shown in Figs. 4.2(a) and 4.3(a)).

Figures 4.11(a)-(b) and 4.12(a)-(b) show the simulated TOA upward SW and LW fluxes, respectively, for Atmosphere 1 (a) and Atmosphere 0 (b). The corresponding LWC and IWC fields are shown on the same latitudinal grid, allowing direct comparison of cloud structure and radiative response.

Fluxes are simulated with MYSTIC and DISORT with assessment domains matching BMA-FLX resolution. No quality filtering is used in order to assess the continuous impact of the atmospheric input on the simulated fluxes. For Atmosphere 1, the operational 1D fluxes from ACM-RT are also included.

Several notable features are evident in Figs. 4.11 and 4.12:

1. Atmosphere 1 shows a more continuous and physically consistent cloud field than Atmosphere 0, without the white ice-cloud “holes” seen in Atmosphere 0 (see lower panels of Figs. 4.11 and 4.12) and with what appears to be a more realistic liquid water structure. In contrast, Atmosphere 0 exhibits elongated liquid clouds extending across nearly half a degree of latitude seen between $3.8\text{--}4.1^\circ$ and $4.5\text{--}5^\circ\text{N}$. The overall cloud structure in Atmosphere 1 is consistent with its formal recognition as the EarthCARE mission’s best estimate (Cole et al., 2023). These issues in ice and liquid clouds in the composite profiles (Atmosphere 0) have been discussed with the ESA EarthCARE validation teams. The ice cloud “holes” are caused by quality filtering, where data flagged as bad by the A-ICE processor are removed, leading to unrealistic clear-sky regions within the cloud layers. Improvements are ongoing, and better consistency within ice cloud regions is already observed, although some artifacts in liquid layers still remain.
2. Generally, more rapidly varying TOA upward SW fluxes are simulated by both MYSTIC (blue line) and DISORT (green line) when using Atmosphere 0 seen in Figs. 4.11(b) and 4.12(b). In contrast, simulations using Atmosphere 1, seen in

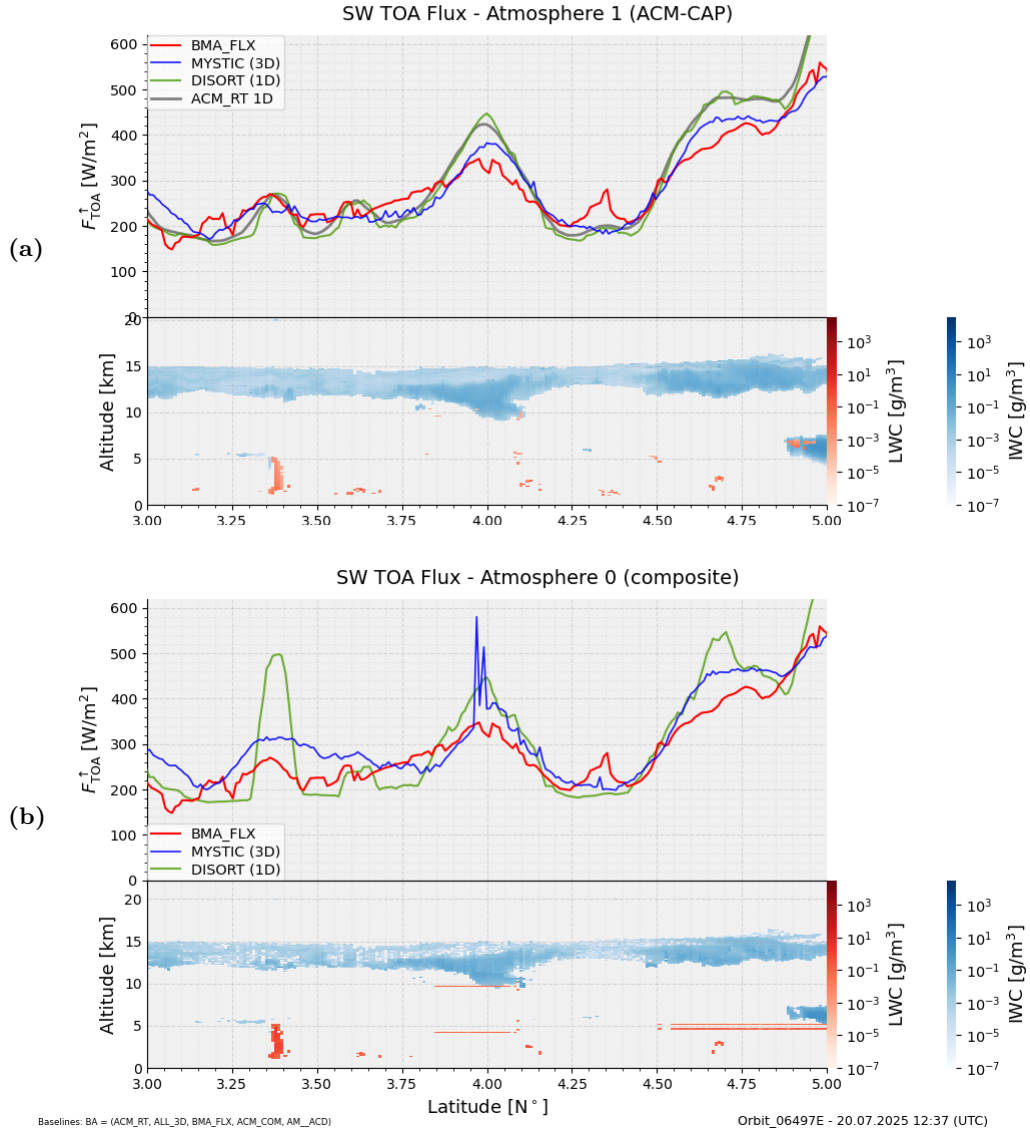


Figure 4.11: TOA upward SW fluxes simulated using a) Atmosphere 1 (ACM-CAP) and b) Atmosphere 0 (composite profiles). BMA-FLX (red), MYSTIC (blue), DISORT (green), and ACM-RT (grey) are shown. The corresponding LWC and IWC fields for each atmosphere are shown in the lower panels.

Figs. 4.11(a) and 4.12(a), show better agreement with the spatial variability of the observed fluxes from BMA-FLX (red line). The rapid flux fluctuations from Atmosphere 0 are assumed to be related to the apparent ice cloud “holes” visible in the lower panels of the respective figures for this atmosphere.

- DISORT overlaps closely with ACM-RT 1D (grey line) for Atmosphere 1 in both the SW region (Fig. 4.11(a)) and in the LW region (Fig. 4.12(a)), which is expected as both 1D RTMs use ACM-CAP profiles (Cole et al., 2023).
- Around 3.4°N in Fig. 4.11(b), a specific high radiative signal is observed for the DISORT simulation using Atmosphere 0, but not for the corresponding MYSTIC

4.4. Sensitivity of TOA Fluxes to ACM-COM Atmospheres

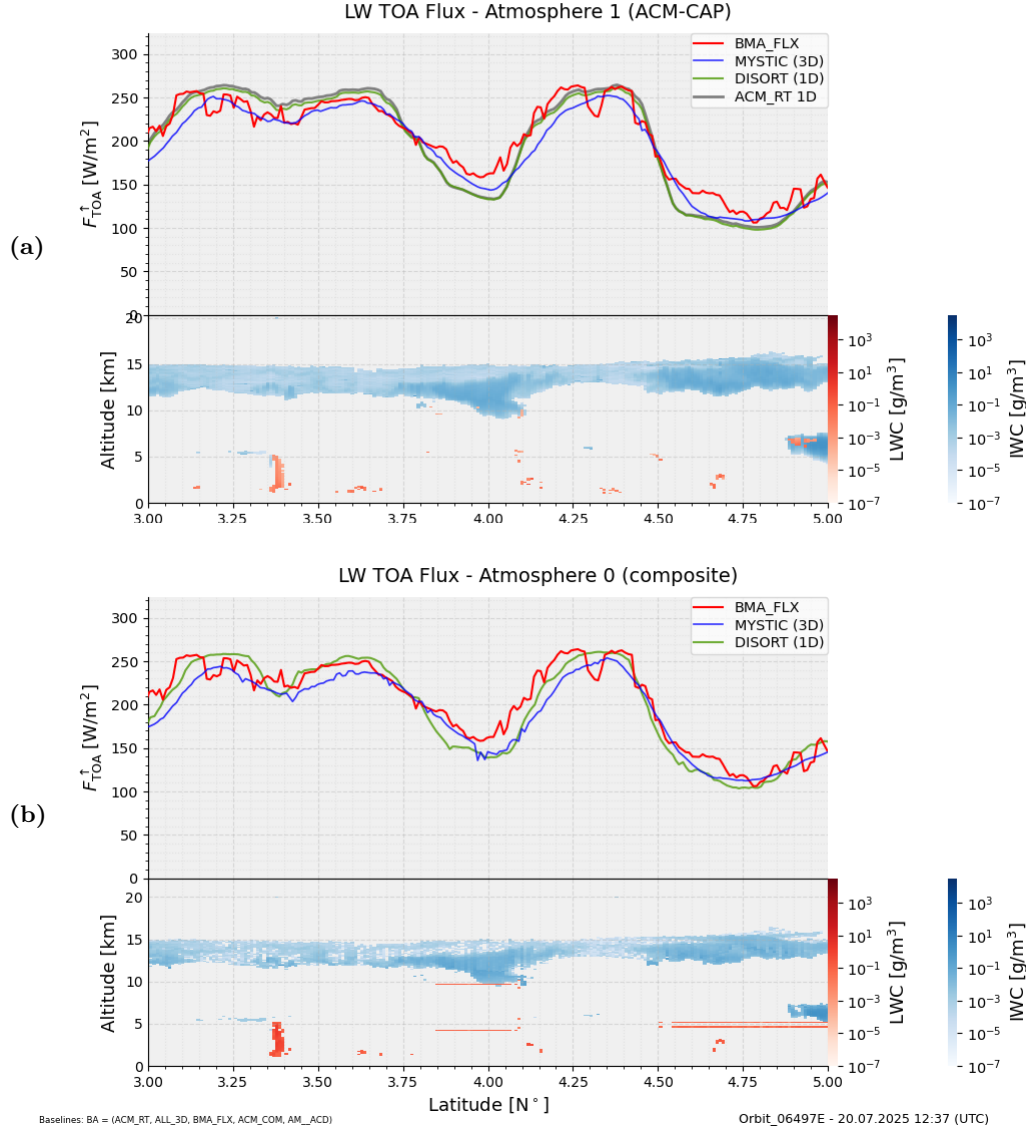


Figure 4.12: TOA upward LW fluxes simulated using a) Atmosphere 1 (ACM-CAP) and b) Atmosphere 0 (composite profiles). BMA-FLX (red), MYSTIC (blue), DISORT (green), and ACM-RT (grey) are shown. The corresponding LWC and IWC fields for each atmosphere are shown in the lower panels.

simulation. This artifact is likely related to the high IWC below 5 km at this latitude. The TOA upward SW fluxes from DISORT are averaged over 10 consecutive along-track pixels, which makes the flux signal extend farther than the horizontal extent of the liquid cloud. The signal is not seen with MYSTIC, likely because MYSTIC also includes across-track pixels. The 11×11 pixel assessment domain in MYSTIC therefore includes about 90% more pixels extending across-track than the DISORT nadir-track averaging domain, which likely reduces the localized signal seen in DISORT.

These results emphasize the importance of accurate atmospheric input to the RTMs and the uncertainties such inputs introduce into computed radiative quantities. In

Tab. 4.4, this is further quantified by showing TOA upward SW flux differences between libRadtran RTMs (MYSTIC and DISORT) and observations from BMA-FLX for the full swath of Orbit 06497E. For direct comparison with BMA-FLX, the simulated fluxes are provided at BMA-FLX resolution, and quality filtering is applied.

Table 4.4: TOA upward SW flux differences between libRadtran RTMs (MYSTIC and DISORT) and BMA-FLX observations for the two ACM-COM atmospheres over the full Orbit 06497E.

ACM-COM Atmosphere	RTM	Flux Difference (RTM - BMA-FLX) [W/m^2]
ACM-CAP (Atm. 1)	MYSTIC	8.8 ± 24.7
	DISORT	8.9 ± 29.4
Composite (Atm. 0)	MYSTIC	43.7 ± 82.1
	DISORT	46.1 ± 101.9

Atmosphere 1 leads to substantially better agreement with BMA-FLX observations for both 3D and 1D RTMs compared to Atmosphere 0, with the bias reduced by approximately a factor of five and the standard deviation by about a factor of three. ACM-COM provides several atmospheric parameters to our RTMs in addition to liquid and ice cloud content (see Tab. 3.2). Variations in all these parameters influence the simulated radiative quantities, and their representation in the different atmospheres remains a direction for future work. Based on the findings described above, we follow EarthCARE’s practice for ACM-RT and use ACM-CAP profiles, which are recognized as the best estimate of atmospheric profiles (Cole et al., 2023).

4.5 RTM Evaluation Against Surface SW Flux Observations

Accurate surface SW fluxes are important because they drive the exchange of energy and moisture between the surface and atmosphere, and are essential for modeling weather, climate, and radiative energy (Huang et al., 2019). Extending the focus of this Master’s thesis from the TOA to the surface provides additional insight into the range of RTM applications and allows further assessment of EarthCARE data. This section includes an assessment of simulated surface downward SW fluxes against ground-based observations from seklima.met.no. We also look at corresponding TOA fluxes to further compare accuracy at the surface and at the TOA (method described in Sect. 3.4.5).

4.5.1 Surface Fluxes

Figure 4.13 shows a scatter plot of surface SW downward fluxes from MYSTIC (blue) and DISORT (green) compared to ground-based pyranometer observations from seklima.met.no. The fluxes are computed for the station locations corresponding to across-track pixels within $\pm 5\text{km}$ of the satellite nadir-track with the use of the Scene Construction Algorithm (SCA) (see Sect. 3.3.3). Simulated fluxes are based on a 1×1 pixel assessment domain with no quality filtering because of the relatively small sample size in this analysis. The correlation is based on 36 valid samples, limited by the number of overpasses (initially 61) with high SZAs.

The correlation between RTM simulations and ground-based observations in Fig. 4.13 is slightly better for 3D RT modeling (MYSTIC; blue) compared to 1D RT (DISORT; green). MYSTIC yields a coefficient of determination of $R^2 = 0.78$ and a mean flux difference of $\Delta F_{\text{SUR}}^{\downarrow} = -28.6 \pm 98.2 \text{ W}/\text{m}^2$, while DISORT shows lower correlation with

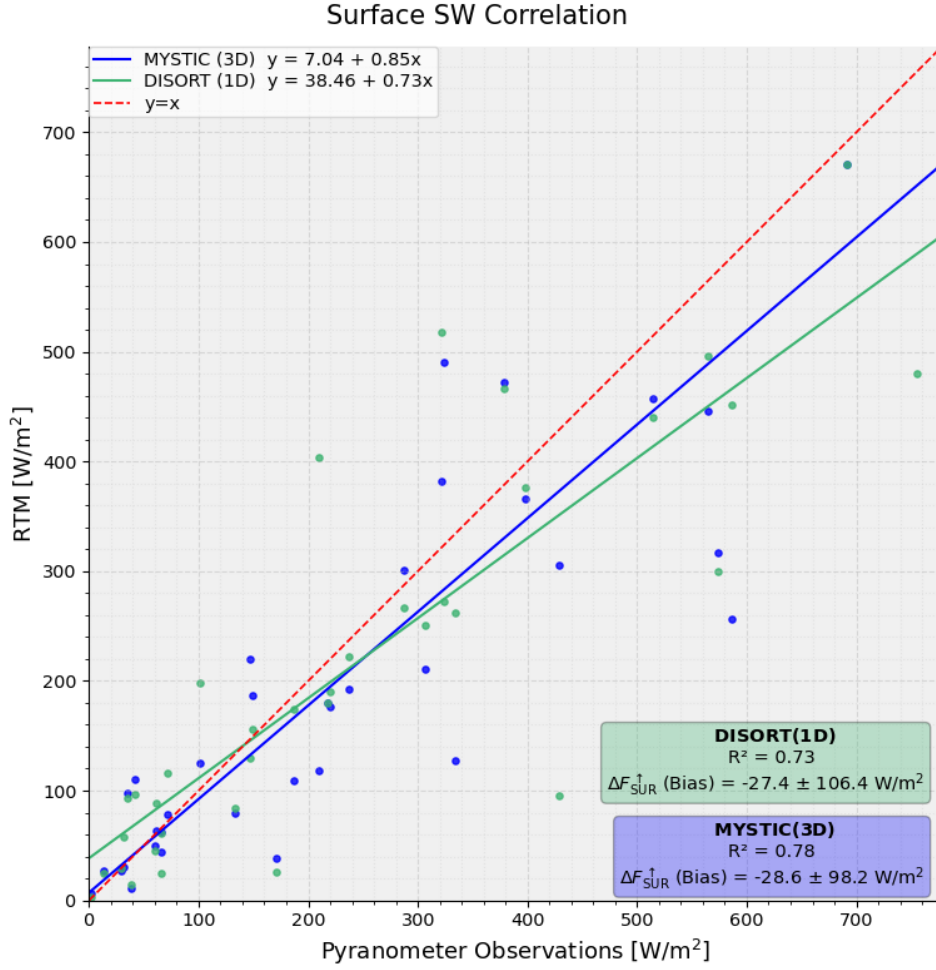


Figure 4.13: Correlation between simulated SW surface downward fluxes from MYSTIC (blue) and DISORT (green) and ground-based pyranometer observations. Linear least-squares fits and associated statistical diagnostics are shown in the respective model colors. The $y = x$ reference line is indicated by a red dotted line.

$R^2 = 0.73$ and a similar mean bias but larger spread, $\Delta F_{\text{SUR}}^{\downarrow} = -27.4 \pm 106.4 \text{ W/m}^2$. In the linear regression, MYSTIC (blue line) shows an intercept closer to zero and a slope closer to unity than DISORT (green).

The results in Fig. 4.13 also suggest that the improved physical realism with a 3D RTM leads to a better least-squares fit compared to the 1D model, along with reduced variability indicated by a smaller standard deviation. However, there is still a large spread between both simulations and the observed fluxes, related to uncertainties in both observations and simulations. Still, the flux differences in Fig. 4.13 are comparable to, but somewhat larger than, the mean surface downward SW flux differences of $15 \pm 61 \text{ W/m}^2$ reported by [Griesche et al. \(2024\)](#). Their comparison was made between a 1D RTM based on ground-based cloud radar and lidar retrievals and observations collected during the PS106 cruise. This provides a useful reference and confirms that the results presented here are within the range reported in the literature.

One difference between 3D and 1D RT is that 3D RT includes slanted shadows from adjacent areas, whereas 1D RT does not. This is particularly relevant for stations at high latitudes and low solar elevations. In MYSTIC, this increases the influence of the

reconstructed atmosphere in the across-track direction from the SCA, because a shadow from an adjacent pixel can propagate to the station location. In contrast, DISORT only considers the nadir atmospheric column and therefore always produces vertical shadows.

Cloud-top height uncertainty in the SCA can affect the 3D RTM more directly than the 1D RTM. A simple example illustrates this: for a cloud-top height error of 1 km and an SZA of 45° , the horizontal displacement of the cloud shadow at the surface is 1 km (assuming flat terrain). At the ~ 1 km horizontal pixel resolution used here, this corresponds to a shift of the shadow into a neighboring pixel relative to the observational site.

Topography and surface albedo also influence RT at the surface. Topographic effects and horizontal variation in surface albedo are only represented in MYSTIC. Several ground-based stations are located in mountain regions with complex terrain and spatially varying albedo, which highlights the limited realism of the 1D RTM. Including topography allows sloped surfaces to be treated more realistically, for example through downward reflection of radiation from snow-covered slopes (Mayer et al., 2025b), a typical feature of Norwegian terrain. An example of this terrain is at the Filefjell station located in a valley that is typically snow-free in late April at the time of observation, while the surrounding mountains remain snow-covered. This example highlights the potential for strong local contrasts in surface albedo that are best captured in MYSTIC.

4.5.2 TOA Fluxes

TOA fluxes are evaluated at the same station locations as in the surface flux comparison (Fig. 4.13). Figure 4.14 shows a scatter plot of TOA SW upward fluxes from MYSTIC (blue) and DISORT (green) compared to BMA-FLX observations. The fluxes are assessed at the satellite nadir pixel closest to the ground-based stations to match the BMA-FLX position. Simulated fluxes use a 1×1 pixel assessment domain, consistent with the surface correlation in Fig. 4.13. BMA-FLX quality filtering is used. ACM-COM quality filtering is not applied to the simulations because the limited sample size, following the same procedure in the surface flux comparison. The correlation is based on 18 valid matchups, limited by the number of overpasses (initially 61), high SZAs, and BMA-FLX quality flags. This is fewer than for the surface comparison (36 matchups; Fig. 4.13).

Figure 4.14 shows that 3D RT modeling with MYSTIC gives improved agreement with observations compared to 1D DISORT simulations. MYSTIC yields a coefficient of determination of $R^2 = 0.91$ and a mean flux difference of $\Delta F_{\text{TOA}}^\uparrow = 2.1 \pm 42.2 \text{ W/m}^2$, while DISORT shows lower correlation with $R^2 = 0.87$ and $\Delta F_{\text{TOA}}^\uparrow = 5.9 \pm 52.9 \text{ W/m}^2$. The regression fits indicate similar overall performance.

There are two limitations in the TOA SW analysis (Fig. 4.14). First, there is a mismatch between the used 1×1 pixel assessment domain and the standard $10 \times 10 \text{ km}^2$ resolution of the BMA-FLX products (Velázquez Blázquez et al., 2024). This choice is motivated by the need to maintain a consistent comparison framework between the surface and TOA analyses, rather than to achieve the best possible agreement at TOA. Second, the statistical robustness of the correlation is limited by the small sample size of only 18 valid TOA samples.

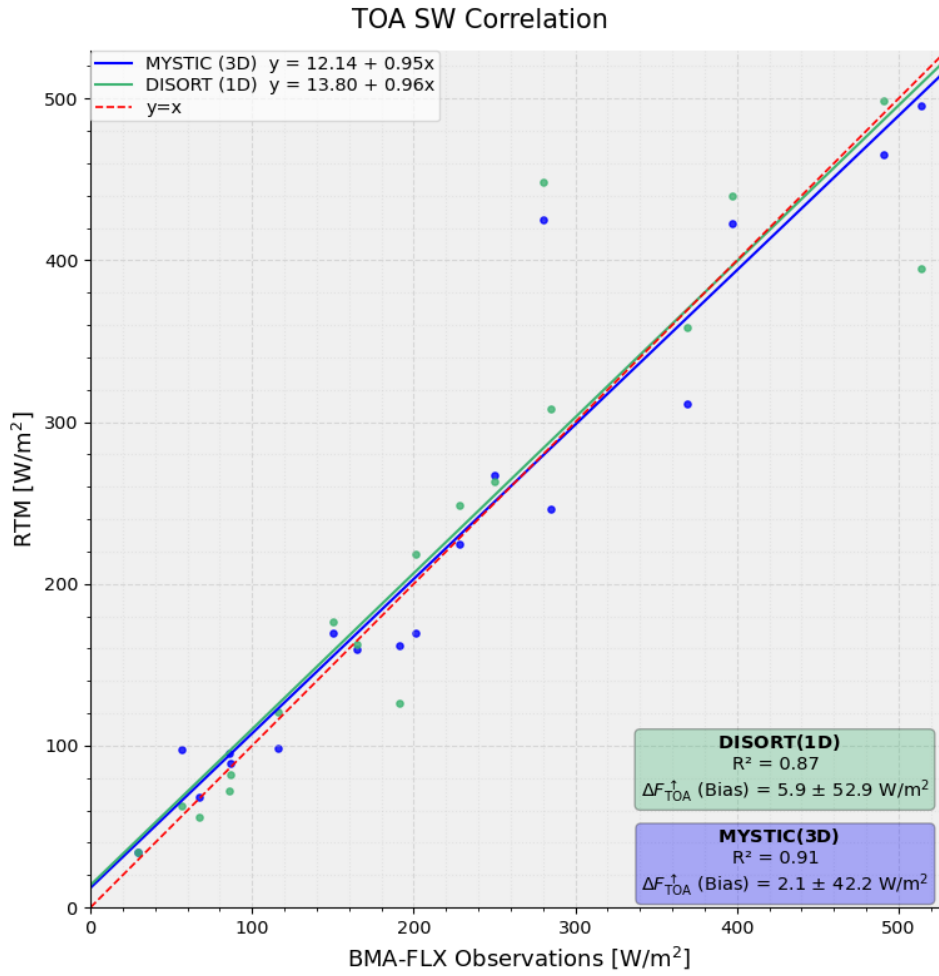


Figure 4.14: Correlation between simulated SW TOA upward fluxes from MYSTIC (blue) and DISORT (green) and BMA-FLX observations. Linear least-squares fits and associated statistical diagnostics are shown in the respective model colors. The $y = x$ reference line is indicated by a red dotted line.

4.5.3 Surface vs. TOA Flux Comparison

The biases in SW fluxes at the surface and TOA relative to observations are summarized in Table 4.5. Only 16 EarthCARE overpasses can be evaluated at both surface and TOA, and only these overpasses are included. This ensures consistent locations and atmospheric conditions. The correlation scatter plots are based on different samples with 18 valid surface samples (Fig. 4.13) and 36 valid TOA samples (Fig. 4.14). These sets do not fully overlap. Therefore, these figures are not suitable for direct comparison between surface and TOA.

Table 4.5 shows that there is a smaller bias and standard deviation at TOA than at the surface. The TOA bias is within EarthCARE's radiative-closure target of $\pm 10 \text{ W/m}^2$ (Cole et al., 2023) for both MYSTIC (3.2 W/m^2) and DISORT (6.0 W/m^2). In contrast, the surface bias for DISORT is much larger and exceeds this target (-26.3 W/m^2). For MYSTIC, the surface bias is within the radiative-closure target (-7.4 W/m^2). The standard deviation is about twice as large at the surface as it is at the TOA. The relatively small sample size of 16 datapoints limits the statistical robustness of these

Table 4.5: SW flux bias for MYSTIC (3D RTM) and DISORT (1D RTM) at the surface and TOA. The bias is the difference between modeled and observed SW fluxes for surface downward and TOA upward flux.

Location	RTM	SW Bias [W/m^2]
Surface	MYSTIC	-7.4 ± 96.2
	DISORT	-26.3 ± 117.6
TOA	MYSTIC	3.2 ± 44.5
	DISORT	6.0 ± 56.1

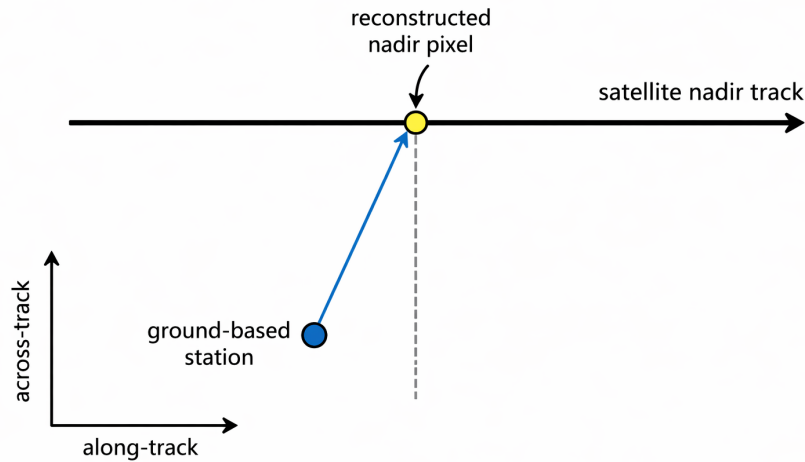


Figure 4.15: Illustration of reconstructing the simulated pixel for a ground-based station with a donor nadir pixel using the SCA.

results. However, both Table 4.5 and the correlation plots in Figs. 4.13 and 4.14 show the same trend: both bias and spread are smaller at TOA than at the surface. This shows that the SW flux agreement between the RTMs and the observations is better at TOA than at the surface.

There are several reasons why the model-observation correlation is better at TOA than at the surface. The maximum horizontal distance between the simulated surface pixel at the ground-based stations and the EarthCARE nadir-track is about ± 5 km. In contrast, the TOA comparison between simulations and observations is made at the satellite nadir. The SCA therefore has a stronger impact on the surface correlation than on the TOA correlation, because the SCA is used to map across-track pixels to a nadir donor pixel. Figure 4.15 shows this process where a ground-based station away from the satellite nadir is reconstructed with the SCA using a donor nadir pixel. The SCA also affects the TOA comparison, but to a smaller extent, through the 3D buffer in the 3D RTM, which extends nadir profiles into full 3D fields (Barker et al., 2011).

It should also be noted that the SCA is a match-and-substitute method that uses EarthCARE's MSI TOA radiances as a constraint (Barker et al., 2011). At TOA, the MSI radiances themselves are used to reconstruct the scene. In contrast, surface radiation is not used in the SCA. This may partly explain the better agreement at TOA than at the surface, as the SCA is more directly constrained at TOA.

The relatively small distance threshold of about ± 5 km between the satellite overpass

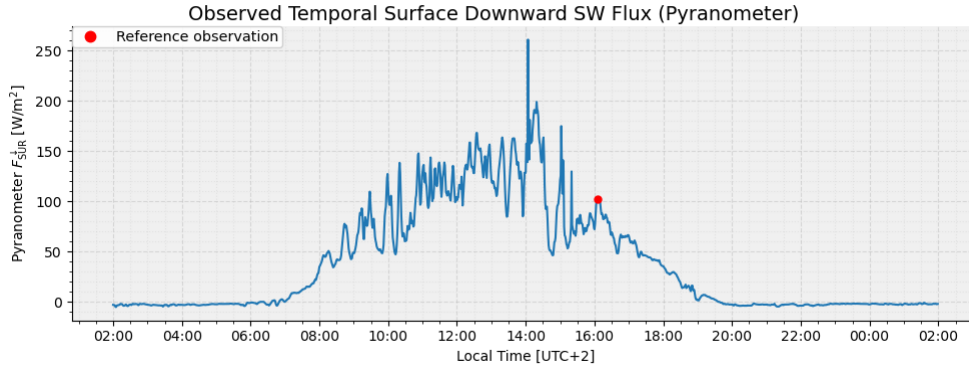


Figure 4.16: Observed surface downward SW flux at the Gratangen-Skjærvika station over the day of the Orbit 01690C overpass. The red dot marks the time of the overpass.

and the ground-based station reduces the impact of the SCA on the analysis, and in particular the uncertainty, which generally increases with distance from the nadir track (Qu et al., 2023). This motivates the use of such a strict threshold, but it also reduces the sample size.

4.5.4 Single-Station Analysis

As demonstrated in the previous sections, modeled surface SW fluxes can be subject to relatively large errors compared to observations. A single-station analysis can be used to investigate this in more detail, here demonstrated by the Gratangen-Skjærvika station during the overpass of Orbit 01690C. This swath was selected because the satellite passed within two across-track pixels of the station and because the MYSTIC simulation shows reasonable agreement with the observation. The observed surface SW flux is 101.7 W/m^2 , compared with 125.0 W/m^2 from MYSTIC and 197.8 W/m^2 from DISORT. This analysis considers both temporal and spatial variation.

The temporal behavior of surface downward SW flux can vary strongly over time. Under cloud-free conditions, the flux changes smoothly and is mainly controlled by solar elevation. In contrast, clouds introduce strong variability and make the temporal behavior more complex. This is shown in Fig. 4.16, which presents the observed flux at the Gratangen-Skjærvika station on the day of the Orbit 01690C overpass, 14 September 2024 at 16:05 local time. The red dot marks the observation time corresponding to the EarthCARE satellite overpass. The flux can vary significantly from second to second, making it crucial to have an exact overpass time and precise time on the ground-based pyranometer measurements.

In addition to temporal variability in SW flux, the spatial structure can be complex, especially with rapidly varying cloud structures. Figure 4.17(a)-(c) show the downward SW flux fields at various altitudes for a) MYSTIC, b) DISORT, and c) the 3D effect. The flux fields are shown for the reconstructed across-track pixel located two pixels from the nadir-track, which corresponds to the ground-based station position. The red dot shows the location of the Gratangen-Skjærvika station. The solar geometry shown in the figure ($\text{SZA} = 74^\circ$, $\phi = 233^\circ$) is defined at the station location. The simulations use a 1×1 pixel assessment domain, representing a single atmospheric column.

Figures 4.17(a) and (b) show how 3D and 1D RT geometry affect the radiation fields, illustrated as slanted shadows in Fig. 4.17(a). Here, the sun direction ($\phi = 233^\circ$) is from the south-west and is approximately parallel to the satellite trajectory, so the shadows

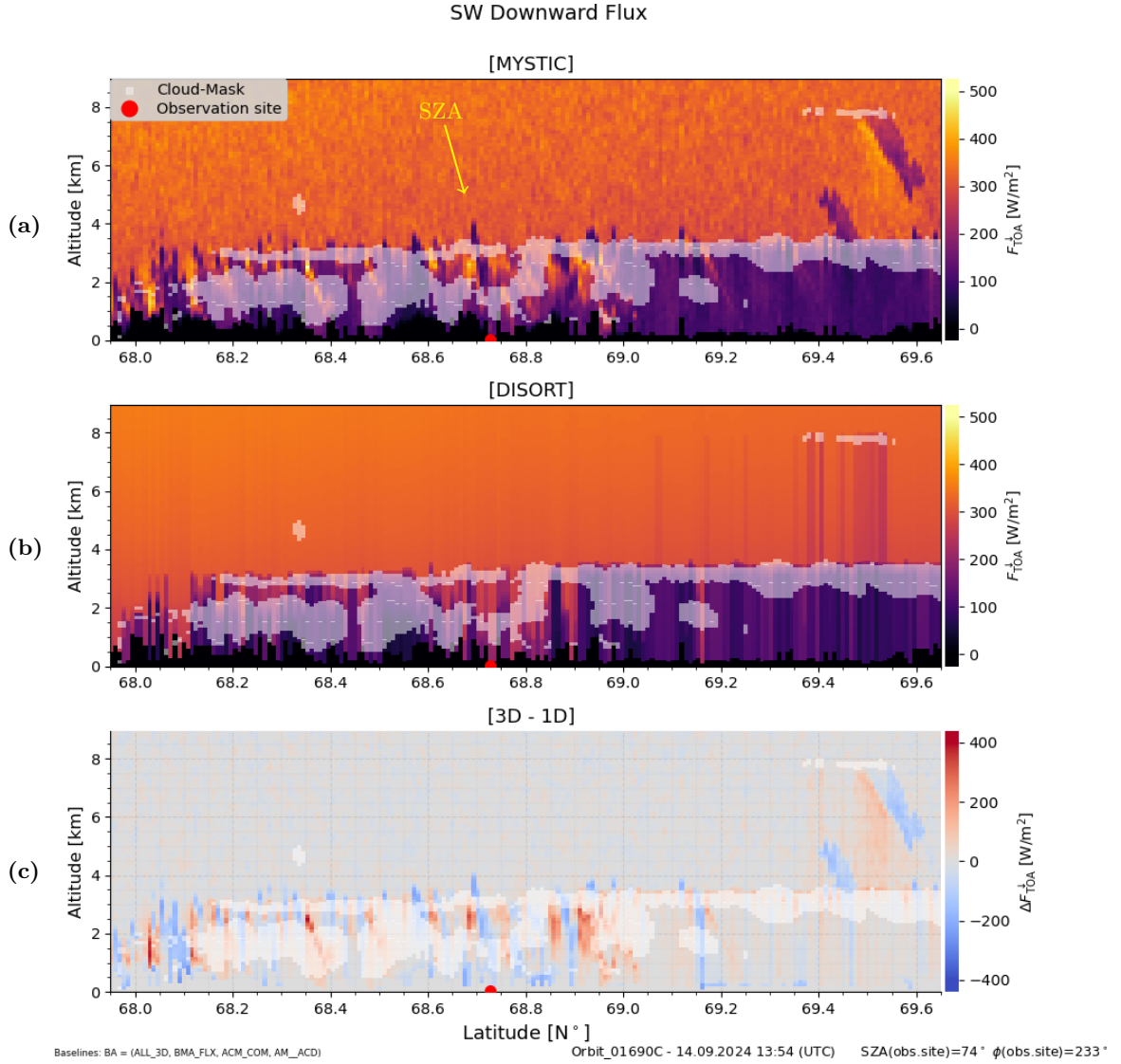


Figure 4.17: Simulated downward SW flux fields around the Gratangen-Skjærvika station (red dot), for a) MYSTIC, b) DISORT, and c) 3D effect. Cloudy pixels are shown in white. The solar zenith angle (SZA) at the station location is illustrated in yellow at the top panel (a).

extend partly along-track. These shadows are not present in Fig. 4.17(b) because the 1D method uses the IPA and only considers single nadir pixels. In this example, agreement between the simulations and the surface observations at the Gratangen-Skjærvika station is better for the 3D RTM than for the 1D RTM. This difference between 3D and 1D RTMs is seen as a negative 3D-1D bias (blue color) at the station location in Fig. 4.17(c).

The noisier flux field from MYSTIC (Fig. 4.17(a)) compared to DISORT (Fig. 4.17(b)) is due to Monte Carlo noise. The noise is larger when fluxes are calculated for all atmospheric levels than for simulations at a single altitude level (e.g., TOA or surface). In addition, the assessment domain is a 1×1 pixel column with no spatial averaging, which reduces the ability to average out the noise. For more information on Monte Carlo noise, see Appendix Sect. 6.2.

An interesting feature of the MYSTIC simulated flux fields in Fig. 4.17(a) is the presence of cloud shadows at altitudes above the cloud mask, and shadows that suddenly

disappear. Shadows start just above the cloud mask, typically on the order of 100 m, throughout the scene. These appear as purple regions above the clouds in MYSTIC and DISORT simulations in Fig. 4.17(a) and (b). One clear case can be seen at 69.4°N in the MYSTIC simulation, where a shadow appears around 5 km altitude without any clouds present. Another example at 69.5°N shows a shadow cast by a cloud at 8 km altitude that disappears around 5 km.

The simulated scene in Fig. 4.17 corresponds to pixels located two across-track pixels from nadir, representative for the Gratangen-Skjærvika station, whereas the cloud mask is shown for the retrieved nadir-track. This leads to differences between the nadir cloud mask and the shadows observed about 2 km across-track because of cloud variability in the across-track direction. This explains why some shadows appear slightly above the cloud mask across the swath in Fig. 4.17(b).

A more pronounced radiative signal occurs where shadows appear clearly above the cloud top, about 1 km or more above it. This is related to the solar azimuth and the 3D RT geometry. The satellite trajectory is not perfectly aligned with the solar azimuth. As a result, the direct solar beam has an across-track component. In 3D geometry, radiation can interact with clouds outside the along-track plane because the 3D buffer allows horizontal photon transport. Adjacent clouds within the 3D buffer can therefore cast slanted shadows that intersect the flux field plane. This can produce shadows without a corresponding cloud in the nadir cross-section (at 69.4°N in Fig. 4.17(a)) or shadows that disappear into the across-track direction (as at 69.5°N). A closer look at the 3D cloud structure shows that across-track pixels around 69.4°N use donor pixels from the high cloud at 8 km altitude. This explains why shadows appear in the flux field at 69.4°N, even though the clouds that cast the shadows are outside the nadir cross-section and are therefore not visible there.

4.6 3D Cloud Radiative Effects

1D RTMs do not capture horizontal photon transport, so 3D RTMs are required for a more realistic representation of radiative interactions with clouds (Várnai and Davies, 1999). In this study, 3D effects are assessed and defined as flux differences between MYSTIC and DISORT (3D - 1D) according to Eq. 2.20 (see Sect. 2.2.3). A 1×1 pixel assessment domain is used, representing the smallest resolution for the flux fields. The resulting 3D effects are therefore local, as no horizontal averaging is applied.

Two swaths are used to study the 3D effects: Orbits 7883D and 6886E, shown in Fig. 4.1. They are briefly described below:

- Orbit 7883D passes over Europe with the satellite trajectory oriented toward the south-west, aligned with the Sun.
- Orbit 6886E passes over Africa along an approximately north-south trajectory with the Sun from the west.

Note that the SZA and solar azimuthal angle ϕ shown in the figures are mean values over the displayed latitudinal region and vary little across the relatively small regions considered.

4.6.1 Cloud-Side Illumination and Shadowing Effects

Using simulated 3D and 1D upward and downward SW flux fields, cloud-side illumination and slanted shadows are visualized in Fig. 4.18(a)-(f). The upper panels (a)-(c) show

downward fluxes, while the lower panels (d)-(f) show upward fluxes. MYSTIC results are shown to the left (a,d), DISORT results in the middle (b,e), and the 3D effect (3D - 1D) is shown to the right (c,f).

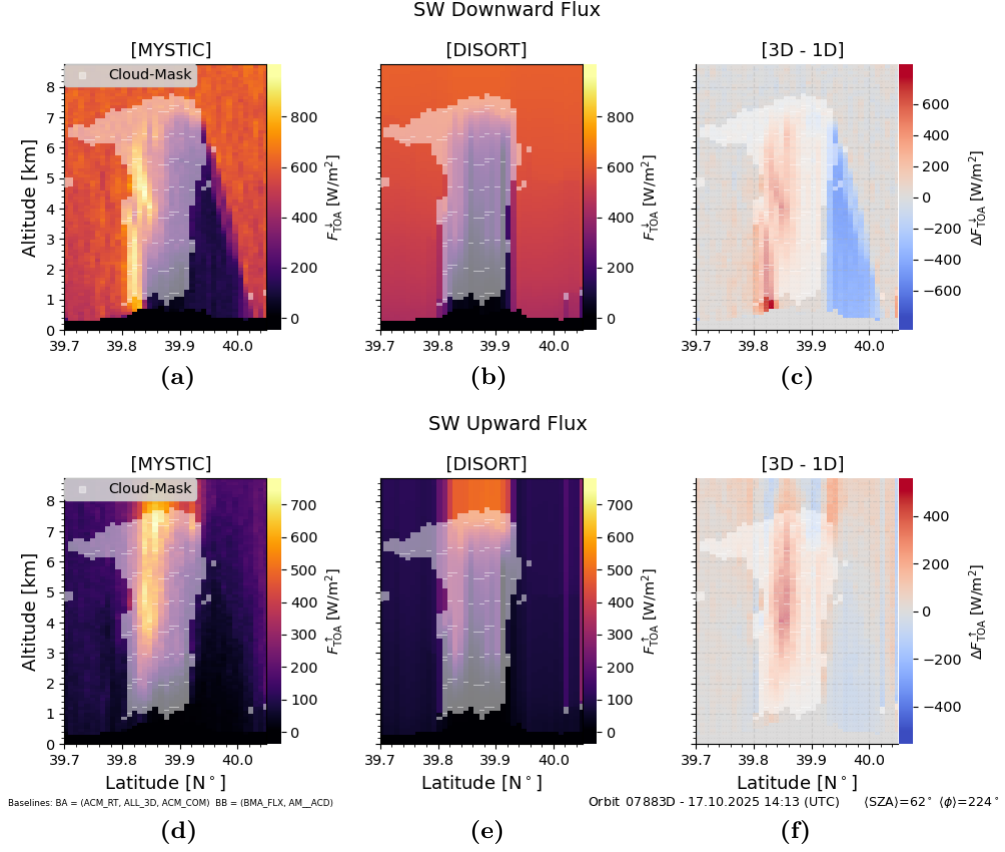


Figure 4.18: Simulated SW flux fields illustrating cloud-side illumination and slanted shadowing effects. Downward fluxes are shown in the upper panels and upward fluxes in the lower panels. The three columns display MYSTIC, DISORT, and the 3D - 1D flux difference from left to right.

As seen in Fig. 4.18(a)-(c), there are larger shadows from side illumination in the downward flux fields from the 3D RTM relative to the 1D RTM. As described in Sect. 2.2.3, cloud-side illumination refers to sunlight entering clouds from the side rather than only from above, as assumed in 1D RT with the IPA (see Fig. 4.18(b)). This increases brightness at the illuminated left cloud edge. This is seen as a positive 3D-1D downward flux bias (red) on the left cloud-side in Fig. 4.18(c), with a downward SW flux difference of about 400-800 W/m². A positive 3D-1D upward flux bias is also observed inside the cloud in Fig. 4.18(f), with an upward SW flux difference of about 200-400 W/m². The upward 3D-1D bias further within the cloud, rather than at the cloud edge, is likely due to multiple scattering, as cloud SW scattering is mainly forward-directed (Hogan and Shonk, 2013), allowing photons to penetrate deeper before being redirected upward in 3D RT.

Elongated slanted shadows extend from clouds at low solar elevation in 3D RT, as seen in Fig. 4.18(a). Here, the shadow spans from approximately 39.92°N to 40.02°N, corresponding to a horizontal distance of about 13 km along the satellite trajectory (azimuth $\approx 220^\circ$). This is consistent with the solar geometry: for SZA = 62° and a vertical height difference $\Delta z \approx 7$ km (surface to cloud top), the horizontal shadow

length L is

$$\begin{aligned} L &= \Delta z \cdot \tan(\text{SZA}) \\ &\approx 7 \text{ km} \cdot \tan(62^\circ) \\ &\approx 13 \text{ km}, \end{aligned} \tag{4.1}$$

with negligible difference due to the relative azimuthal offset of $\sim 4^\circ$ between the solar azimuth and the satellite trajectory.

Figure 4.18(c) shows that slanted shadows are not present in the 1D RTM using the IPA. Only vertically aligned (directly downward) shadowing is represented, as the model considers only the nadir column. The slanted shadow to the right of the cloud in 3D RT is instead dominated by direct solar radiation in 1D RT, which appears as a negative 3D-1D bias (blue) on the right of the cloud. The 3D effect in the right shadow region (~ 39.9 - 40°N) is on the order of -400 W/m^2 (Fig. 4.18(c)). The DISORT downward SW flux in the corresponding slanted shadow region is approximately 600 W/m^2 (Fig. 4.18(b)), resulting in an estimated 60-70% reduction in SW downward flux in 3D relative to 1D RT due to slanted shadows.

With illumination also able to propagate to the side of the cloud, 3D RT allows a larger portion of the cloud to intercept direct solar radiation, which increases reflection (Hogan et al., 2019). The upward reflected SW flux in Figs. 4.18(d) and (e) shows no clear slanted radiation signature. This is caused by diffuse reflection from the cloud. However, differences are observed in the 3D - 1D upward flux difference field in Fig. 4.18(f), with alternating positive and negative values above the cloud. This pattern is likely caused by local cloud inhomogeneities that produce upward trapping (UT) and upward escape (UE) within the cloud (see Sect. 2.2.3). Although this inhomogeneity is not resolved in the cloud mask, radiation moving from thinner to thicker cloud regions can be affected by UT, while radiation moving from thicker to thinner regions can be affected by UE (Várnai and Davies, 1999). It is therefore unclear whether total cloud reflection increases in 3D RT in this case. A domain-averaged assessment is needed to determine this, but is not performed here and is left for future work.

This cloud scene was selected because the single isolated cloud has a large vertical extent. Combined with the low sun angle and a satellite trajectory aligned with the solar azimuth, shadows from the direct sunlight extend along the along-track direction and are visible in the flux fields. This setup provides a clear illustration of shadowing effects in 3D RT compared to 1D RT.

4.6.2 Downward Escape

Increased downward flux in 3D RT relative to 1D due to radiation moving from thicker to thinner cloud regions, i.e., downward escape (DE; see Sect. 2.2.3) (Várnai and Davies, 1999), is shown in Fig. 4.19(a)-(c). This is illustrated by the downward SW flux fields from a) MYSTIC, b) DISORT, and c) the 3D-1D difference field. The DE mechanism is located to the right of the lower cloud around 5.2°N and appears as a positive 3D-1D bias (red) in Fig. 4.19(c). The flux bias is on the order of 200 W/m^2 , corresponding to an estimated 50% increase in SW flux in 3D RT relative to 1D.

The observed DE is consistent with radiation escaping through cloud sides in 3D RT, which is often directed downward due to predominantly forward SW cloud scattering (Hogan and Shonk, 2013). However, the presence of an overlying cloud layer at approximately 10-14 km altitude adds complexity and may also influence the

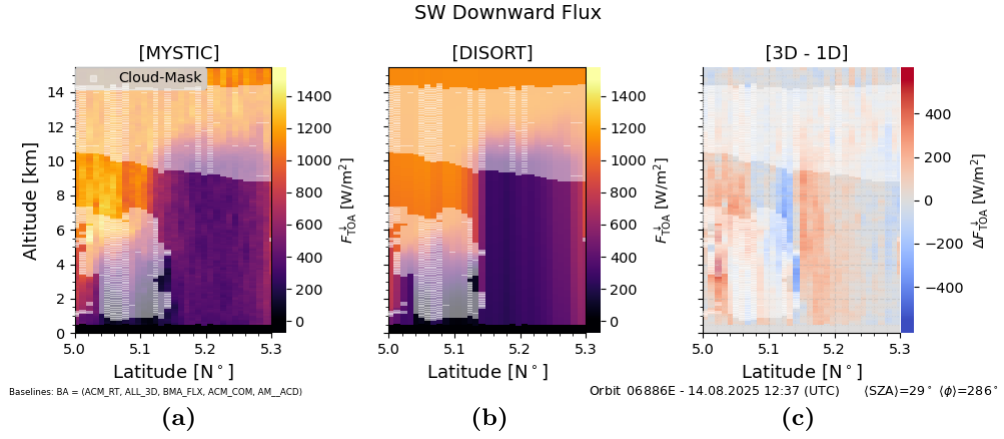


Figure 4.19: Simulated downward SW flux fields illustrating downward escape. The three panels display a) MYSTIC, b) DISORT, and c) the 3D - 1D flux difference.

observed signal. Nevertheless, this case is selected because it clearly shows increased SW downward flux in 3D RT at the side of the lower cloud in Fig. 4.19.

4.6.3 Upward Trapping and Upward Escape

Decreased upward flux in 3D relative to 1D RT can occur when radiation moves from thinner to thicker cloud regions, called upward trapping (UT). Increased upward flux can occur when radiation moves from thicker to thinner cloud regions. This is called upward escape (UE; see Sect. 2.2.3; [Várnai and Davies, 1999](#)). Both mechanisms are observed in Fig. 4.20(a)-(c). From the SW upward flux fields for a) MYSTIC, b) DISORT, and c) their difference, UT appears above the cloud at latitudes larger than 10°N as reduced upward flux in 3D RT relative to 1D RT, shown in blue color in Fig. 4.20(c). UE is located just to the left between 9.94 - 10°N as increased upward flux in 3D RT seen in red color in Fig. 4.20(c). Additional insight is provided in Fig. 4.21 showing the upward SW flux field from MYSTIC together with the corresponding LWC and IWC fields. While the cloud appears homogeneous in Fig. 4.20, the cloud consists of a mixture of optically thick and thin regions, as well as both liquid and ice cloud phases, as seen in Fig. 4.21.

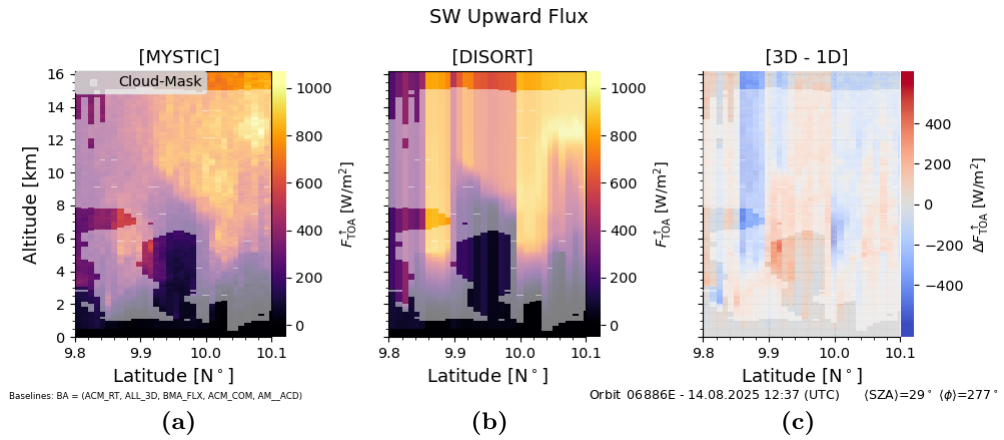


Figure 4.20: Simulated upward SW flux fields illustrating upward trapping and upward escape. The three panels display a) MYSTIC, b) DISORT, and c) the 3D - 1D flux difference.

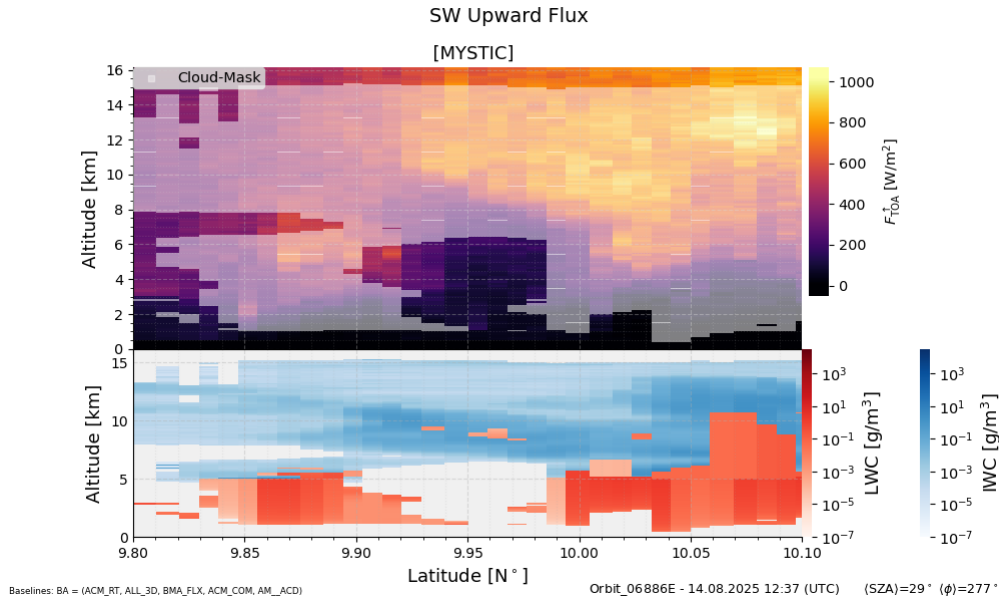


Figure 4.21: Simulated SW upward flux fields from MYSTIC illustrating upward trapping and upward escape together with the corresponding LWC and IWC fields over the same latitudinal range.

Horizontal transport in 3D RT smooths the flux field (Fig. 4.20(a)) compared to the 1D case (Fig. 4.20(b)). Radiation can propagate horizontally from optically thinner regions to thicker cloud regions, where it is more likely to be attenuated and leading to UT with reduced upward flux in 3D compared to 1D RT. This structure is reflected in Fig. 4.21, with generally thicker ice cloud regions (dark blue) located to the right of 10°N, while thinner regions (light blue) are found to the left.

Radiation in thicker cloud regions can propagate horizontally into thinner regions in 3D RT, where it has a higher probability of escaping the cloud, i.e., UE (Várnai and Davies, 1999). This results in increased upward flux, as seen in Fig. 4.20(c) above the cloud between 9.94-10°N as positive 3D-1D bias.

4.6.4 Entrapment

Entrapment occurs when upward-traveling radiation in 3D RT is intercepted by overlying clouds and is redirected downward or absorbed, while in 1D RT it escapes more easily, thereby reducing cloud reflectance in 3D compared to 1D RT (Hogan et al., 2019) (see Sect. 2.2.3). Since DE and UT were originally proposed as mechanisms contributing to entrapment (Hogan et al., 2019), they are identified here to show the presence of entrapment.

Entrapment is shown in Fig. 4.22(a)-(f). The upper panels (a)-(c) show downward fluxes, and the lower panels (d)-(f) show upward fluxes. MYSTIC results are shown to the left (a, d), DISORT results in the middle (b, e), and their difference (3D-1D) is shown to the right (c, f). Because entrapment affects both upward and downward radiation, both flux fields are presented.

Consistent with reduced cloud reflectance due to entrapment (Hogan et al., 2019), MYSTIC simulates lower upward SW flux above the cloud-top at around 11 km than DISORT. This appears as a negative 3D-1D bias (blue) above the cloud in Fig. 4.22(f).

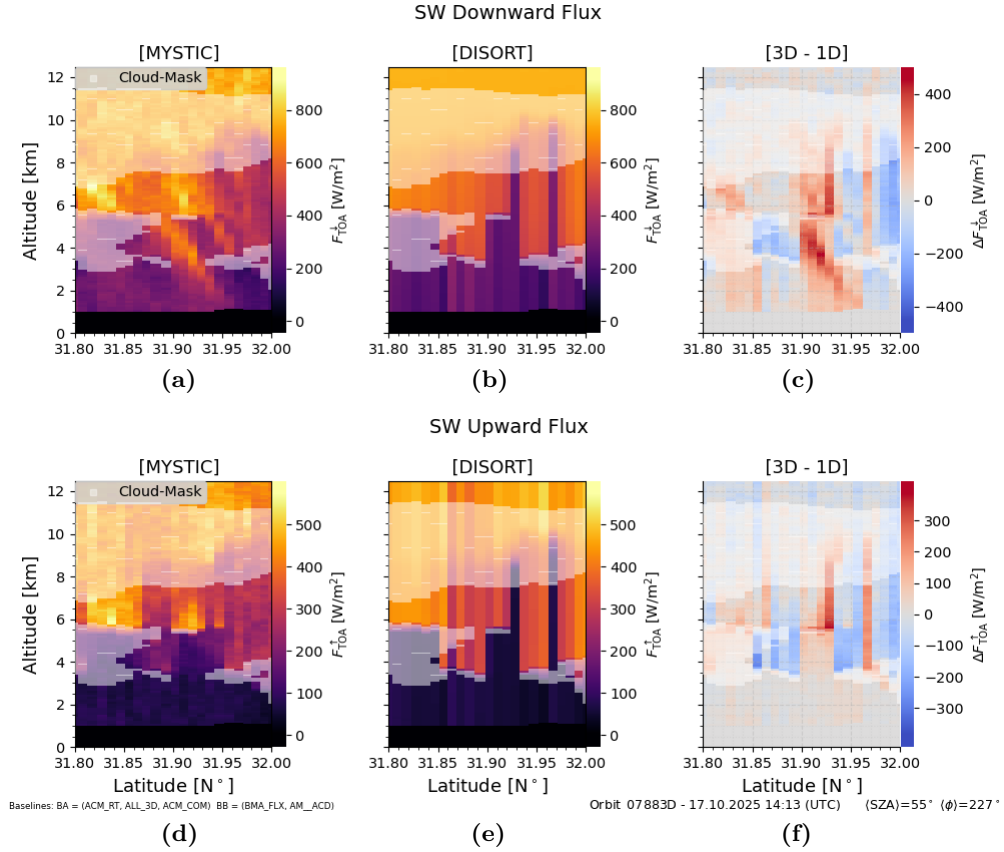


Figure 4.22: Simulated SW flux fields illustrating entrapment. Downward fluxes are shown in the upper panels and upward fluxes in the lower panels. The three columns display MYSTIC, DISORT, and the 3D - 1D flux difference from left to right.

This reduction implies that parts of the radiation that would escape upward in 1D RT are instead trapped in 3D RT.

A reduced cloud-top 3D upward flux indicates the presence of UT, which decreases reflectance (Várnai and Davies, 1999). However, DE can also contribute, as in 3D RT, radiation can escape horizontally through cloud sides and reach the surface, whereas in 1D RT it remains confined within the column and is more likely to be scattered back to space (Hogan et al., 2019). Indeed, increased downward SW fluxes in MYSTIC, compared to DISORT, are observed between the cloud layers. This is seen as positive 3D-1D biases (red) in Fig. 4.22(c), consistent with DE.

Both increased downward flux with 3D RT from DE and reduced upward flux from UT are considered in this example to describe entrapment. The increased downward radiation below the upper cloud layer in MYSTIC, together with reduced cloud-top reflection, indicates that radiation is both redirected downward and less likely to escape to space. Given the multilayer cloud structure, where entrapment is particularly relevant (Hogan et al., 2019), these features support the presence of entrapment in the flux field shown in Fig. 4.22.

4.7 Future Prospects

Several open questions and possible improvements have appeared during this work, pointing to directions for future study. These relate both to the RTM itself and to

the radiative assessment. The first set of possible improvements concerns the RTM setup and the physical processes represented in the simulations. These are listed below:

- **Surface BRDF** can be included to describe the angular dependence of surface reflectance, which is beneficial because surface reflection is often not Lambertian (Mayer, 2009). In libRadtran, BRDF is not currently supported together with topography in the 3D solver (Mayer et al., 2025b). Surface BRDF is therefore mainly applicable to the 1D solver, or to ocean cases in the 3D solver where topography can be turned off.
- **Aerosol type specification** would allow a more realistic representation of how aerosols interact with radiation by assigning different optical properties to different aerosol types.
- **The pseudospherical approximation** can improve the 1D solver at high SZA. As described in Mayer and Kylling (2005), plane-parallel geometry is generally a good approximation for $\text{SZA} < 70^\circ$ (Dahlback and Stamnes, 1991). For lower solar elevations, the pseudospherical approximation is an appropriate solution, treating the direct solar beam in spherical geometry to account for Earth’s curvature, while multiple scattering remains in plane-parallel geometry.

In general, including more swaths would improve the statistical robustness of the results. Further analysis of challenging areas, such as Arctic regions specified in the NEVAR project (NILU, 2022–2027), would also be beneficial.

Throughout this Master’s thesis, different assessment domains are used, defined as horizontal pixel areas over which fluxes are averaged. These include single-pixel domains (1×1) and domains matching the BMA-FLX resolution using 11×11 pixels for MYSTIC and averaging over 10 consecutive pixels for DISORT. EarthCARE’s ACM-RT processor uses a different setup with a 21×5 pixel domain elongated along-track to emphasize the evaluation of the retrieved nadir cross-section while limiting reliance on the SCA in the across-track direction (Qu et al., 2023). The square 11×11 pixel domain leads to a stronger dependence on the SCA with about 91% of pixels relying on reconstructed across-track information compared to 80% for EarthCARE. This can include less reliable profiles as the SCA performance generally decreases with distance from nadir (Qu et al., 2023), and may in some cases be worse than cyclic boundary conditions. To keep flexibility in the simulation framework, all simulated fluxes are stored within the 21×21 pixel 3D buffer. This allows for future testing of different assessment domains, for example to compare with the 21×5 pixel domain of ACM-RT 3D (Qu et al., 2023) and BMA-FLX assessment resolution (Velázquez Blázquez et al., 2024).

Further work regarding the 3D buffer remains an interesting direction for future work. This includes testing the impact of the 3D buffer on periodic horizontal boundary conditions using the outer pixel frame with only a gaseous atmosphere and comparing different assessment domains to look into how uncertainties related to the SCA affect the results.

More comparison with products from the ACM-RT processor, including both 3D and 1D simulations, can provide an additional reference to observations. For ACM-RT 3D, the assessment domains need to be modified. ACM-RT 1D is useful due to its single-pixel flux resolution, and with the SCA, it can also be used in the surface analysis for comparison with across-track pixels.

Another potential extension of this work is an analysis of the cloud modification factor (CMF). It is defined as the ratio between cloudy and cloud-free surface downward

SW fluxes,

$$\text{CMF} = \frac{F_{\text{cloud}}}{F_{\text{clear-sky}}}, \quad (4.2)$$

where F_{cloud} is the flux under cloudy all-sky conditions and $F_{\text{clear-sky}}$ is the corresponding flux for a cloudless atmosphere (Schwander et al., 2002). Note that $F_{\text{cloud}} = F_{\text{clear-sky}}$ under cloud-free conditions. CMF can be calculated using observed or simulated all-sky fluxes together with simulated clear-sky fluxes. The clear-sky fluxes can be obtained from libRadtran simulations or from EarthCARE ACM-RT data, which provide both all-sky and clear-sky fluxes (Cole et al., 2023). Comparison with observations allows systematic identification of biases between simulations and observations under cloudy conditions, $\text{CMF} < 1$, and under clear-sky conditions, $\text{CMF} = 1$.

A useful extension of the surface analysis is to examine how the bias and standard deviation between simulated surface fluxes and observations vary with across-track distances from stations to satellite nadir. Such an analysis provides further insight into how the SCA performs as the distance from the nadir-track increases. A larger across-track distance threshold between EarthCARE overpasses and ground-based stations would also give a larger dataset of surface observations. This would allow a more accurate comparison with surface data, which is crucial for validation activities.

3D effects are assessed in this Master's thesis using flux fields along the nadir-track. The SCA allows this analysis to be extended to the across-track dimension, enabling fully 3D flux fields. This is technically and computationally feasible and remains an interesting direction for future work.

Chapter 5

Conclusion

In this study, multiple radiative assessments were performed using simulated fluxes from 3D and 1D RTMs with the libRadtran software package based on EarthCARE satellite data. The initial objective was to evaluate the 3D and 1D RTMs, MYSTIC (Mayer, 1999) and DISORT (Stamnes et al., 2000), using EarthCARE input data and observations of TOA fluxes. Further analyses included model-observation comparisons using surface measurements and model-model comparisons to assess sensitivities to aerosol loading and ice crystal habits. In total, several radiative assessments were performed, including 1) comparisons with EarthCARE radiometer data, 2) sensitivity analysis of aerosols, 3) sensitivity analysis of ice crystal habits, 4) evaluation of atmospheric input data, 5) surface flux evaluation, and 6) analysis of 3D effects.

The comparison of simulated and observed TOA fluxes for eight swaths (Sect. 4.1) shows that both 3D and 1D RTMs meet the EarthCARE radiative-closure target of $\pm 10 \text{ W/m}^2$ (Cole et al., 2023). The 3D RTM consistently reduces biases relative to the 1D RTM in both the SW and LW regions. Overall agreement is better in the LW region than in the SW region.

The analysis of the LW sensitivity to aerosol loading (Sect. 4.2) shows that changes in AOD lead to noticeable and non-negligible variations in TOA LW flux. Despite this sensitivity, the bias relative to BMA-FLX observations remains within the $\pm 10 \text{ W/m}^2$ EarthCARE target. The largest differences are found for over Africa, where the inclusion of coarse-mode absorbing aerosols yields a mean flux difference of approximately 6 W/m^2 . Local differences, however, exceed 25 W/m^2 in areas with high dust concentrations.

The sensitivity of TOA fluxes to ice crystal habits (Sect. 4.3) is mainly evident in the SW, while differences in the LW remain small. Relative to the general habit mixture, mean SW flux differences reach up to about 11 W/m^2 for rough-aggregate, while LW differences remain below about 1 W/m^2 for solid-column. All mean differences lie within or close to the EarthCARE radiative-closure target of $\pm 10 \text{ W/m}^2$. Local SW deviations range from $18\text{--}60 \text{ W/m}^2$, whereas LW deviations are limited to $2\text{--}5 \text{ W/m}^2$. The habit-induced differences depend on SZA and ice water path (IWP). Increasing SZA reduces the flux differences between habits, whereas increasing IWP reduces the differences only for $\text{IWP} \gtrsim 10^2 \text{ g/m}^2$, consistent with multiple scattering suppressing habit-specific features (Wendisch et al., 2005).

The atmospheric input data from ACM-COM to the RTMs come in two sets of atmospheric profiles: one based on the ACM-CAP “best-estimate” atmosphere and one representing the Level 2a composite atmosphere. Simulated TOA SW and LW fluxes for a single swath over Africa (Sect. 4.4) show that the ACM-CAP atmosphere gives

more realistic results and better agreement with observed TOA fluxes from BMA-FLX. The composite atmosphere produces rapid fluctuations in simulated fluxes and, in some places, large deviations from observations. In contrast, the ACM-CAP atmosphere shows good agreement with observations for both 3D and 1D RTMs, with biases reduced by a factor of about five and standard deviations by a factor of about three compared with the composite atmosphere.

The SW radiative assessment at the surface (Sect. 4.5), using ground-based pyranometer observations across Norway (seklima.met.no) and reconstructed across-track pixels using the scene construction algorithm (SCA) from EarthCARE, shows that both 3D and 1D RTMs exceed the EarthCARE $\pm 10 \text{ W/m}^2$ target, with biases in the 25-30 W/m^2 range. A corresponding TOA analysis using BMA-FLX observations shows better agreement, with reduced bias and standard deviation compared to the surface. However, the robustness of this comparison is limited by the small sample size, with only 16 overlapping cases between TOA and surface analyses.

The analysis of 3D effects on SW radiation due to clouds (Sect. 4.6) shows how the 3D RTM (MYSTIC) represents RT in inhomogeneous cloud scenes compared with the 1D RTM (DISORT). Observed 3D effects include side illumination, shadowing, upward and downward escape, upward trapping (Várnai and Davies, 1999), and entrapment (Hogan et al., 2019). Consistent non-zero 3D-1D biases in vertical flux profiles along the satellite nadir-track demonstrate the importance of horizontal photon transport in affecting the fluxes. Quantitatively, slanted shadows lead to differences of about 30-40% in SW downward fluxes, while downward escape can produce differences of up to about 50% between 3D and 1D simulations. Although the analysis focuses on local features at scales of around 1 km and does not quantify domain-averaged impacts, the results highlight the need for 3D SW RT at km scales for realistic simulations of atmospheric RT. This may seem overly optimistic, but Monte Carlo RT methods are becoming more possible for solving 3D RT in atmospheric models, although this remains a long-term perspective (Villefranque et al., 2023).

Overall, the results show that atmospheric RT is complex and depends on several factors, including cloud structure, aerosol loading, and microphysical properties. The consistent improvement of 3D RT simulations over 1D RT simulations, when compared with flux observations, highlights the added value of representing atmospheric inhomogeneity and horizontal radiation transport. The biases between models and observations, particularly at the surface, underline the importance of accurate atmospheric input data. The sensitivity to atmospheric constituents, such as aerosols and cloud microphysics, further shows that their representation in RTMs is important for accurate atmospheric modeling. In addition, the analysis of 3D effects shows that horizontal transport of radiation can strongly affect local fluxes. Together, these findings show that realistic radiative transfer modeling is constrained both by the ability to prescribe accurate atmospheric input data and by the ability to simulate radiation across all three dimensions.

Chapter 6

Appendix

6.1 Computational Performance

Computational performance is reported to support reproducibility and to quantify the magnitude of the computational workload. Since the MYSTIC simulations account for the largest share of this workload in this Master’s project, an *effective processing time per pixel per processor*, T_{eff} [minutes], is reported. The metric is based on the number of simulated libRadtran input files, i.e., the sample grids that form the 3D domains/3D buffers. These grids correspond to the EarthCARE satellite nadir pixels for which fluxes are simulated. Thus, one pixel corresponds to one 3D domain in this context. T_{eff} quantifies the wall-clock time required to process one such pixel (3D domain) per processor, providing a measure of computational workload:

$$T_{\text{eff}} = \frac{t_{\text{wall}}}{N_{\text{along-track}}/N_{\text{processors}}}, \quad (6.1)$$

where t_{wall} is the wall-clock runtime, $N_{\text{along-track}}$ is the number of simulated 3D domains, and $N_{\text{processors}}$ is the number of processors used. This metric includes all overhead and therefore reflects user-perceived performance, not only photon propagation over the 3D domains. It provides a swath-averaged measure of performance because the computational time in MYSTIC depends on scene properties, such as clouds, which affect the number of scattering events and therefore the computational time ([Mayer, 2009](#)).

Before presenting the observed performance, it is useful to summarize the simulation setup to which these results apply. All computations were carried out with MYSTIC with $N_{\text{photons}} = 3.2 \cdot 10^7$ photons in the SW region and $N_{\text{photons}} = 1.2 \cdot 10^4$ photons in the LW region. Both SW and LW calculations were executed together within the same simulation job. The 3D buffer size was 21×21 pixels, as used throughout this Master’s project.

All simulations presented within this section were executed on the NILU HPC cluster using one compute node with 24 processors. The performance metric T_{eff} is reported per processor because the number of processors used for individual simulation jobs depended on processor availability on the NILU cluster. Depending on availability, the jobs ran on nodes equipped with Intel Xeon (Cascade Lake) CPUs and either 512 GB or 1 TB of memory.

Table 6.1 shows that the effective processing time per pixel per processor T_{eff} is consistently about 2.7-2.9 minutes across the four swaths. This means that simulating a full swath of approximately 5000 nadir pixels in both the SW and LW regions takes

Table 6.1: Effective processing time per pixel per processor T_{eff} for selected swaths.

Swath	T_{eff} [minutes]
Orbit 07883D	2.72
Orbit 06888C	2.88
Orbit 07277C	2.73
Orbit 06331C	2.67

around 460 hours, or about 20 days, on a single processor. These results show that MYSTIC is the largest contributor to the total computational workload in this Master’s project and that Monte Carlo RTMs can have substantial computational costs from tracing large numbers of photons through cloudy atmospheres.

6.2 Monte Carlo Noise and 3D Buffer Size

With Monte Carlo RTMs, such as MYSTIC, the results are inherently noisy (Mayer, 2009). The following discussion explains the reasoning behind the 3D RTM setup applied in this study, highlighting how Monte Carlo noise in the context of 3D buffer size influence the radiative outputs.

The Monte Carlo noise, quantified here by the Monte Carlo standard deviation, depends on the sample grid size, i.e., the horizontal pixel extent of the 3D buffer, with N_x pixels across-track and N_y pixels along-track. The precision of a radiative quantity decreases as $1/\sqrt{N_x N_y}$ (Mayer et al., 2025b). Monte Carlo noise also depends on the number of photons (Mayer, 2009), so a larger 41×41 pixel 3D buffer was deemed impractical for simulations using a total of $3.2 \cdot 10^7$ photons in SW and $1.2 \cdot 10^4$ photons in LW.

Test runs with a larger 3D buffer of 41×41 pixels yield Monte Carlo standard deviations of roughly 20 W/m^2 in the SW region and 2 W/m^2 in the LW region. Using the smaller 21×21 pixel 3D buffer, as applied throughout this Master’s project, reduces the Monte Carlo standard deviations to approximately 6 W/m^2 in the SW region and 0.5 W/m^2 in the LW region.

6.3 Conversion of Atmospheric Composition to Number Densities for RTMs

The ACM-COM processor reports gaseous profiles as volume mixing ratios. These were converted to molecular number densities for use in the libRadtran RTMs, following the theoretical basis given in Lamb and Verlinde (2011). This procedure is described below.

Assuming ideal gas behavior, the air molecular number density n_{air} was derived from pressure and temperature as

$$n_{\text{air}} = \frac{A_v P}{RT}, \quad (6.2)$$

where P denotes pressure, T temperature, R the universal gas constant, and A_v Avogadro’s number. Molecular number densities n_x of the gaseous constituents x were subsequently computed from the obtained gaseous volume mixing ratio y_x (mole fraction [mol/mol]) according to

$$n_x = y_x n_{\text{air}}. \quad (6.3)$$

6.3. Conversion of Atmospheric Composition to Number Densities for RTMs

In this study, x represents O_3 , O_2 , and CO_2 .

Water vapor requires separate treatment, as it is reported as specific humidity in the ACM-COM product. Accordingly, specific humidity values [kg/kg] were converted to volume mixing ratios [mol/mol] before applying the same procedure outlined above. The specific humidity q_v is related to the volume mixing ratio of water vapor y_v through

$$q_v \equiv \frac{\rho_v}{\rho_{\text{air}}} = \frac{n_v M_v}{n_d M_d + n_v M_v} = \frac{\varepsilon y_v}{1 - y_v(1 - \varepsilon)} \approx \varepsilon y_v, \quad (6.4)$$

where $\varepsilon = M_v/M_d = 0.622$ is the ratio of the molar masses of water vapor, M_v , and that of dry air, M_d . ρ_v and ρ_{air} denote the mass densities [kg/m³] of water vapor and moist air, respectively, and n_d and n_v represent the molar number densities [mol/m³] of dry air and water vapor, respectively. The final approximation was adopted such that the water vapor volume mixing ratio was obtained as $y_v = q_v/\varepsilon$

Table 6.2: Glossary of symbols.

Symbol	Name
I_ν	Spectral radiance (monochromatic intensity)
I_ν^+	Upward radiance
I_ν^-	Downward radiance
F_ν	Spectral flux (irradiance)
F	Broadband flux (SW or LW)
$F_{\text{TOA}}^\uparrow$	Upward flux at TOA
$F_{\text{SUR}}^\downarrow$	Downward flux at the surface
ν	Wavenumber
Ω	Propagation direction (solid angle)
θ	Zenith angle
ϕ	Azimuth angle
μ	Cosine of zenith angle ($\mu = \cos \theta$)
κ_{ext}	Volume extinction coefficient
τ_ν	Spectral optical depth
τ_z	Vertical optical depth
σ_a	Absorption coefficient
σ_s	Scattering coefficient
ω	Single-scattering albedo
p_ν	Scattering phase function
g	Asymmetry parameter ($\langle \cos \Theta \rangle$)
S_ν	Spectral source function in RTE
S_{th}	Thermal source term
S_{sc}	Scattering source term
B_ν	Planck function
T	Temperature
ϵ	Directional emittance
α	Directional absorptance
ρ	Surface reflectance (albedo)
w_j	Quadrature weight
u_j	Discrete ordinate

Table 6.3: List of abbreviations.

Abbreviation	Meaning
RT	Radiative Transfer
RTM	Radiative Transfer Model
RTE	Radiative Transfer Equation
SW	Shortwave
LW	Longwave
TOA	Top of Atmosphere
SZA	Solar Zenith Angle
AOD	Aerosol Optical Depth
IPA	Independent Pixel Approximation
ICA	Independent Column Approximation
GHM	General Habit Mixture (ice habit)
SC	Solid Column (ice habit)
RA	Rough Aggregate (ice habit)
LWC	Liquid Water Content
IWC	Ice Water Content
IWP	Ice Water Path
CMF	Cloud Modification Factor
BRDF	Bidirectional Reflectance Distribution Function
DEM	Digital Elevation Model
ESM	Earth System Model
GCM	Global Climate Model
ATLID	ATmospheric LIDar
CPR	Cloud-Profiling Radar
MSI	MultiSpectral Imager
BBR	BroadBand Radiometer
BMA-FLX	EarthCARE - Flux products (Velázquez Blázquez et al., 2024)
ACM-RT	EarthCARE - RT products (Cole et al., 2023)
ACM-COM	EarthCARE - Atmospheric data (Cole et al., 2023)
ACMB-3D	EarthCARE - 3D product (Qu et al., 2023)
AM-ACD	EarthCARE - Aerosol properties (Haarig et al., 2023)
SCA	Scene Construction Algorithm (Barker et al., 2011)
CERES	Clouds and the Earth's Radiant Energy System
MODIS	Moderate Resolution Imaging Spectroradiometer
RRTMG	Rapid Radiative Transfer Model for GCMs
NCCS	Norwegian Centre for Climate Services

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